

Construction of Arithmetic Teichmüller Spaces I

Preliminary version for comments

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Then let winged Fancy wander
Through the thought still spread beyond her:
Open wide the mind's cage-door,
She'll dart forth, and cloudward soar.
O sweet Fancy! let her loose;

John Keats [John Keats]

1 Introduction

The present series of papers ([Joshi, 2020a, 2021a, 2022, 2023b,a]) lays the foundations of the Theory of Arithmetic Teichmüller Spaces of Algebraic Varieties over Number Fields.

As is well-known, from [Dedekind and Weber, 1882 (Translation 2012)], one has an analogy between Riemann surfaces and number fields in which the field, $\mathbb{C}(\Sigma)$, of meromorphic functions on a connected, compact Riemann surface Σ corresponds to a number field L :

$$\mathbb{C}(\Sigma) \longleftrightarrow L.$$

This analogy has played a central role in the evolution of Number Theory and Algebraic Geometry over the past hundred and fifty or so years since its inauguration.

Classical Teichmüller Theory ([Ahlfors, 1987], [Lehto, 1987], [Nag, 1988], [Imayoshi and Taniguchi, 1992]) provides a refined view of Riemann surfaces, extending Riemann's moduli Theory (some of the salient results of Classical Teichmüller Theory are recalled here in Section 2). Therefore, it is not unreasonable to ask if a Teichmüller Theory exists for number fields.

That such a Teichmüller theory exists was first asserted, but with far lesser quantitative precision, by Shinichi Mochizuki in [Mochizuki, 2021a,b,c] (the relationship with Mochizuki's Theory is discussed in § 10 and in substantial detail in [Joshi, 2024c] which provides a 'Rosetta Stone' for [Mochizuki, 2021a,b,c,d]). Earliest indication, independent of Mochizuki's work, that such a theory does indeed exist came from [Joshi, October 2020] which approached the problem from an analytic function theory *optik*. The present paper and [Joshi, 2022] (older version [Joshi, October 2020]) deal with the arithmetic Teichmüller Spaces of varieties over p -adic fields and number fields and are an elaboration of the ideas of [Joshi, October 2020]; [Joshi, 2023a] deals with the construction of Arithmetic Teichmüller Space of a fixed number field. Other papers: [Joshi, 2023b], [Joshi, 2024c], [Joshi, 2024b] deal with the principal application of this theory envisaged by Mochizuki in [Mochizuki, 2021a,b,c,d].

The goal of the present series of papers is to create, for each number field L (assumed for simplicity to have no real embeddings) and for each geometrically connected, smooth, quasi-projective variety X/L over a number field L , and for each place $v \in \mathbb{V}_L$,

- (1) a v -adic analog of the Classical Teichmüller Space, more precisely a category $\mathfrak{J}(X, L_v)$ (Definition 5.1.1 for $v \in \mathbb{V}_L^{non}$ and Definition 5.2.1 for $v \in \mathbb{V}_L^{arc}$; in dimension one, for $v \in \mathbb{V}_L^{arc}$, one obtains classical Teichmüller spaces), of X/L_v , such that objects of $\mathfrak{J}(X, L_v)$ can be distinguished from each other (by means of their *Arithmetic Holo-morphic Structures* (Definition 4.1.1) and in general the v -adic Berkovich analytic space

structure of functions changes in a manner similar to classical Teichmüller space (Theorem 5.3.1); for $v \in \mathbb{V}_L^{arc}$, the Banach structure of rings of functions varies (Theorem 2.5.1, Theorem 5.2.5). But all objects of $\mathfrak{J}(X, L_v)$ have isomorphic tempered fundamental groups (Proposition 5.5.1 for $v \in \mathbb{V}_L^{non}$ and Theorem 5.2.5 for $v \in \mathbb{V}_L^{arc}$) and hence these, and other results proved here, justify the moniker ‘*Arithmetic Teichmüller space of X/L_v .*’

- (2) The construction of (1) provides an Adelic Arithmetic Teichmüller Space i.e. an adelic category $\widetilde{\mathfrak{J}(X, L)}$ (Theorem 9.3.1) built from the local Arithmetic Teichmüller Spaces $\mathfrak{J}(X, L_v)$ (for each $v \in \mathbb{V}_L$) together with global compatibilities, forced by the constructions of (3), which justify the moniker ‘*Adelic Arithmetic Teichmüller space of X/L .*’
- (3) As a special case, taking $X = \text{Spec}(L)$, the construction of (1,2) provides a topological space, with a certain list of properties, which justify the moniker ‘*Arithmetic Teichmüller Space of the number field L .*’ [This construction is detailed in [Joshi, 2023a].]

In the texts on Teichmüller Theory, Classical Teichmüller Spaces are constructed by considering quasi-conformal mappings between Riemann surfaces instead of conformal mappings. A key property of quasi-conformal mappings is that they are not necessarily holomorphic mappings, and secondly under such mappings, locally i.e. in a neighborhood a point, metrics are scaled or distorted ([Ahlfors, 1987]). In the context of ((1), (2)) above, there is an analogous notion of v -adic quasi-conformal mappings and under such mappings, local i.e. v -adic metrics are distorted or scaled relative to each other (for each $v \in \mathbb{V}_L$). This is proved here in Theorem 5.4.1, Corollary 5.4.2.

In the adelic context, global arithmetic of a number field enters through the validity of the product formula of [Artin and Whaples, 1945]. In the present case, the global compatibility of (2,3) is forced (in [Joshi, 2023a]) by the requiring that the (global) product formula holds for normalized v -adic metrics (for all $v \in \mathbb{V}_L$). [This strategy of using the product formula to globalize the adelic theory is also employed in [Mochizuki, 2021a,b,c].] However a key feature of the theory is that one cannot uniformly fix the global normalization at all points of the Arithmetic Teichmüller Space i.e. the choice of normalized v -adic metrics (for all $v \in \mathbb{V}_L$) cannot be uniformly fixed for all objects parameterized by the Arithmetic Teichmüller Space $\widetilde{\mathfrak{J}(X, L)}$ [Mochizuki’s assertion to this effect is [Mochizuki, 2021a, Example 3.5, Remark 3.5.1]]. In particular, after [Joshi, 2023a], it is possible to quantify the phrase “there are many distinct *avatars* or versions of the arithmetic of a fixed number field L ,” (this is claimed in [Mochizuki, 2021a, Page 24]) and also to precisely quantify that in any two avatars, relative to each other, L appears stretched or shrunk in a precise topological sense (at some or all of its primes). Since number theorists have always viewed any given number field as a rigid object on which the drama of Diophantine Geometry unfolds, the existence of such stretchings or shrinkings may seem quite tantalizing but is mathematically quite robust [Joshi, 2023a] (and this is an important feature common with Classical Teichmüller Theory). In particular, after the present paper and [Joshi, 2023a], the arithmetic of a fixed number field L itself (in the precise sense of [Joshi, 2023a]) can be treated as a dynamical variable, and this opens the possibility of applying techniques of Arithmetic Dynamics [Silverman, 2007] to this dynamical variable.

As noted earlier, the possibility that an Arithmetic Teichmüller Theory of the type (1,2,3) exists is suggested by Shinichi Mochizuki for the special case of elliptic curves in [Mochizuki, 2021a,b,c]. His proof of the *abc*-conjecture in [Mochizuki, 2021d] remarkably uses the idea of averaging over such Teichmüller data (for each fixed number field L). From the standpoint of this paper, Mochizuki's strategy of proof of the *abc*-conjecture may be understood as averaging over the adelic Arithmetic Teichmüller Space or a suitably chosen subset of it. [The central and foundational difficulty with [Mochizuki, 2021a,b,c,d] has been that of unambiguously ascertaining the claimed existence of Teichmüller data. A detailed discussion of this can be found in [Joshi, 2024c, § 1.11], [Joshi, 2024a, § 1].] It should be noted that averaging over Classical Teichmüller spaces is a quite well-established paradigm [Wright, 2019]. The present series of papers independently establishes, as special cases, the claims of [Mochizuki, 2021a,b,c] and thus allowing one to arrive at the proof of the *abc*-conjecture [Mochizuki, 2021d], [Joshi, 2024b].

It should be noted that Mochizuki neither suggests nor constructs Arithmetic Teichmüller Spaces in his work. Mochizuki attempts a purely group theoretic approach in establishing his theory. My approach to establishing the Theory of Arithmetic Teichmüller Spaces is based on entirely different set of ideas from those espoused by Mochizuki and is grounded in holomorphic function theory, wherein group theory manifests itself as symmetries of spaces on which the function live. Thus my approach puts the theory on par with classical Teichmüller Theory from the very beginning.

On the other hand, let me also say that, regardless of how one wants to approach the issue of establishing the existence of Arithmetic Teichmüller Spaces using either

- (1) the present series of papers ([Joshi, 2020a, 2021a, 2022, 2023b,a]), or
- (2) *Mochizuki's Key Principle of Inter-Universality* [Mochizuki, 2021a, § I3, Pages 25–26] (which underlies his theory, but treated correctly as discussed in § 10.2), or
- (3) the theory of tempered fundamental groupoids (this is essentially (2)), or
- (4) the Theory of Diamonds [Scholze, 2017] (Section 4.3),

all point to the existence of a unique (Arithmetic) Teichmüller Theory which is Arithmetic-Geometric in origin, and is as established in the present series of papers ([Joshi, 2020a, 2021a, 2022, 2023b,a]).

Readers may also find [Joshi, 2025], [Joshi, June, 2024], [Joshi, 2023c], [Joshi, 2021b] discussing [Mochizuki, 2021a,b,c,d] and [Scholze and Stix, 2018] useful.

§ 1.1 The present paper is a substantially revised version of all the previous release of this paper—notably, the definition of arithmetic holomorphic structure (Definition 4.1.1) (resp. Arithmetic Teichmüller Space, Definition 5.1.1) is now aligned with those given in [Joshi, 2022, Definition 4.1] ([Joshi, 2022, Definition 5.1]). The archimedean case (for varieties of higher dimensions) given here in Definition 5.2.1 is far more precise than that given in the previous versions.

§ 1.2 Acknowledgments The theory presented here is inspired by Mochizuki's work on anabelian geometry and on p -adic Teichmüller Theory so my intellectual debt to Shinichi Mochizuki cannot be overstated. *Notably this paper could not have existed without his work.* I thank Peter Scholze for some correspondence on early versions of Theorem 3.10.1 and Theorem 3.16.1. Thanks are due to Yuichiro Hoshi for answering many of my elementary questions about tempered fundamental groups and [Mochizuki, 2021a,b,c,d]; also to Taylor Dupuy and Anton Hilado for some conversations about [Mochizuki, 2021a,b,c,d] during the early days of this work and for providing early versions [Dupuy and Hilado, 2020b], [Dupuy and Hilado, 2020a] and [Dupuy, 2021]. I thank Kiran Kedlaya for correspondence, comments and encouragement. Another expert with whom I corresponded, has chosen to remain anonymous, but our correspondence is acknowledged here with pleasure. I thank Jacob Stix for providing a correction to § 3.9 and for answering a question regarding [Schmidt and Stix, 2016]. I also thank Emmanuel Lepage for some correspondence in the context of [Lepage, 2010].

2 Classical Teichmüller Theory

§ 2.1 Classical Teichmüller Theory in a Nutshell Since the principal result of this paper is the demonstration of the existence of a p -adic Arithmetic Teichmüller Theory, it will be useful to recall the salient theorems of Classical Teichmüller Theory (these are organized in Theorem 2.1.1 stated below). This list of results, and others, be gleaned from standard texts on Classical Teichmüller Theory ([Lehto, 1987], [Nag, 1988], [Imayoshi and Taniguchi, 1992]). This theorem is the one which my work imitates in the arithmetic case.

Theorem 2.1.1. *Let Σ be a connected, compact Riemann surface of topological type $g \geq 0$ and let $\mathcal{T}_g = \mathcal{T}(\Sigma)$ be the Teichmüller space of Σ and let $\mathcal{M}_g$ be the moduli of Riemann surfaces. Then*

- (1) $\mathcal{T}_g$ parameterizes, up to a suitable equivalence hereafter dubbed Teichmüller equivalence, pairs (Σ', f) consisting of a compact Riemann surface Σ' and a non-constant quasi-conformal mapping $f : \Sigma' \rightarrow \Sigma$ [Lehto, 1987, Chapter V, 2.1]. Note that, in general, f need not be a holomorphic mapping (and hence may be a non-algebraic mapping), but f is a homeomorphism, so Σ' has the same topological type as Σ . Importantly, $\mathcal{T}_g$ is not an algebraic variety or a scheme.]
- (2) The space $\mathcal{T}_g = \mathcal{T}(\Sigma)$ can also be described as the set of equivalence classes (the equivalence relation is not recalled here) of pairs (Σ, ds^2) consisting of the Riemannian manifold Σ equipped with an arbitrary Riemannian metric ds^2 ([Imayoshi and Taniguchi, 1992, Theorem 1.8]).
- (3) For $(\Sigma', f) \in \mathcal{T}_g$, let $[\Sigma']$ denote the bi-holomorphism class i.e. the isomorphism class of the Riemann surface Σ' . Then one has a natural, surjective mapping $\mathcal{T}_g \rightarrow \mathcal{M}_g$ given by $(\Sigma', f) \mapsto [\Sigma']$.
- (4) There exists an infinite discrete group, denoted Γ_g and called the Teichmüller modular group or the mapping class group, which acts freely on $\mathcal{T}_g$ and $\mathcal{T}_g \rightarrow M_g$ is the quotient mapping $\mathcal{T}_g \rightarrow \mathcal{T}_g/\Gamma_g = \mathcal{M}_g$ ([Imayoshi and Taniguchi, 1992, Chapter 6]).

- (5) For $g = 1$ (elliptic curves), $\mathcal{T}_1 = \mathfrak{H}$ is the upper half-plane, $\Gamma_1 = \mathrm{PSL}_2(\mathbb{Z})$, and $\mathcal{M}_1 = \mathfrak{H}/\mathrm{PSL}_2(\mathbb{Z}) = \mathbb{C}$ [Imayoshi and Taniguchi, 1992, Chapter 1].
- (6) The fiber of the natural morphism $\mathcal{T}_g \rightarrow \mathcal{M}_g$ over the isomorphism class of $[\Sigma] \in \mathcal{M}_g$ provides a collection of pairs $(\Sigma', f) \in \mathcal{T}_g$ for which $[\Sigma'] = [\Sigma]$.
- (7) There exists a Virasoro symmetry i.e. the Virasoro Uniformization Theorem (of [Kontsevich, 1987], [Beilinson and Schechtman, 1988]) holds (see [Frenkel and Ben-Zvi, 2001, Theorem 17.3.2] for an accessible exposition). [For more on this Virasoro symmetry see section 5.11.]
- (8) For every $(\Sigma', f) \in \mathcal{T}_g$, one has an isomorphism $\pi_1(\Sigma') \xrightarrow{\cong} \pi_1(\Sigma)$ between the fundamental groups of Σ' and Σ given by the homeomorphism underlying the quasi-conformal mapping $f : \Sigma' \rightarrow \Sigma$.

Remark 2.1.2. It should be noted that Theorem 2.1.1(7) is generally not a part of Classical Teichmüller Theory textbooks [Lehto, 1987, Nag, 1988, Imayoshi and Taniguchi, 1992]. On the other hand, one of the important aspect of this paper is the demonstration of the existence of the analog of the Virasoro action Theorem 2.1.1(7) in the theory of Arithmetic Teichmüller Spaces. [While, the underlying property (which I prove) appeared under the name of ‘Indeterminacy Ind2’ in Mochizuki’s work [Mochizuki, 2021c, Theorem 3.11 and Corollary 3.12], I recognized it as being the analog of the Virasoro Uniformization Theorem, and gave a precise formulation and proof in Theorem 5.21.1.]

§ 2.2 Classical Teichmüller Theory as a functor in Banach Algebras Classical Teichmüller Spaces admits description as a functor in infinite dimensional Banach algebras (see Theorem 2.5.1 below). This rests on a remarkable Banach ring theoretic characterization of quasi-conformal and conformal mappings of Riemann surfaces due to [Nakai, 1959, 1960, Nakai and Sario, 1970].

In particular, this implies that the p -adic theory of this paper based on the idea that the p -adic Banach algebra structure can be made to vary (Theorem 3.16.1, Theorem 5.3.1), has an archimedean counterpart in classical Teichmüller Theory. This puts the p -adic Arithmetic Teichmüller Theory of this paper on par with classical Teichmüller Theory. [I discovered Mitsuru Nakai’s aforementioned works after I had established [Joshi, October 2020].]

§ 2.3 Let Σ be a connected Riemann surface (open or closed). Recall (from [Royden, 1953], [Nakai, 1960]) that the Royden algebra $\mathcal{R}_\Sigma$ of a connected Riemann surface Σ is the $\mathbb{C}$ -algebra of all complex valued functions $f : \Sigma \rightarrow \mathbb{C}$ satisfying the following three properties:

- (1) f is absolutely continuous on Σ in the Tonelli sense (see [Nakai and Sario, 1970, Chap. III, 1A, Page 147] for the definition),
- (2) $|f|$ is bounded on Σ , and
- (3) the Dirichlet integral

$$D[f] = \int \int_{\Sigma} \left(\left| \frac{\partial^2 f}{\partial x^2} \right| + \left| \frac{\partial^2 f}{\partial y^2} \right| \right) dx \cdot dy < \infty.$$

For $f \in \mathcal{R}_\Sigma$, define

$$|f|_\Sigma = \sup_\Sigma |f| + \sqrt{D[f]}.$$

The following is proved in [Royden, 1953] [Nakai, 1959].

Theorem 2.3.1. *Let Σ be any connected Riemann surface.*

- (1) *The mapping $|-|_\Sigma : \mathcal{R}_\Sigma \rightarrow \mathbb{R}$ given by*

$$f \in \mathcal{R}_\Sigma \mapsto |f|_\Sigma$$

defines a norm on $\mathcal{R}_\Sigma$, and

- (2) *the normed algebra $(\mathcal{R}_\Sigma, |-|_\Sigma)$ is a commutative, complex Banach algebra.*

Remark 2.3.2.

- (1) Note that $\mathcal{R}_\Sigma$ is an infinite dimensional, commutative Banach algebra even if Σ is a compact Riemann surface.
- (2) Note that $\mathcal{R}_\Sigma$ is also equipped with several different topologies other than its norm-topology. The norm topology is the strongest among these topologies ([Nakai, 1960]).

§ 2.4 Let T_Σ be the Teichmüller space of Σ and let $\mathcal{T}_\Sigma$ be the category whose objects are the points of T_Σ i.e. its objects are Riemann surfaces $(\Sigma', f : \Sigma' \rightarrow \Sigma) \in T_\Sigma$. A morphism between $(\Sigma', f : \Sigma' \rightarrow \Sigma) \rightarrow (\Sigma'', g : \Sigma'' \rightarrow \Sigma)$ is a quasi-conformal mapping $h : \Sigma' \rightarrow \Sigma''$.

§ 2.5 The next theorem is my reformulation of main theorem of [Nakai, 1959, 1960] it provides a remarkable algebraic characterization of quasi-conformal mappings (the theory of [Scholze, 2019] espouses an algebraization of analysis in greater generality):

Theorem 2.5.1. *Let Σ, Σ' be two connected two Riemann surfaces. Then*

- (1) *Σ, Σ' are quasi-conformal if and only if one has a $\mathbb{C}$ -algebra isomorphism*

$$\mathcal{R}_\Sigma \simeq \mathcal{R}_{\Sigma'}$$

between their Royden algebras.

- (2) *Any $\mathbb{C}$ -algebra isomorphism $\mathcal{R}_\Sigma \simeq \mathcal{R}_{\Sigma'}$ is a homeomorphism for the respective norm-topologies.*
- (3) *Moreover, Σ and Σ' are conformally equivalent (i.e. the Riemann surfaces are isomorphic) if and only if*

$$(\mathcal{R}_\Sigma, |-|_\Sigma) \simeq (\mathcal{R}_{\Sigma'}, |-|_{\Sigma'})$$

is an isometry of Banach algebras.

- (4) *The construction of $(\mathcal{R}_\Sigma, |-|_\Sigma)$ is functorial for quasi-conformal mappings between Riemann surfaces.*

(5) In particular one has a functor

$$(\Sigma', f : \Sigma' \rightarrow \Sigma) \mapsto (\mathcal{R}_{\Sigma'}, |-\|_{\Sigma'})$$

from the Teichmüller category $\mathcal{T}_{\Sigma}$ to the category of infinite dimensional complex Banach algebras.

Proof. The first assertion is [Nakai, 1960, Theorem III.1], the second assertion is [Nakai, 1960, Theorem III.4], the third assertion is [Nakai, 1960, Theorem III.4(i)]. The fourth assertion is [Nakai, 1960, Theorem III.3(i)]. The last assertion should be clear from the previous four assertions. $\square$

Remark 2.5.2.

- (1) Thus quasi-conformality of Riemann surfaces has a purely algebraic characterization and conformality of Riemann surfaces has a purely Banach algebra theoretic characterization.
- (2) Note that by Theorem 2.5.1(5), $\mathcal{T}_{\Sigma}$ may be considered as an archimedean version of a Banach-Colmez Space of [Colmez, 2002].

Remark 2.5.3. Now fix a connected Riemann surface Σ and let $\mathcal{R} = \mathcal{R}_{\Sigma}$ be the Royden algebra of Σ . Let $f : \Sigma' \rightarrow \Sigma$ be a quasi-conformal mapping. Then using the isomorphism $\mathcal{R}_{\Sigma'} \rightarrow \mathcal{R}_{\Sigma}$ induced by f one can pull-back the norm $|-\|_{\Sigma'}$ of $\mathcal{R}_{\Sigma'}$ and write $(\mathcal{R}_{\Sigma}, |-\|_{\Sigma'})$ for the norm on $\mathcal{R}_{\Sigma}$ given using (Σ', f) . Thus for each $(\Sigma', f) \in T_{\Sigma}$, one has a Banach algebra norm on the fixed topological ring $\mathcal{R} = \mathcal{R}_{\Sigma}$. In particular classical Teichmüller theory provides a distinguished collection

$$\{(\mathcal{R}, |-\|_{\Sigma'}) : \Sigma' \in T_{\Sigma}\}$$

of Banach algebra norms on the fixed topological ring $\mathcal{R}$. Thus one may view classical Teichmüller Theory as deformations of Banach algebra structure on the fixed $\mathbb{C}$ -algebra $\mathcal{R}$ i.e. classical Teichmüller Theory is a variation of Banach ring structures. Notably, this observation brings the classical theory closer to the p -adic theory of Banach-Colmez Spaces [Colmez, 2002].

The constructions of the present paper can be viewed in a similar manner: as changes in the Banach ring structure of functions as one deforms the valued field $\mathbb{C}_p$.

3 The Absolute Grothendieck Conjecture is false for Berkovich Spaces

By Theorem 2.1.1(8), one knows that each $(\Sigma', f) \in \mathcal{T}_{\Sigma}$ has topological fundamental group isomorphic to that of Σ , and hence the fundamental group does not determine the datum $(\Sigma, f) \in \mathcal{T}_{\Sigma}$ nor does it determine the complex structure of Σ' . One can say, therefore, that the Absolute Grothendieck Conjecture is trivially false for Riemann surfaces because of the existence of distinct Teichmüller data (Σ', f) and this non-validity is a manifestation of the existence of distinct complex analytic structures.

The p -adic approach (to Arithmetic Teichmüller Spaces) of this paper began with a demonstration that the Absolute Grothendieck Conjecture is false in the category of Berkovich analytic spaces while the tempered fundamental groups remain isomorphic. This is established in Theorem 3.16.1. [A brief version of this result appeared in [Joshi, October 2020].] This leads to a precise definition of Arithmetic Holomorphic Structures in Section 4 and to the construction of Arithmetic Teichmüller Spaces in Section 5.

§ 3.1 *All valuations on base fields considered in this paper will be rank one valuations.* For the theory of tempered fundamental groups see [André, 2003a,b] or [Lepage, 2010]. As is noted in [André, 2003a], tempered fundamental groups are natural in the p -adic analytic contexts because they capture finite étale coverings *and* discrete coverings such as those arising from Tate or Mumford Uniformization available in the p -adic contexts. Berkovich spaces (see [Berkovich, 1990] and [Berkovich, 1993]) will be strictly analytic (and mostly will arise as analytifications of geometrically connected smooth quasi-projective varieties).

§ 3.2 In what follows I will work with algebraically closed, perfectoid fields of characteristic zero. A typical example of such a field is the completed algebraic closure $\mathbb{C}_p$ of $\mathbb{Q}_p$. Such fields can also be characterized in many different ways. For the convenience of the readers unfamiliar with perfectoid fields, the following lemma (immediate from [Scholze, 2012, Definition 3.1]), provides a translation of this condition into more familiar hypothesis.

Lemma 3.2.1. *Let K be a valued field and let $R \subset K$ be the valuation ring and assume that $|p|_K < 1$. The following conditions are equivalent:*

- (1) *K is an algebraically closed field, complete with respect to a rank one non-archimedean valuation with residue characteristic $p > 0$.*
- (2) *K is an algebraically closed, perfectoid field.*

Proof. A perfectoid field has residue characteristic $p > 0$ and is complete with respect to a rank one valuation. So (2) $\implies$ (1) is trivial. So it is enough to prove that (1) $\implies$ (2). I claim that Frobenius $\phi : R/pR \rightarrow R/pR$ is surjective. let $\bar{x} \in R/pR$ and suppose $x \in R$ is an arbitrary lift of $\bar{x}$. Then as K is algebraically closed, $x^{1/p} \bmod pR$ provides a lift of $\bar{x}$. As K is complete with respect to a rank one valuation and Frobenius is surjective on R/pR , so K is perfectoid by [Scholze, 2012, Definition 3.1] and by my hypothesis K also algebraically closed. This proves (1) $\implies$ (2). $\square$

§ 3.3 For a perfectoid algebraically closed field K as above, one has naturally associated field $K^\flat$, algebraically closed, perfectoid of characteristic $p > 0$, called the *tilt* of K and K is called an untilt of $K^\flat$ (see [Scholze, 2012, Lemma 3.4]).

§ 3.4 Fix an algebraically closed field, perfectoid F of characteristic $p > 0$ (see [Scholze, 2012]). For example readers can simply assume, without any loss of generality, that $F = \mathbb{C}_p^\flat$ as this case is quite adequate for my purposes.

§ 3.5 By an E -untilt of F , I will mean the data $(E \hookrightarrow K, K^\flat \simeq F)$ where $E \hookrightarrow K$ is an isometric embedding of a p -adic field E into a perfectoid field K , of characteristic zero, with an isometry $K^\flat \simeq F$. If $E = \mathbb{Q}_p$, I will simply say an untilt of F instead of $\mathbb{Q}_p$ -untilt of F .

Note that by [Scholze, 2012, Proposition 3.8] K is algebraically closed as its tilt $K^\flat = F$ is algebraically closed (by my hypothesis). If $F = \mathbb{C}_p^\flat$ then $K^\flat$ is isometric with $\mathbb{C}_p^\flat$. By the theory of [Fargues and Fontaine, 2018] untilts K of F exist and are parametrized by Fargues-Fontaine curves.

§ 3.6 Let E be a p -adic field which is fixed for the present discussion. I will work with untilts K , of F , equipped with continuous embeddings $E \hookrightarrow K$ with the valuation of K providing a valuation on E which is equivalent to the natural p -adic valuation on E . By [Fargues and Fontaine, 2018] for a given pair (F, E) , such fields $K \hookrightarrow E$, exist and are parametrized by Fargues-Fontaine curves (denoted here by $\mathcal{X}_{F,E}$). Without further mention, all untilts K will be assumed to be of this type (for our chosen p -adic field E).

§ 3.7 Crucial point for this paper is that *there exist untilts of $\mathbb{C}_p^\flat$ which are not topologically isomorphic*. This is the main result of [Kedlaya and Temkin, 2018, Theorem 1.3] (also see [Matignon and Reversat, 1984]). Note that all characteristic zero untilts of $\mathbb{C}_p^\flat$ have the cardinality of $\mathbb{C}_p$ and are complete and algebraically closed fields and hence are abstractly isomorphic fields but may not be topologically isomorphic after [Kedlaya and Temkin, 2018, Theorem 1.3] (also see [Matignon and Reversat, 1984]).

§ 3.8 Now fix a geometrically connected, smooth quasi-projective variety X/E , with E a p -adic field. Let X^{an}/E be the strictly analytic space associated to X/E . Let

$$(3.8.1) \quad \Pi_{X/E}^{temp} = \pi_1^{temp}(X^{an}/E)$$

be the tempered fundamental group of the strictly E -analytic space associated to X/E in the sense of [André, 2003b] or [André, 2003a].

(Note that my notation $\Pi_{X/E}^{temp}$ suppresses the passage to the analytification X^{an}/E for simplicity of notation. The theory of (tempered) fundamental groups also requires a choice of base point which will be suppressed from my notation for the moment.)

§ 3.9 The relationship between the tempered fundamental groups and the étale fundamental groups of geometrically connected, smooth quasi-projective varieties as follows.

Let X/E be a geometrically connected, smooth, quasi-projective variety over a p -adic field E . Then one has a natural homomorphism ([André, 2003b, Proposition 4.4.1], [André, 2003a, Section 2.1.4]):

$$\Pi_{X/E}^{temp} \rightarrow \pi_1^{et}(X/E),$$

which is injective if $\dim(X) = 1$ ([André, 2003a, Chapter III, Prop. 2.1.6]). In any dimension, the image of the above homomorphism is dense and moreover $\pi_1^{et}(X/E)$ is the profinite completion ([André, 2003b, Proposition 4.4.1]):

$$\widehat{\Pi_{X/E}^{temp}} = \pi_1^{et}(X/E).$$

§ 3.10 Let E'/E be a finite extension of E with a continuous embedding $E' \hookrightarrow K$ (as K is algebraically closed, valued field containing E , such E' exists). One can consider $X_{E'} = X \times_E E'$ (similarly $X_K = X \times_E K$). Then one has an exact sequence by [André, 2003a, Prop.

2.1.8]

$$1 \rightarrow \Pi_{X_{E'}/E'}^{temp} \rightarrow \Pi_{X/E}^{temp} \rightarrow \text{Gal}(E'/E) \rightarrow 1.$$

Let $\overline{E} \subseteq K$ be the algebraic closure of E contained in K .

By varying E' over all finite extensions of $E \hookrightarrow K$ one obtains (see [André, 2003b, Section 5.1]) an exact sequence of topological groups:

$$1 \rightarrow \varprojlim_{E'/E} \Pi_{X_{E'}/E'}^{temp} \rightarrow \Pi_{X/E}^{temp} \rightarrow \text{Gal}(\overline{E}/E) \rightarrow 1.$$

Theorem 3.10.1. *Let F be an algebraically closed perfectoid field of characteristic $p > 0$ (for example $F = \mathbb{C}_p^\flat$). Let E be a p -adic field. Let K, K_1, K_2 be arbitrary untilts of F with continuous embedding $E \hookrightarrow K$ (resp. into K_1 and K_2). Let $\overline{E}$ (resp. $\overline{E}_1, \overline{E}_2$) be the algebraic closure of E in K (resp. in K_1, K_2). Let X/E be a geometrically connected, smooth, quasi-projective variety over E . Then one has the following:*

(1) *a continuous isomorphism*

$$\Pi_{X/K}^{temp} \simeq \varprojlim_{E'/E} \Pi_{X/E'}^{temp},$$

where the inverse limit is over all finite extensions E' of E contained in K , and a

(2) *a short exact sequence of topological groups*

$$1 \rightarrow \Pi_{X/K}^{temp} \rightarrow \Pi_{X/E}^{temp} \rightarrow G_E \rightarrow 1,$$

and

(3) *In particular for any two untilts K_1, K_2 of F , one has a continuous isomorphism*

$$\Pi_{X/K_1}^{temp} \simeq \Pi_{X/K_2}^{temp}.$$

Proof. The assertion (1) is true assuming only that K is a complete algebraically closed field containing E isometrically and is due to [Lepage, 2010]. My own proof of (1), before I found the assertion in [Lepage, 2010], was by reworking of [André, 2003b, Prop. 5.1.1] for any K algebraically closed perfectoid field, and I was interested in proving (1) because I wanted to establish (3) (whose importance will become clear in Theorem 3.16.1 below). Here I provide an approach to the proof of (1) via the reduction to the principle of invariance of fundamental groups under extension of algebraically closed fields (also due to [Lepage, 2010]), for completeness. So (3) is the new and important observation here—from the point of view of Theorem 3.16.1 below.

Let me remind the reader that my hypothesis on K, K_1, K_2 imply that K, K_1, K_2 are algebraically closed and complete with respect to a rank one valuation.

Let me prove (1), this will also lead to (2). Since K is algebraically closed, it follows that K contains an algebraic closure $\overline{E}$ of E . Let $\tilde{E} \subseteq K$ be the closure (with respect to valuation topology of K) of $\overline{E}$.

It is clear that $\tilde{E} \supset \overline{E}$ is complete and algebraically closed field and $\tilde{E}$ contains the algebraic closure $\overline{E} \subset K$ of E contained in K as a dense subfield. In particular $\tilde{E}$ is the completion of

$\overline{E}$ with respect to the induced valuation. In other words $\widetilde{E}$ is a copy of the completion of an algebraic closure of E (usually denoted $\widehat{\overline{E}}$) equipped with an isometric embedding $\iota : \widehat{\overline{E}} \hookrightarrow K$ with $\iota(\widehat{\overline{E}}) = \widetilde{E}$. Hence $K/\widetilde{E}$ is an isometric extension of algebraically closed, complete valued fields (with rank one valuations).

Now one can apply the principle of invariance of fundamental groups under passage to extensions of algebraically closed fields. This principle is well-known for étale fundamental groups of proper varieties (see [Grothendieck, 1971, Exposé X, Corollaire 1.8]). For tempered fundamental groups (and X not necessarily proper) this principle is proved in [Lepage, 2010, Proposition 2.3.2]. Thus applying [Lepage, 2010, Proposition 2.3.2] to the extension $K/\widetilde{E}$ one has an isomorphism of topological groups

$$\Pi_{X_K}^{temp} \simeq \Pi_{X_{\widetilde{E}}}^{temp}.$$

On the other hand by [André, 2003b, Proposition 5.1.1], as $\widetilde{E}$ is the completion of the algebraic closure of $\overline{E} \subset K$ of E , one has an isomorphism

$$(3.10.2) \quad \Pi_{X/\widetilde{E}}^{temp} \simeq \varprojlim_{E'/E} \Pi_{X_{E'}/E'}^{temp}$$

and an exact sequence of topological groups

$$1 \rightarrow \Pi_{X/K}^{temp} \simeq \varprojlim_{E'/E} \Pi_{X_{E'}/E'}^{temp} \rightarrow \Pi_{X/E}^{temp} \rightarrow \text{Gal}(\overline{E}/E) \rightarrow 1.$$

This proves the assertions (1), (2) as claimed.

Let me now prove (3). The claimed isomorphism $\Pi_{X/K_1}^{temp} \simeq \Pi_{X/K_2}^{temp}$ follows from the fact that both the groups can be identified with $\varprojlim_{E'/E} \Pi_{X_{E'}/E'}^{temp}$ where the inverse limit is over all finite extensions of E'/E contained in K_1 (resp. K_2) and the fact that there is an equivalence between categories of finite extensions of E contained in K_1 and the category of finite extensions of E contained in K_2 , since finite extensions of E are given by adjoining roots of polynomials with coefficients in E and this data is independent of the embedding of E in K_1 or K_2 and moreover any abstract isomorphism of finite extensions of a complete discretely valued field is in fact an isometry—i.e. given a finite extension of E , $E' \hookrightarrow K_1$ contained in K_1 , there is an isometry $E' \hookrightarrow K_2$ and vice versa. $\square$

§ 3.11 The importance of working with algebraically closed perfectoid fields K_1, K_2 with isometric tilts $K_1^b \simeq K_2^b$ (i.e. with untilts of a fixed algebraically closed perfectoid field of characteristic $p > 0$) will become clear from Theorem 5.21.1 which will be proved later. Note that if K_1, K_2 are arbitrary algebraically closed perfectoid fields, then K_1^b and K_2^b need not be isometric. A simple example of this is given as follows. Let $\mathbb{C}_p^{\max} \supset \mathbb{C}_p$ be a maximally (i.e. spherically) complete field (i.e. a maximal immediate extension of $\overline{\mathbb{Q}}_p$) (see [Kaplansky, 1942], [Poonen, 1993] for the construction of $\mathbb{C}_p^{\max}$), then $\mathbb{C}_p^{\max}$ and $\mathbb{C}_p$ do not have isometric tilts other wise $\mathbb{C}_p^b \simeq \mathbb{C}_p^{\max^b}$ is also spherically (i.e. maximally) complete, which is certainly not the case. [The field extension $\mathbb{C}_p^{\max} \supset \mathbb{C}_p$ is of uncountable transcendence degree (both the fields have the same cardinality).]

§ 3.12 The passage from $\mathbb{C}_p$ to $\mathbb{C}_p^{\max}$ should be considered as a “vertical variation” of the algebraically closed perfectoid field because it involves extension of their tilts $\mathbb{C}_p^{\max b} \supset \mathbb{C}_p^b$ (because this is also of uncountable transcendence degree). On the other hand [Kedlaya and Temkin, 2018] shows that there is also a “horizontal” or an “iso-tilted or equi-tilted variation” possible in which the tilts stay fixed isometrically.

§ 3.13 Let E be a p -adic field. In [Berkovich, 1990, Section 2.3, Section 3.1], Berkovich constructs the category of *analytic spaces over E* (or more simply the category of *Berkovich spaces over E*) (a similar theory is also sketched in [Berkovich, 1993]). While I refer the reader to these references for the general case, let me recall what this means in the context I will use. By [Berkovich, 1990, Section 3.1] an *analytic space over E* is a K -analytic space for some valued field $K \supseteq E$ (with a rank one valuation) and equipped with an isometric embedding $E \hookrightarrow K$. Let X/E be a quasi-projective variety and let X^{an} denote analytification of X/E (in the sense of [Berkovich, 1990]). Thus the K -analytic space $X_K^{an} = X^{an} \times_E K$ is an analytic space over E . Let K_1, K_2 be two valued fields (with rank one valuation) containing E (isometrically). Let $X_{K_1}^{an} = X^{an} \times_E K_1$ and similarly define $X_{K_2}^{an}$. So one has two analytic spaces over E . By [Berkovich, 1990, Section 2.3, Section 3.1] one can consider the notion of (iso)morphisms $X_{K_1}^{an} \xrightarrow{\simeq} X_{K_2}^{an}$ of analytic spaces over E . Specifically, this reduces to defining the notion of (iso)morphisms between affinoid spaces over E . This is done as follows (in the notation of [Berkovich, 1990, Section 1.2]): if $(\mathcal{M}(A_i), A_i)$ are K_i -affinoid spaces over E , for $i = 1, 2$, then an (iso)morphism between them is given by a bounded (iso)morphism, $f : A_1 \xrightarrow{\simeq} A_2$, of Banach rings compatible with their structure as normed algebras over the valued field E (and the corresponding continuous (iso)morphism between the semi-norm spectra $\mathcal{M}(A_i)$). In particular if $E \supseteq \mathbb{Q}_p$ then one can consider (iso)morphisms of analytic spaces over $\mathbb{Q}_p$. Thus an isomorphism $X_{K_1}^{an} \xrightarrow{\simeq} X_{K_2}^{an}$ of analytic spaces over $\mathbb{Q}_p$ makes sense and is the p -adic analytic analog, of the notion of isomorphisms of schemes over $\mathbb{Z}$. [Note that this can be obviously formulated more generally, without assuming that X/E is quasi-projective, but I have restricted myself to the case I will use in Theorem 3.16.1 given below.]

§ 3.14 Let X/E be a geometrically connected, projective variety over a p -adic field and $E \hookrightarrow K$ an isometric embedding into a complete valued field with a rank one valuation. Then one has the (projective) analytic spaces X^{an}/E and X^{an}/K . Projectivity (though not essential for my argument) ensures, by [Berkovich, 1990, Chap 3], that a number of adjectives which may be applied to an analytic space, can be applied to both of these spaces: both are proper (hence separated, so quasi-separated), strict, good, compact (hence quasi-compact) i.e. covered by a finite number of affinoid open subsets and the construction below applies to analytic spaces which enjoy some of these properties (but not necessarily projectivity).

By definition of an analytic space, X_K^{an} is equipped with an atlas of affinoid opens. This data can be used to equip X_K^{an} with a sheaf of Banach algebras $\mathcal{O}_{X_K^{an}}$ (to be precise this means that for any quasi-compact open, $U \subset X_K^{an}$, the algebra $\mathcal{O}_{X_K^{an}}(U)$ is a Banach algebra which is functorial in such U with the following properties: (1) if $U = \mathcal{M}(A)$ is an affinoid open then $\mathcal{O}_{X_K^{an}}(U) = A$ and (2) if U is any quasi-compact open with $U = \cup_i U_i$ a finite cover by affinoids then $\mathcal{O}_{X_K^{an}}(U) \rightarrow \prod_i \mathcal{O}_{X_K^{an}}(U_i)$ is a closed embedding of Banach algebras (note that the Banach norms provided in this construction are not claimed to be unique (locally) but equivalent). This construction is detailed in [Temkin, 2015, Section 3.3.2, Section 4.1.2]. The

important point here is not the sheaf itself, but the fact that the spaces of local analytic functions acquire a Banach space structure, which agrees with the norm on constant functions i.e. on our field K , in a manner that is compatible with gluing of local analytic functions and independent of the gluing data. The most succinct way of expressing all this is to say that one has a sheaf of Banach algebras $\mathcal{O}_{X_K^{an}}$ on X_K^{an} for a suitable Grothendieck topology on X_K^{an} . This implies, in particular, that the ring $\Gamma(X_K^{an}, \mathcal{O}_{X_K^{an}})$ of global analytic functions on X_K^{an} is naturally a Banach algebra.

§ 3.15 Let me briefly sketch a proof of the above claims. Readers familiar with the construction of such a sheaf may skip this paragraph. By definition, an analytic space is equipped with an atlas of affinoid open subsets and some gluing data and the analytic space can be equipped with a Grothendieck topology using this data i.e. one restricts the notion of open subsets for the purpose of constructing sheaves. The construction of $\mathcal{O}_{X_K^{an}}$ uses this datum. The key tool in the construction of $\mathcal{O}_{X_K^{an}}$ is Tate's Acyclicity Theorem [Temkin, 2015, 3.3.2.1]. The following general facts about Banach algebras and Banach modules over Banach algebras will be useful to remember:

- (1) If A is any K -affinoid algebra then A is equipped with a norm (and even a power multiplicative norm, if one assumes additionally that A is reduced, which is certainly true in the case which I am concerned with here [Bosch et al., 1984, 6.2.4, Theorem 1], but the existence of some norm on A can always be inferred from the Gauss norm), equipping A with a structure of a Banach algebra (i.e. A is complete with respect to this norm), and any two norms on A are equivalent and moreover the restriction of this norm to $K \hookrightarrow A$ is the valuation norm on K . If $K \supseteq E$ is a complete valued subfield, then one can think of a K -affinoid algebra as an E -Banach algebra.
- (2) If A, B are Banach E -algebras then any E -linear homomorphism $f : A \rightarrow B$ is continuous if and only if it is bounded [Bosch et al., 1984, 2.1.8, Proposition 2].
- (3) Product of E -Banach algebras $A_1, \dots, A_n$, is also a E -Banach algebra with the obvious definition of a norm.

The sheaf $\mathcal{O}_{X_K^{an}}$ is constructed as follows (see [Temkin, 2015, Section 3.3.2, Section 4.1.2]). First consider the case of an affinoid open set. If $U = \mathcal{M}(A) \subset X_K^{an}$ is an affinoid open subset then $\mathcal{O}_{X_K^{an}}(U) = A$ is evidently a Banach algebra over E . If U is covered by a finite number of affinoid opens $U = \bigcup_{i=1}^n U_i$, with $U_i = \mathcal{M}(A_i)$ and $U_i \cap U_j = \mathcal{M}(A_{i,j})$ then using Tate's Acyclicity Theorem [Bosch et al., 1984, 8.2.1, Theorem 1] or [Temkin, 2015, 3.3.2.1]) one obtains the equality of Banach algebras $A = \ker(\prod A_i \rightarrow \prod A_{i,j})$ so one can indeed define the sheaf $\mathcal{O}_{X_K^{an}}$ using the rule $\mathcal{O}_{X_K^{an}}(U) = A$ on affinoids (provided in the atlas and the net of affinoids defining the analytic space X_K^{an}). Moreover, Tate's acyclicity theorem also shows, that this gives a sheaf of Banach algebras on $U = \mathcal{M}(A)$ which is independent of the choice of the covering.

Now suppose $U \subseteq X_K^{an}$ is an arbitrary quasi-compact open subset of X_K^{an} . Choose a finite covering of $U = \{U_i\}_{i=1}^n$ by affinoids with $U_i = \mathcal{M}(A_i)$, with $U_i \cap U_j = \mathcal{M}(A_{i,j})$. Then $\mathcal{O}_{X_K^{an}}(U)$ is the equalizer of the two restriction arrows

$$\prod A_i \rightrightarrows \prod_{i,j} A_{i,j}.$$

So $\mathcal{O}_{X_K^{an}}(U)$ is closed in the product and hence carries a natural structure of Banach algebra. This is independent of the choice of the covering: any two such covers of U have a common refinement and provide isomorphisms between the three possible Banach structures on $\mathcal{O}_{X_K^{an}}(U)$ via [Bosch et al., 1984, 2.8.1 Banach's Open Mapping Theorem]. So one gets independence of coverings and also natural compatibility of the Banach norms on $\mathcal{O}_{X_K^{an}}(U)$.

Since X_K^{an} is covered by a finite number of affinoids, this constructs $\mathcal{O}_{X_K^{an}}$ as a sheaf of E -Banach algebras for the Grothendieck topology of X_K^{an} given by the net of compact analytic domains in X . Moreover, affinoid locally on X_K^{an} , the norm on $\mathcal{O}_{X_K^{an}}$, on constant functions K , coincides with the valuation. The construction of the sheaf of Banach algebras $\mathcal{O}_{X_K^{an}}$ is functorial for morphisms of analytic spaces described above. Moreover one also sees from this local description that the ring of global functions $\Gamma(X_K^{an}, \mathcal{O}_{X_K^{an}}) = H^0(X_K^{an}, \mathcal{O}_{X_K^{an}})$, on X_K^{an} , is a Banach E -algebra and on the constant functions, this norm is equivalent to the one given by the valuation.

§ 3.16 Now let me prove the following important observation:

Theorem 3.16.1. *Let X/E be a geometrically connected, smooth projective variety. Let K_1, K_2 be two untilts of $\mathbb{C}_p^\flat$ which contain E . Suppose that K_1, K_2 are not topologically isomorphic. Then*

(1) *one has an isomorphism of topological groups*

$$\Pi_{X/K_1}^{temp} \simeq \Pi_{X/K_2}^{temp},$$

(2) *but the analytic spaces X^{an}/K_1 and X^{an}/K_2 are not isomorphic as analytic spaces over $\mathbb{Q}_p$ (in the sense of § 3.13).*

(3) *In particular the Absolute Grothendieck Conjecture fails in the category of Berkovich spaces over perfectoid fields of characteristic zero.*

Proof. After Theorem 3.10.1, only (2) needs to be proved as (2) $\implies$ (3). The hypothesis of Theorem 3.16.1 are non-vacuous—by [Kedlaya and Temkin, 2018], fields K_1, K_2 exist.

Assume that $X/E, K_1, K_2$ are as in my hypothesis and that X is geometrically connected, smooth and projective over E . Suppose, if possible, that X^{an}/K_1 and X^{an}/K_2 are isomorphic as analytic spaces over $\mathbb{Q}_p$. Then one has a bounded isomorphism of Banach rings

$$K_1 \simeq H^0(X^{an}/K_1, \mathcal{O}_{X^{an}/K_1}) \simeq H^0(X^{an}/K_2, \mathcal{O}_{X^{an}/K_2}) \simeq K_2.$$

I claim that this is in fact an isomorphism of valued fields. Write K_1° (resp. K_2°) for the respective subrings of power bounded elements of K_1 (resp. K_2) (for the respective norms). By [Bosch et al., 1984, 1.2.5, Proposition 4], the above isomorphism induces an isomorphism of $K_1^\circ \xrightarrow{\simeq} K_2^\circ$. Further, as the norm $|\cdot|_{K_i}$ on K_i arises from the valuation of these fields, so one sees that the norms are power-multiplicative. This implies, by [Bosch et al., 1984, 1.3.1, Proposition 4], that one has the equality

$$K_i^\circ = \{x \in K_i : |x|_{K_i} \leq 1\},$$

i.e. K_i° is the valuation subring of K_i and thus the valued fields K_i have isomorphic valuation rings and hence K_1 and K_2 are therefore isomorphic as valued fields. Thus one has arrived at a contradiction. $\square$

Remark 3.16.2. For the existence of uncountably many topologically non-isomorphic perfectoid fields see the construction of inequivalent valuations given by [Schmidt, 1933] (sketched here in Section 11) or [Kaplansky, 1942, Theorem 8].

Remark 3.16.3. As an aside let me remark that the proof of [Kedlaya and Temkin, 2018] (also see [Matignon and Reversat, 1984, Théorème 2 and §3 Remarque 2]) provides an uncountable collection of perfectoid fields K_1, K_2 with tilts isometric to $\mathbb{C}_p^b$ and such that K_1, K_2 are not topologically isomorphic.

§ 3.17 Let me introduce some terminology from [Joshi, 2020a]. I will say that two geometrically connected varieties X/E and X'/E' over fields E, E' are *anabelomorphic* (resp. *tempered anabelomorphic*) if one has a topological isomorphism of their étale fundamental groups (resp. tempered fundamental groups if E, E' are p -adic fields):

$$\alpha : \pi_1(X/E) \simeq \pi_1(X'/E') \text{ resp. } \alpha : \Pi_{X/E}^{\text{temp}} \simeq \Pi_{X'/E'}^{\text{temp}}$$

and in this situation I will write $\alpha : X/E \rightsquigarrow X'/E'$ for this *anabelomorphism* (resp. *tempered anabelomorphism*). I will say that an anabelomorphism (resp. *tempered anabelomorphism*) $\alpha : X/E \rightsquigarrow X'/E'$ is a *strict anabelomorphism* (resp. *strict tempered anabelomorphism*) if $X/E, X'/E'$ are anabelomorphic but not isomorphic (resp. anabelomorphic but not isomorphic analytic spaces).

In this terminology, Theorem 3.16.1 says there exist perfectoid fields $K, K' \supset E$ and a strict tempered anabelomorphism

$$(3.17.1) \quad X^{an}/K \rightsquigarrow X^{an}/K'.$$

Note that anabelomorphism defines an equivalence relation on geometrically connected, smooth, quasi-projective varieties and it makes perfect sense to talk about the anabelomorphism class of a variety.

Let me remark that in [Joshi, 2020a] I show that anabelomorphy of p -adic fields changes important invariants of p -adic fields such as discriminants and more importantly it also impacts geometric invariants of varieties such as minimal discriminants of elliptic curves.

Another notion introduced in [Joshi, 2020a] is that of amphoricity: a quantity, a property or an algebraic structure associated to X/E is said to be *amphoric* if it is an invariant of the anabelomorphism class of X/E .

§ 3.18 As is well-known from [Mochizuki, 1997], a p -adic field E is not amphoric [Mochizuki, 1997] i.e. G_E does not determine the isomorphism class of E , but as was shown in [Mochizuki, 1997], G_E equipped with its upper numbering ramification filtration $G_E^\bullet$ determines E . Now let X/E be a geometrically connected, smooth, quasi-projective variety. Let $\Pi = \Pi_{X/E}^{\text{temp}}$ and let $\Pi \supset \overline{\Pi}$ be the geometric tempered fundamental group. Mochizuki has shown that the quotient $\Pi_{X/E} \rightarrow G_E$ is amphoric [Mochizuki, 2004, Lemma 1.3.8] and hence its kernel $\overline{\Pi} \subset \Pi$ is an

amphoric subgroup i.e. determined by the isomorphism class of $\Pi = \Pi_{X/E}^{temp}$. In the light of this and Theorem 3.10.1 and Theorem 3.16.1 one can ask the following question:

Question 3.18.1. Let K be a complete, algebraically closed valued field containing an isometric embedding of E . Is there some filtration by normal subgroups $\Pi_X^\bullet \subset \Pi_X$ which determines the pair of analytic spaces (X_E^{an}, X_K^{an}) up to an isomorphism?

Mochizuki's Theorem that $\overline{\Pi}$ is amphoric should be considered as the analog of the assertion (of [Mochizuki, 1997]) that the inertia subgroup $I_E \subset G_E$ is amphoric. If $G \simeq G_E$ is an isomorph of G_E then one may equip G with many different inertia filtrations corresponding to anabelomorphisms $G_E \simeq G \simeq G_{E'}$. Similarly if Π is an isomorph of $\Pi_{X/E}$ then above suggests remarkably that there are many different filtrations $\Pi^\bullet$ each corresponding to an algebraically closed, complete valued field $K \supset E$. At least when X/E is a hyperbolic curve this question is quite reasonable.

4 Arithmetic Holomorphic Structures

The results of Section 3 namely, Theorem 3.10.1 and Theorem 3.16.1 provide us with a way of defining Arithmetic Holomorphic Structures (see Definition 4.1.1) and allow one to assert with impunity that this notion of Arithmetic Holomorphic Structures is fully compatible with our conventional way of thinking about distinct classical holomorphic structures on Riemann surfaces. As I note in Theorem 7.4.3, my definition of arithmetic holomorphic structures also provides arithmetic holomorphic structures in sense this term is used in [Mochizuki, 2021a,b,c,d]. In particular, in the p -adic contexts one can retain both: our ideas from classical theory complex holomorphic functions and also preserve ideas from anabelian geometry. Thus this work provides a natural solution to the problem (raised in the context of [Mochizuki, 2021a,b,c,d]) of producing *geometrically, and canonically labeled copies of the tempered fundamental groups* in a manner compatible with the classical case (Theorem 2.1.1). This aspect is detailed in Section 7.

While the term Arithmetic Holomorphic Structures was coined and used by Mochizuki in [Mochizuki, 2021a,b,c,d], my definition Definition 4.1.1 is far more precise and Theorem 3.16.1(2) shows that it is a holomorphic structure in the sense of holomorphic functions, secondly this holomorphic structure is dependent on the arithmetic and topological properties of the coefficient field and hence the nomenclature 'Arithmetic Holomorphic Structure' is quite apt. Arithmetic Holomorphic Structures defined here also provide arithmetic holomorphic structures in Mochizuki's sense. *This is the starting point of the Teichmüller Theory presented in the the present series of papers ([Joshi, 2020a, 2021a, 2022, 2023b,a]).* [Mochizuki does not take this Berkovich analytic space approach which I have taken here.]

Once one has defined Arithmetic Holomorphic Structures, by imitating the classical theory, one defines the Arithmetic Teichmüller Space (Definition 5.1.1) as the category of all Arithmetic Holomorphic Structures with a fixed tempered fundamental group.

Let me also point out that the approach taken here for the p -adic case involves deforming (infinite dimensional) Banach spaces of (Berkovich) analytic functions. As should be clear from Section 2.2, especially Theorem 2.5.1, that Banach space approach also exists in the

classical case (but not so well-known—I discovered Nakai’s works after [Joshi, October 2020] was posted on the arxiv). At any rate, one now has a complete parallel between the p -adic and archimedean theory.

§ 4.1 Definitions Let $\mathcal{M}(A)$ be the Berkovich spectrum [Berkovich, 1990] of any Banach algebra A , and let X_E^{an}, X_K^{an} be the analytic spaces arising from X/E etc. (as before X/E is a geometrically connected, smooth, quasi-projective variety over a p -adic field E). It will be useful to recall the definition of geometric base-points in the context of [André, 2003a].

Definition 4.1.1. Let X/E be a geometrically connected, smooth, quasi-projective variety over a p -adic field E . A (pointed) *arithmetic holomorphic structure*

$$(X/E, (K \supset E, K^\flat \simeq F), *_{\mathcal{K}} : \mathcal{M}(K) \rightarrow X_E^{an})$$

on X/E consists of

- (1) a choice of an untilt $(K \supset E, K^\flat \simeq F)$ for some algebraically closed, perfectoid field F of characteristic $p > 0$, and
- (2) a choice of a morphism of Berkovich analytic spaces $*_{\mathcal{K}} : \mathcal{M}(K) \rightarrow X_E^{an}$.

Morphisms of arithmetic holomorphic structures are be defined as morphisms of the data $(X/E, (K \supset E, K^\flat \simeq F), *_{\mathcal{K}} : \mathcal{M}(K) \rightarrow X_E^{an})$ of the arithmetic holomorphic structure.

Definition 4.1.2. Following see [André, 2003a, Chapter III, 1.2.2], a K -geometric base-point (or less precisely, a *geometric base-point*) of the analytic space X_E^{an} is an algebraically closed, complete (rank one) valued field K and an isometric extension of valued fields K/E and a morphism of analytic spaces

$$*_{\mathcal{K}} : \mathcal{M}(K) \rightarrow X_E^{an}$$

.

Remark 4.1.3.

- (1) Let me remark that [André, 2003a] works in the context of [Berkovich, 1990], and hence any geometric base-point as defined here and in [André, 2003a, Chapter III, 1.2.2] provides an algebraically closed, valued field which is complete with respect to a rank one valuation. Hence, especially that, a K -geometric base-point in Definition 4.1.2 provides an algebraically closed, complete (rank one) valued field K containing $\mathbb{Q}_p$ isometrically. Thus, by Lemma 3.2.1, a geometric base-point provides an algebraically closed, perfectoid field $K \supset E \supset \mathbb{Q}_p$.
- (2) The discussion of [Mochizuki, 2021a, §I3, Page 21] makes it clear that [Mochizuki, 2021a,b,c,d] requires arbitrary geometric base-points (and so the valued field $K \supset E$ must be arbitrary). *This is how algebraically closed, perfectoid fields enter [Mochizuki, 2021a,b,c,d] (see [Joshi, 2024c]).*

Lemma 4.1.4. *Each arithmetic holomorphic structure*

$$(X/E, (K \supset E, K^\flat \simeq F), *_{\mathcal{K}} : \mathcal{M}(K) \rightarrow X_E^{an})$$

is equipped with a K -geometric base-point, namely the morphism $*_K : \mathcal{M}(K) \rightarrow X_E^{an}$.

Proof. This is clear from the definitions. $\square$

Here is one immediate consequence of Definition 4.1.1 and Lemma 4.1.4.

Proposition 4.1.5. *Let $(X/E; (K \supset E, K^\flat \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an})$ be an arithmetic holomorphic structure on X/E . Then the geometric base-point provided by Lemma 4.1.4 provides the exact sequence of topological groups*

$$(4.1.6) \quad 1 \rightarrow \Pi_{X/K}^{temp} \rightarrow \Pi_{X/E}^{temp} \rightarrow G_{E;K} \rightarrow 1.$$

The isomorphism class each of the groups and the isomorphism class of this exact sequence of topological groups is independent of the choice of the arithmetic holomorphic structure on X/E and also independent of the choice geometric base-point.

Proof. This is immediate from Theorem 3.10.1 and properties of tempered fundamental groups established in [André, 2003b]. $\square$

Let me record the following which makes it clear that distinct i.e. non-isomorphic arithmetic holomorphic structures exist.

Proposition 4.1.7. *Let F be any algebraically closed perfectoid field of characteristic $p > 0$. Let $((K_1 \supset E, K_1^\flat \simeq F), *_K : \mathcal{M}(K_1) \rightarrow X_E^{an})$, $((K_2 \supset E, K_2^\flat \simeq F), *_K : \mathcal{M}(K_2) \rightarrow X_E^{an})$ be two arithmetic holomorphic structures. Then the following assertions hold:*

- (1) *The valued fields K_1, K_2 have characteristic zero, the same cardinality, the same residue fields and the same value groups as that of F .*
- (2) *There exist arithmetic holomorphic structures $((K_1 \supset E, K_1^\flat \simeq F), *_K : \mathcal{M}(K_1) \rightarrow X_E^{an})$, $((K_2 \supset E, K_2^\flat \simeq F), *_K : \mathcal{M}(K_2) \rightarrow X_E^{an})$ which are not isomorphic.*

Furthermore, if $F = \mathbb{C}_p^\flat$, then K_1, K_2 need not even be topologically isomorphic fields (and hence need not be isomorphic as valued fields) and, if X/E is projective, then by Theorem 3.16.1, $X_{K_1}^{an}$ and $X_{K_2}^{an}$ need not be $\mathbb{Q}_p$ -isomorphic analytic spaces.

Proof. The first assertion is an immediate consequence of [Fargues and Fontaine, 2018], [Scholze, 2012]. By [Fargues and Fontaine, 2018, Corollaire 2.2.22], one sees that for any given perfectoid field F of characteristic $p > 0$ and any given p -adic field E , there exist many non-isomorphic untilts and this together with Definition 4.1.1, implies that there exist many non-isomorphic arithmetic holomorphic structures. The last assertion for $F = \mathbb{C}_p^\flat$ is a consequence of Theorem 3.16.1 and the fundamental theorem of [Kedlaya and Temkin, 2018]. $\square$

Remark 4.1.8. Theorem 3.10.1 makes it clear that one is dealing with holomorphic structures in the familiar sense of complex analytic function on the other hand, this theorem and Proposition 4.1.7 also makes it clear that this phenomenon arises from topological changes in arithmetic of the fields one is dealing with here. This is the reason why the term Arithmetic Holomorphic Structures is appropriate here.

Remark 4.1.9. Let me remark importantly that by [Kedlaya and Temkin, 2018] that there exist algebraically closed, perfectoid fields with an isometrically embedded $\mathbb{Q}_p$ and with tilts isometric to $\mathbb{C}_p^\flat$, but such that $\mathbb{Q}_p$ is not necessarily dense in these fields. So one cannot hope to work (solely) over $\mathbb{Q}_p$ once one recognizes that arbitrary algebraically closed perfectoid fields are required in [Mochizuki, 2021a,b,c,d] via its requirement of arbitrary geometric base-points.

Proposition 4.1.10. *From the category whose objects are of the form $(X/E, (K \supset E, K^\flat \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an})$, there are natural forgetful functors, to various categories, given as*

$$\begin{array}{ccccc}
 & & \pi_1^{temp}(X_E^{an}, *_K : \mathcal{M}(K) \rightarrow X_E^{an}) & & \\
 & & \uparrow & & \\
 & & (X_E^{an}, *_K : \mathcal{M}(K) \rightarrow X_E^{an}) & & \\
 & & \uparrow & & \\
 \text{follows:} & & (X/E, (K \supset E, K^\flat \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an}) & & \\
 & \swarrow & \downarrow & \searrow & \\
 X/E & & (K \supset E, K^\flat \simeq F) & & X_K^{an} \rightarrow X_E^{an} \\
 & \swarrow & \downarrow & \searrow & \\
 K & & & & K^\flat
 \end{array}$$

Proof. The functor

$$(X/E, (K \supset E, K^\flat \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an}) \mapsto (X_K^{an} \rightarrow X_E^{an})$$

is obtained by considering the pair of spaces (X_E^{an}, X_K^{an}) and the morphism $X_K^{an} \rightarrow X_E^{an}$ obtained by base change via the given isometric inclusion $E \hookrightarrow K$.

The morphism $*_K : \mathcal{M}(K) \rightarrow X_E^{an}$ is a geometric base-point of the analytic space X_E^{an} and the functor

$$(X_E^{an}, *_K : \mathcal{M}(K) \rightarrow X_E^{an}) \mapsto \pi_1^{temp}(X_E^{an}, *_K : \mathcal{M}(K) \rightarrow X_E^{an})$$

is the tempered fundamental group functor constructed in [André, 2003b] using this geometric base-point given in the datum of the object $(X/E, (K \supset E, K^\flat \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an})$.

The functor

$$(X_E^{an}, *_K : \mathcal{M}(K) \rightarrow X_E^{an}) \mapsto \pi_1^{temp}(X_E^{an}, *_K : \mathcal{M}(K) \rightarrow X_E^{an})$$

is thus the composite functor obtained as

$$(X_E^{an}, *_K : \mathcal{M}(K) \rightarrow X_E^{an}) \mapsto (X_E^{an}, *_K : \mathcal{M}(K) \rightarrow X_E^{an}) \mapsto \pi_1^{temp}(X_E^{an}, *_K : \mathcal{M}(K) \rightarrow X_E^{an}).$$

The other functors are clear. □

Remark 4.1.11. Suppose one is given any K -geometric base point $*_K : \mathcal{M}(K) \rightarrow X_K^{an}$, then by composition with the base-change morphism $X_K^{an} \rightarrow X_E^{an}$ provided by the given isometric

embedding $E \hookrightarrow K$, one obtains a K -geometric base-point of $*_K : \mathcal{M}(K) \rightarrow X_E^{an}$ (note the conflation of notation). In particular, in defining $\mathfrak{J}(X, E)$, one can work instead with the datum

$$((K \supset E, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow X_K^{an})$$

instead of

$$((K \supset E, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an}).$$

§ 4.2 Notational Contractions In this paper, for simplicity of notation, I may contract the notation

$$(X/E; (K \supset E, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an})$$

to

$$(X/E; (K \supset E, K^b \simeq F)).$$

This may give the impression that the geometric base-point datum is not) is unimportant, but let me caution the reader that this is certainly not the case. I may also further contract the notation to

$$(X/E; E \hookrightarrow K).$$

This may also give the impression that the tilting data (i.e. the isometry $K^b \simeq F$ and the geometric base-point datum is not) is unimportant, but let me caution the reader that this is certainly not the case.

Remark 4.2.1.

- (1) The tilting datum $K^b \simeq F$ provided by an untilt allows us to compute arithmetic degrees in one fixed location namely the value group of F even as the untilt moves (see [Joshi, 2023b] for how this gets used and its relevance for [Mochizuki, 2021c, Corollary 3.12]).
- (2) As is established in [Joshi, 2024c, § 8.3], in the parlance of [Mochizuki, 2021a,b,c,d], the perfectoid field K provides “étale-Picture” given by group theoretic data of the isomorphisms $\Pi_{X/E;K}^{temp}$ and $G_{E;K}$ of the topological groups $\Pi_{X/E}^{temp}$ and G_E respectively. While the tilting data is “Frobenius-Picture.” In fact, after Definition 4.1.1, Mochizuki’s term “Frobenius-like” can be understood in the literal sense as the construction of K^b requires p^{th} -powers i.e. Frobenius morphism.

§ 4.3 Relationship with the Diamond $X_E^\diamond$ Let X/E be a geometrically connected, smooth, quasi-projective variety over a p -adic field E . Write $X_E^\diamond = (X_E^{an})^\diamond$ for the diamond associated to the analytic space X_E^{an} by [Scholze, 2017, Section 15]. In this section I want to record an important observation which makes it clear that the theory of the the present series of papers ([Joshi, 2020a, 2021a, 2022, 2023b,a]), the theory of tempered fundamental groups (and hence [Mochizuki, 2021a,b,c,d]) are inseparably linked to Scholze’s Theory of Diamonds described in [Scholze, 2017]. This remark was first pointed out in [Joshi, 2022].

Proposition 4.3.1. *There is a natural functor from the category of objects of the form*

$$(X/E, (K \supset E, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an})$$

to the diamond $X_E^\diamond$ given by the rule

$$\begin{array}{c} (X/E, (K \supset E, K^\flat \simeq F), *_{\mathcal{K}} : \mathcal{M}(K) \rightarrow X_E^{an}) \\ \downarrow \\ ((K \supset E, K^\flat \simeq F), *_{\mathcal{K}} : \mathrm{Spa}(K, \mathcal{O}_K) \rightarrow X_E^{ad}) \in X_E^\diamond(F). \end{array}$$

Proof. This is immediate from the results of [Huber, 1994], [Scholze, 2017, Section 15]: the K -geometric base-point i.e. the morphism $*_{\mathcal{K}} : \mathcal{M}(K) \rightarrow X_E^{an}$ of Berkovich analytic spaces provides, by the means of the fully-faithful functor from Berkovich analytic spaces to adic spaces constructed by [Huber, 1994, Proposition 4.3 and Proposition 4.5], to a morphism $*_{\mathcal{K}} : \mathrm{Spa}(K, \mathcal{O}_K) \rightarrow X_E^{ad}$ and conversely. $\square$

5 Construction of Arithmetic Teichmüller Spaces

§ 5.1 Definition in the non-archimedian case Classical Teichmüller, Theorem 3.10.1, the definition of Arithmetic Holomorphic Structures on X/E Definition 4.1.1 and Theorem 7.4.3 suggests a natural definition of Arithmetic Teichmüller Space of X/E : namely, this space should be the category whose objects are arithmetic holomorphic structures all of which have tempered fundamental groups isomorphic to the tempered fundamental group of X/E . Here is a more precise version:

Definition 5.1.1. Let E be a p -adic field and let X/E be a geometrically connected, smooth, quasi-projective variety over E . Then the *Arithmetic Teichmüller Space* $\mathfrak{J}(X, E)$ of X/E is a category whose objects are arithmetic holomorphic structures

$$(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_{\mathcal{K}} : \mathcal{M}(K) \rightarrow Y_{E'}^{an})$$

where

- (1) E' is a p -adic field,
- (2) F is some algebraically closed perfectoid field of characteristic $p > 0$,
- (3) $(E' \hookrightarrow K, K^\flat \simeq F)$ is an E' -untilt of F ,
- (4) Y/E' is a geometrically connected, smooth, quasi-projective variety such that
- (5) one has

$$\dim(Y) = \dim(X),$$

- (6) and

- (a) Y/E' is tempered anabelomorphic to X/E i.e. one has an anabelomorphism

$$\Pi_{Y/E'}^{temp} \simeq \Pi_{X/E}^{temp},$$

- (b) and any such anabelomorphism induces an anabelomorphism of their geometric tempered fundamental subgroups

$$\Pi_{Y/K}^{temp} \simeq \Pi_{X/\mathbb{C}_p}^{temp}.$$

Morphisms in $\mathfrak{J}(X, E)$ are morphisms of the data

$$(Y/E', (E' \hookrightarrow K, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{an}).$$

Remark 5.1.2. In the principal case of hyperbolic curves over p -adic fields which occurs in the Anabelian Geometry literature, the requirement (4)(b) is implied by (4)(a) (by [Mochizuki, 2004, Lemma 1.3.8]). I do not know if this is still true in the general case considered here.

Remark 5.1.3. In recent correspondence, Kiran Kedlaya pointed out to me the following consequence of [Kedlaya and Liu, 2019, Proposition 8.8.9]: *every deformation of an analytic space (arising from the analytification of a quasi-projective variety) over a perfectoid field arises from a deformation of the perfectoid field*. So the idea of moving the algebraically closed perfectoid field (considered here) is, in a rather precise sense (of [Kedlaya and Liu, 2019]), optimal.

§ 5.2 Definition in the Archimedean Case This section still needs some cleanup in case $\dim(X) > 1$. This section may be skipped by readers who are interested only in the case of smooth, hyperbolic curves i.e. the dimension one case: in the dimension one case one simply takes classical Teichmüller space [Imayoshi and Taniguchi, 1992] as the Archimedean Teichmüller Space.

Now suppose that E is an algebraically closed, archimedean complete local field. So $E \simeq \mathbb{C}$ and think X is a complex, smooth, quasi-projective variety. I will refer to this as the archimedean case. I will write $X_{\mathbb{C}}^{an}$ for the complex analytic space i.e. the complex manifold given by the scheme X .

In the archimedean case, if the complex variety $X/\mathbb{C}$ has $\dim(X/\mathbb{C}) = 1$, then classical Teichmüller Theory provides a robust theory, but if $\dim(X/\mathbb{C}) > 1$ then there is no good definition of (archimedean) Teichmüller space and hence one has some latitude in defining what it should be.

Here I provide a reasonable working definition of (archimedean) Teichmüller space which works for all $\dim(X/\mathbb{C}) \geq 1$ and which coincides with the classical case of $\dim(X/\mathbb{C}) = 1$ (Riemann surfaces). Further work is, of course needed, if $\dim(X/\mathbb{C}) > 1$.

The notion of quasi-conformal mappings extends to higher dimensional Riemannian manifolds, but on the other hand for Riemannian manifolds of dimension greater than two, the notion of quasi-conformal mappings is not adequate for a satisfactory theory. So an alternative approach is needed to build the theory.

My approach here is based on [Nakai, 1959], [Nakai, 1960], [Nakai, 1972] and [Nakai and Sario, 1970, Appendix: Higher Dimensions]. As is detailed in Section 2.2, for $\dim(X/\mathbb{C}) = 1$, the results of [Nakai, 1959], [Nakai, 1960] using Royden algebras of functions are closest to the Banach space approach of this paper and allow us to view classical Teichmüller Theory as a functor in infinite dimensional Banach algebras (as is done here in the p -adic case). For $\dim(X/\mathbb{C}) \geq 2$, [Nakai, 1972] suggests a natural generalization (again using results of [Nakai,

1972] on Royden algebras) which is compatible with $\dim(X/\mathbb{C}) = 1$ case. In particular, for all dimensions $\dim(X/\mathbb{C}) \geq 1$ one has a good category equipped with a functor to infinite dimensional Banach algebras.

To adopt [Nakai, 1972] approach one must work with Riemannian manifolds. Write $X^{an} = X_{\mathbb{C}}^{an}$ for the associated complex manifold. Since $X/\mathbb{C}$ is a complex, smooth, quasi-projective variety, X^{an} can be equipped with a Riemannian metric. For example, using any embedding $X \hookrightarrow \mathbb{P}^N$ in projective space and considering the restriction of the Fubini-Study metric on $\mathbb{P}^N$ to X^{an} , one sees that X^{an} is a Riemannian manifold. In this subsection, I will view X^{an} for any complex, smooth, quasi-projective variety as a Riemannian manifold.

A *Nakai quasi-isometry* between Riemannian manifolds is a quasi-isometry of Riemannian manifold as defined in [Nakai, 1972, Paragraph 2]. A Nakai quasi-isometry is a quasi-conformal mapping and also a homeomorphism. By [Nakai, 1972]. If $\dim(X/\mathbb{C}) = 1$ (i.e. the underlying real manifold $X_{\mathbb{C}}^{an}$ is of dimension two) then one sees that a Nakai quasi-isometry is the same as a quasi-conformal mapping. Hence one recovers the classical Teichmüller space if $\dim(X/\mathbb{C}) = 1$. For $\dim(X/\mathbb{C}) \geq 1$, the notion of Nakai quasi-isometry appears to be the correct generalization of quasi-conformal mappings which is compatible with the Banach space approach of this paper by the results of [Nakai, 1972].

Definition 5.2.1. Let $X/\mathbb{C}$ be a complex, smooth, quasi-projective variety, considered as being equipped with a Riemannian metric. The archimedean *pre-Teichmüller Space* of $X/\mathbb{C}$ consists of pairs $(Y^{an}, f : Y^{an} \rightarrow X^{an})$ where Y is a connected, smooth, quasi-projective variety over complex numbers and $f : Y^{an} \rightarrow X^{an}$ is a Nakai quasi-isometry of Riemannian manifolds. Note that this includes pairs of the form $(X^{an}, f : X^{an} \rightarrow X^{an})$ and hence the set of such pairs is not vacuous as soon as $X \neq \emptyset$.

The Teichmüller space $\mathfrak{J}(X, \mathbb{C})$ of X is then the pre-Teichmüller Space of $X/\mathbb{C}$ subject to a suitable equivalence.

Definition 5.2.2. Two pairs $(Y^{an}, f : Y^{an} \rightarrow X^{an})$ and $(Z^{an}, g : Z^{an} \rightarrow X^{an})$ as in Definition 5.2.1 are equivalent if $f \circ g^{-1}$ is homotopic to a mapping which induces an isometry between Royden algebras of Y^{an} and Z^{an} . The archimedean Teichmüller Space of $X/\mathbb{C}$ is the set of equivalence classes of pairs $(Y^{an}, f : Y^{an} \rightarrow X^{an})$ in the pre-Teichmüller Space of Definition 5.2.1.

Remark 5.2.3. By [Nakai, 1960], if $\dim(X/\mathbb{C}) = 1$, then two pairs $(Y^{an}, f : Y^{an} \rightarrow X^{an})$ and $(Z^{an}, g : Z^{an} \rightarrow X^{an})$ as in Definition 5.2.1 are equivalent if $f \circ g^{-1}$ is homotopic to a conformal mapping. So $\mathfrak{J}(X, \mathbb{C})$ is the classical Teichmüller Space if $\dim(X/\mathbb{C}) = 1$.

The following result is immediate from [Nakai, 1972]:

Proposition 5.2.4. Suppose $f : Y^{an} \rightarrow X^{an}$ is a Nakai quasi-isometry of Riemannian manifolds. Then the real manifolds Y^{an}, X^{an} are homeomorphic and hence have the same real dimension and hence one has $\dim(X/\mathbb{C}) = \dim(Y/\mathbb{C})$ and have isomorphic topological fundamental groups.

The next theorem is my reformulation of main theorem of [Nakai, 1972] it provides an algebraic characterization of Nakai quasi-isometries similar to the case of Riemann surfaces discussed in Section 2.2:

Theorem 5.2.5. *Let X, Y be connected, smooth, quasi-projective varieties over $\mathbb{C}$. Let $\Sigma = X_{\mathbb{C}}^{an}$, $\Sigma' = Y_{\mathbb{C}}^{an}$ be the underlying complex manifolds considered as being equipped with a Riemannian metrics. Then*

- (1) *A homeomorphism $\Sigma' \rightarrow \Sigma$ is a Nakai quasi-isometry if and only if one has a $\mathbb{C}$ -algebra isomorphism*

$$\mathcal{R}_{\Sigma} \simeq \mathcal{R}_{\Sigma'}$$

between their Royden algebras.

- (2) *Any $\mathbb{C}$ -algebra isomorphism $\mathcal{R}_{\Sigma} \simeq \mathcal{R}_{\Sigma'}$ is a homeomorphism for the respective norm-topologies.*
- (3) *The construction of $(\mathcal{R}_{\Sigma}, |-\|_{\Sigma})$ is functorial for Nakai quasi-isometries between Riemann manifolds.*
- (4) *In particular one has a functor*

$$(\Sigma', f : \Sigma' \rightarrow \Sigma) \longmapsto (\mathcal{R}_{\Sigma'}, |-\|_{\Sigma'})$$

from the Teichmüller category $\mathcal{T}_{\Sigma}$ to the category of infinite dimensional complex Banach algebras.

- (5) *Moreover, if Σ, Σ' are Nakai quasi-isometric then*

- (a) *$\dim(X/\mathbb{C}) = \dim(Y/\mathbb{C})$ as complex varieties, and*
- (b) *$\pi_1^{top}(\Sigma') \simeq \pi_1^{top}(\Sigma)$ and hence*
- (c) *$\pi_1^{et}(Y) \simeq \pi_1^{et}(X)$ is an isomorphism of topological groups, and*
- (d) *hence X, Y are anabelomorphic complex varieties.*

Proof. The assertions (1)–(4) are immediate from [Nakai, 1972] and [Nakai and Sario, 1970, Appendix: Higher Dimensions]. The assertion (5)(a,b) follows from the fact that the Riemannian manifolds Σ' and Σ are Nakai quasi-isometric and hence homeomorphic. The assertion (5)(c) follows from a well-known theorem of Grothendieck [Grothendieck, 1971]. The assertion (5)(d) is immediate from the definition of anabelomorphic varieties. $\square$

Remark 5.2.6. Let me remark that in the context of Arakelov Theory (and hence in Diophantine geometry) one works with the existence of Green's functions. An important reason why one works with Nakai quasi-isometries is that Nakai quasi-isometries (i.e. quasi-conformal mappings if $\dim(X/\mathbb{C}) = 1$) preserve the property of the possession of non-constant Green's functions i.e. if $Y \rightarrow X$ is a Nakai quasi-isometry then either both X, Y possess a non-constant Green's function or neither one does. This is established in [Nakai, 1972], [Nakai and Sario, 1970].

§ 5.3 Variation of Berkovich Analytic Structure As observed in Theorem 2.5.1, Theorem 5.2.5, in the archimedean case, a variation of Banach spaces of functions underlies classical Teichmüller spaces. For E non-archimedean, the category $\mathfrak{J}(X, E)$ shares this important property with the classical Teichmüller Space. Namely, the Berkovich analytic space structure provided by $(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_{K_1} : \mathcal{M}(K_1) \rightarrow Y_{E'_1}^{an})$ varies as $(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_{K_2} : \mathcal{M}(K_2) \rightarrow Y_{E'_2}^{an})$ varies over objects of $\mathfrak{J}(X, E)$. This is recorded in the following consequence of Theorem 3.16.1. This is one of the properties of $\mathfrak{J}(X, E)$ which justifies the term Arithmetic Teichmüller Space $\mathfrak{J}(X, E)$.

Theorem 5.3.1. *Let X/E be a geometrically connected, smooth, projective variety over a p -adic field E .*

- (1) *Each $(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_{K_1} : \mathcal{M}(K_1) \rightarrow Y_{E'_1}^{an}) \in \mathfrak{J}(X, E)$ provides the analytic spaces $Y_{E'_1}^{an}, Y_{K_1}^{an}$ and the morphism $Y_{K_1}^{an} \rightarrow Y_{E'_1}^{an}$ of analytic spaces provided by the given isometric inclusion $E' \hookrightarrow K$.*
- (2) *Two objects $(Y_1/E'_1, (E'_1 \hookrightarrow K_1, K_1^\flat \simeq F), *_{K_1} : \mathcal{M}(K_1) \rightarrow Y_{E'_1}^{an})$ and $(Y_2/E'_2, (E'_2 \hookrightarrow K_2, K_2^\flat \simeq F), *_{K_2} : \mathcal{M}(K_2) \rightarrow Y_{E'_2}^{an})$ of $\mathfrak{J}(X, E)$ need not be isomorphic even if $Y_1/E'_1 = Y_2/E'_2$.*
- (3) *If Y_1/E'_1 and Y_2/E'_2 are both projective then $(Y_{E'_1}^{an}, Y_{K_1}^{an}, Y_{K_1}^{an} \rightarrow Y_{E'_1}^{an})$ need not be isomorphic to $(Y_{E'_2}^{an}, Y_{K_2}^{an}, Y_{K_2}^{an} \rightarrow Y_{E'_2}^{an})$ even if $Y/E'_1 = Y/E'_2$.*

Proof. The first assertion is clear. The second assertion is immediate from the fact that the two given untilts, $(E'_1 \hookrightarrow K_1, K_1^\flat \simeq F)$, $(E'_2 \hookrightarrow K_2, K_2^\flat \simeq F)$ of F need not be isomorphic. The third assertion is immediate from Theorem 3.16.1. $\square$

Remark 5.3.2. One way of understanding the above result is that the Berkovich analytic space structure i.e. the analytic space datum provided by objects of $\mathfrak{J}(X, E)$ varies over $\mathfrak{J}(X, E)$.

§ 5.4 Existence of distortions of p -adic metrics A key property classical Teichmüller spaces, grounded in the basic theory of quasi-conformal mappings of Riemann surfaces is that metrics appear scaled relative to each other (these scalings are encoded in the Beltrami parameter) and local geometries are distorted relative to each other [Imayoshi and Taniguchi, 1992, Chapter 1 and Chapter 4].

In the context of Berkovich analytic spaces [Berkovich, 1990], there is a natural notion of metrics namely, the local Banach norms provided in the datum of a Berkovich analytic space. The following theorem demonstrates that for this notion of Banach norms, objects of $\mathfrak{J}(X, E)$ share this property of local metric scalings with the objects of classical Teichmüller Space ([Imayoshi and Taniguchi, 1992, Theorem 1.7]).

Theorem 5.4.1. *Let X/E be geometrically connected, smooth, quasi-projective variety over a p -adic field E . Let $(Y/E', (E' \hookrightarrow K_1, K_1^\flat \simeq F), *_{K_1} : \mathcal{M}(K_1) \rightarrow Y_{E'_1}^{an})$ and $(Y/E', (E' \hookrightarrow K_2, K_2^\flat \simeq F), *_{K_2} : \mathcal{M}(K_2) \rightarrow Y_{E'_2}^{an})$ be two objects of $\mathfrak{J}(X, E)$. Let $(E', | \cdot |_{K_1})$ ($E', | \cdot |_{K_2}$) be the restriction of the absolute value of K_1 to E' (resp. of K_2 to E'). Then there exists a real number $\alpha \in \mathbb{R}$ such that*

$$(E', | \cdot |_{K_2}) = (E', | \cdot |_{K_1}^\alpha).$$

However, there is no uniform choice of α for all pairs of objects of $\mathfrak{J}(X, E)$.

Proof. One has two embeddings $\iota_1 : E' \hookrightarrow K_1$ and $\iota_2 : E' \hookrightarrow K_2$. Write $E'_1 = \iota_1(E') \subset K_1$ and similarly define E'_2 i.e. view E' as a (discretely) valued subfield of K_j via ι_j . Abstractly, $E' \simeq E'_1 \simeq E'_2 \simeq E'$ and any abstract isomorphism $E'_1 \simeq E'_2$ is in fact an isomorphism of discretely valued fields, and the two valuations on E' induced from K_1, K_2 on E' are equivalent. Hence existence of $\alpha \in \mathbb{R}$ as asserted is clear. The second assertion is now immediate from the existence of Fargues-Fontaine curves [Fargues and Fontaine, 2018]. $\square$

Corollary 5.4.2. *In particular, in the notation of Theorem 5.4.1 and its proof, the p -adic metrics (i.e. the local Banach norms) of $Y_{E'_2}^{an}$ are scaled relative to that of $Y_{E'_1}^{an}$. However, there is no uniform choice of α for all pairs of objects of $\mathfrak{J}(X, E)$.*

Proof. Any isomorphism of valued fields $E'_1 \simeq E'_2$ (as in the above proof) also gives isomorphism of analytic spaces $Y_{E'_1}^{an} \simeq Y_{E'_2}^{an}$. The theorem asserts that valuation of E'_1 is scaled relative to E'_2 . The remaining assertion is immediate from the above theorem and the existence of Fargues-Fontaine curve $\mathcal{Y}_{F, E'}$ established in [Fargues and Fontaine, 2018, Chapitre 2, 6]. $\square$

Remark 5.4.3. In [Mochizuki, 2021a, Remark 3.9.3], Mochizuki asserts the existence of local distortions Corollary 5.4.2 as a key point in his theory.

Remark 5.4.4. In the archimedean case, the existence of distortions of metrics is built into the theory of Nakai quasi-isometries in all dimensions by [Nakai, 1972].

§ 5.5 Invariance of the tempered fundamental groups in Arithmetic Teichmüller Spaces

Here is an important but trivial consequence of the definition of $\mathfrak{J}(X, E)$:

Proposition 5.5.1. *Let X/E be a geometrically connected, smooth, quasi-projective variety over a field E .*

- (1) *If E is a p -adic field, then for every triple $(Y/E', E' \hookrightarrow K) \in \mathfrak{J}(X, E)$ one has an anabelomorphism*

$$\Pi_{Y/E'}^{temp} \simeq \Pi_{X/E}^{temp},$$

- (2) *inducing an isomorphism of their geometric tempered fundamental subgroups*

$$\Pi_{Y/K}^{temp} \simeq \Pi_{X/\mathbb{C}_p}^{temp},$$

and hence an isomorphism

$$G_{E'; K} \simeq G_{E; \mathbb{C}_p}$$

of absolute Galois groups (i.e. E' and E are anabelomorphic p -adic fields).

- (3) *In the archimedean case (i.e. $E = \mathbb{C}$), for every object $(Y, f : Y \rightarrow X)$, one has an isomorphism of topological fundamental groups*

$$\pi_1^{top}(Y) \simeq \pi_1^{top}(X)$$

and hence of their profinite completions

$$\pi_1^{et}(Y) \simeq \pi_1^{et}(X).$$

Proof. The proof is clear from the definition of $\mathfrak{J}(X, E)$ in both the cases. $\square$

Remark 5.5.2. This together with Theorem 5.3.1, Theorem 5.4.1, Corollary 5.4.2 shows why $\mathfrak{J}(X, E)$ has properties similar to classical Teichmüller Spaces.

§ 5.6 The Subcategories $\mathfrak{J}(X, E)_F$ of $\mathfrak{J}(X, E)$ Since F is an arbitrary, algebraically closed perfectoid field of characteristic $p > 0$, the category $\mathfrak{J}(X, E)$ is too big in practice. Hence one works with the following full subcategories of $\mathfrak{J}(X, E)$.

Proposition 5.6.1. *There is a natural functor*

$$(Y/E', E' \hookrightarrow K, \iota : K^\flat \simeq F) \longmapsto F$$

from $\mathfrak{J}(X, E)$ to the category of perfectoid fields of characteristic $p > 0$, with isometries of such fields as morphisms.

Proof. The proof is clear. $\square$

Definition 5.6.2. Let E be p -adic field and fix F to be an algebraically closed perfectoid field of characteristic $p > 0$. Then let

$$\mathfrak{J}(X, E)_F$$

be the full subcategory of $\mathfrak{J}(X, E)$ whose objects are

$$\{(Y/E', E' \hookrightarrow K, \iota : K^\flat \simeq F') \in \mathfrak{J}(X, E) : F' \text{ isometric to } F\}.$$

Thus $\mathfrak{J}(X, E)_F$ is the fiber, over F , of the above functor.

§ 5.7 Arithmetic Teichmüller Spaces With Auxillary Conditions For greater flexibility and with a view to applications, it is useful to consider variants of the construction of $\mathfrak{J}(X, E)$. Let Σ be a (finite) set of geometric or arithmetic conditions one can impose on the data $(Y/E', E' \hookrightarrow K)$ (all objects of $\mathfrak{J}(X, E)$ need not satisfy the conditions Σ).

Let

$$\mathfrak{J}^\Sigma(X, E)$$

be the full subcategory of $\mathfrak{J}(X, E)$ consisting of objects of $\mathfrak{J}(X, E)$ which satisfy Σ :

$$(5.7.1) \quad \mathfrak{J}^\Sigma(X, E) = \{(Y/E', E' \hookrightarrow K) \in \mathfrak{J}(X, E) \text{ and } (Y/E', E' \hookrightarrow K) \text{ satisfies } \Sigma\}.$$

§ 5.8 Two important examples of Σ which are useful from the point of view of [Mochizuki, 2021a,b,c,d], are the following:

(1) Let

$$\Sigma_{hyp} = \{Y/E' \text{ is hyperbolic}\}.$$

(2) If $\dim(X) = 1$, then let

$$\Sigma_{Belyi} = \{Y \text{ hyperbolic and also of strict Belyi Type}\}.$$

Note that $\dim(X) = 1$ implies (by Definition 5.1.1) that $\dim(Y) = 1$; for the definition of Strict Belyi Type see [Mochizuki, 2013, Definition 3.5].

For Σ_{hyp} , I will write $\mathfrak{J}^{hyp}(X, E)$ instead of $\mathfrak{J}^{\Sigma_{hyp}}(X, E)$. Hence

$$\mathfrak{J}^{hyp}(X, E) = \{(Y/E', E' \hookrightarrow K) \in \mathfrak{J}(X, E) : Y/E' \text{ is hyperbolic variety}\},$$

and for Σ_{Belyi} , I will write

$$\mathfrak{J}^{SB}(X, E) = \{(Y/E', E' \hookrightarrow K) \in \mathfrak{J}(X, E) : Y/E' \text{ is of Strict Belyi Type}\}.$$

Proposition 5.8.1. *Let X/E be a geometrically connected, smooth, hyperbolic curve over a p -adic field E and let $(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_{\mathcal{K}} : \mathcal{M}(K) \rightarrow Y_{E'}^{an}) \in \mathfrak{J}^{hyp}(X/E)$ be any object. Then*

- (1) Y/E' has the same topological type=(genus, number of punctures) as that of X/E .
- (2) any anabelomorphism

$$\Pi_{Y/E'}^{temp} \xrightarrow{\simeq} \Pi_{X/E}^{temp}$$

provides an anabelomorphism

$$G_{E'} \simeq G_E$$

i.e. the p -adic fields E' and E are anabelomorphic.

Proof. The first assertion is [Mochizuki, 2004, Lemma 1.3.9]. The second assertion is immediate from [Mochizuki, 2004, Lemma 1.3.8]. $\square$

Proposition 5.8.2. *Let X/E be a geometrically connected, smooth, hyperbolic curve over a p -adic field E . Suppose X/E is also of Strict Belyi Type. Then*

$$\mathfrak{J}^{SB}(X, E) = \mathfrak{J}^{hyp}(X, E).$$

Proof. To prove this it is enough to show that if X/E is of Strict Belyi Type, then every hyperbolic curve Y/E' occurring as a part of the datum of any object of $\mathfrak{J}^{hyp}(X, E)$ is also of Strict Belyi Type. This is immediate from Proposition 5.5.1, the proof of [Mochizuki, 2007b, Theorem 2.10], and the definition of Strict Belyi Type hyperbolic curves [Mochizuki, 2007b, Definition 2.9]. $\square$

Remark 5.8.3. Thus for a geometrically connected, smooth, hyperbolic curve, $\mathfrak{J}^{hyp}(X/E)$ has properties similar to classical Teichmüller Spaces: namely local scalings (Theorem 5.4.1, Corollary 5.4.2), the topological types of all objects and the tempered fundamental group (and hence étale fundamental group) is fixed by Proposition 5.8.1.

§ 5.9 For hyperbolic curves of Strict Belyi Type one has the following:

Proposition 5.9.1. *Let E be a p -adic field and X/E be a geometrically connected, smooth, quasi-projective, hyperbolic curve of Strict Belyi Type over E . Then*

- (1) For every $(Y/E', E' \hookrightarrow K) \in \mathfrak{J}^{SB}(X, E)$ one has an isomorphism of schemes (over $\mathbb{Z}$)

$$Y \simeq X.$$

(2) Hence one has a natural action of $\text{Aut}(\Pi)$ on $\mathfrak{J}^{SB}(X, E)$ via its action on

$$\text{Isom}(Y, X) \simeq \text{Isom}^{\text{out}}(\pi_1(Y), \pi_1(X)).$$

(3) However two such triples $(Y/E', E' \hookrightarrow K), (Y''/E'', E'' \hookrightarrow K') \in \mathfrak{J}^{SB}(X, E)$ may not be isomorphic in general.

Proof. The first two assertions are immediate from Mochizuki's proof of the Absolute Grothendieck Conjecture for hyperbolic curves over p -adic fields of Strict Belyi Type (see [Mochizuki, 2007b, Corollary 2.12]) and the construction of $\mathfrak{J}^{SB}(X, E)$. The second assertion is immediate from Proposition 4.1.7. $\square$

§ 5.10 Let me also record the following proposition in the special case of hyperbolic curves of strict Belyi Type.

Proposition 5.10.1. *Let E be a p -adic field. Let X/E be a geometrically connected, smooth, quasi-projective, hyperbolic curve over E of strict Belyi type. Then there is a natural action of $\text{Out}(\Pi)$ where $\Pi = \Pi_{X/E}^{\text{temp}}$ on $\mathfrak{J}_{SB}(X, E)$.*

Proof. This is immediate from Proposition 5.9.1: for any $(Y/E', E' \hookrightarrow K)$ one has an isomorphism of schemes $Y \simeq X$. $\square$

§ 5.11 **Existence of Virasoro Type Uniformization** The data

$$(Y/E', (E' \hookrightarrow K, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_E^{\text{an}}) \in \mathfrak{J}(X, E)$$

is quite rigid but the data

$$(Y/E', (E' \hookrightarrow K, K^b \simeq F))$$

has symmetries stemming from symmetries of both Y/E' and $(E' \hookrightarrow K, K^b \simeq F)$. Hence along with the data $(Y/E', (E' \hookrightarrow K, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_E^{\text{an}}) \in \mathfrak{J}(X, E)$ it is also convenient to work with the category of data $(Y/E', (E' \hookrightarrow K, K^b \simeq F))$ which omits the geometric base-point $*_K : \mathcal{M}(K) \rightarrow Y_E^{\text{an}}$. This leads to the following definition:

Definition 5.11.1. Let E be a p -adic field and let X/E be a geometrically connected, smooth, quasi-projective variety over E . Let $\underline{\mathfrak{J}}(X, E)$ be the category whose objects are pairs

$$(Y/E', (E' \hookrightarrow K, K^b \simeq F))$$

where

- (1) E' is a p -adic field, and
- (2) F is some perfectoid field of characteristic $p > 0$.
- (3) $(E' \hookrightarrow K, K^b \simeq F)$ is an E -untilt of F .
- (4) Y/E' is anabelomorphic to X/E ,
- (5) $\dim(Y) = \dim(X)$; and

Morphisms in $\underline{\mathfrak{J}}(X, E)$ are morphisms of the pairs $(Y/E', (E' \hookrightarrow K, K^\flat \simeq F))$.

Proposition 5.11.2.

(1) *There is a forgetful functor*

$$\mathfrak{J}(X, E) \rightarrow \underline{\mathfrak{J}}(X, E)$$

given by forgetting the geometric base-point:

$$(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_E^{an}) \longmapsto (Y/E, (E \hookrightarrow K, K^\flat \simeq F)).$$

(2) *Given any*

$$(Y/E', (E' \hookrightarrow K, K^\flat \simeq F)) \in \underline{\mathfrak{J}}(X, E),$$

each choice of a geometric base-point

$$*_K : \mathcal{M}(K) \rightarrow Y_{E'}^{an}$$

provides an object

$$(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{an}) \in \mathfrak{J}(X, E).$$

(3) *There is a functor*

$$\underline{\mathfrak{J}}(X, E) \ni (Y/E', (E' \hookrightarrow K, K^\flat \simeq F)) \longmapsto (Y_K^{an} \rightarrow Y_{E'}^{an})$$

from $\underline{\mathfrak{J}}(X, E)$ to the category of morphisms of analytic spaces $\mathbb{Q}_p$ -analytic spaces (with morphisms defined in the obvious way).

(4) *Given $(Y/E', (E' \hookrightarrow K, K^\flat \simeq F)) \in \underline{\mathfrak{J}}(X, E)$, any choice of K -geometric base-point $*_K : \mathcal{M}(K) \rightarrow Y_{E'}^{an}$ gives tempered fundamental group*

$$\Pi_{Y/E'}^{temp} \quad (\simeq \Pi_{X/E}^{temp}).$$

Proof. All the assertions are clear. □

Definition 5.11.3.

(1) I will refer to $(Y_K^{an} \rightarrow Y_{E'}^{an})$ as the arithmetic holomorphic structure associated to

$$(Y/E', (E' \hookrightarrow K, K^\flat \simeq F)) \in \underline{\mathfrak{J}}(X, E).$$

(2) Let $\underline{\mathfrak{J}}(X, E)_E$ be the full subcategory of $\underline{\mathfrak{J}}(X, E)$ consisting of the objects of the form $(Y/E, (E \hookrightarrow K, K^\flat \simeq F)) \in \underline{\mathfrak{J}}(X, E)$. Evidently $(X/E, (E \hookrightarrow K, K^\flat \simeq F)) \in \underline{\mathfrak{J}}(X, E)_E$ and hence $\underline{\mathfrak{J}}(X, E)_E$ is not an empty category.

§ 5.12 The goal of the next few paragraphs is to demonstrate that there are natural symmetries which transform the arithmetic holomorphic structures associated to $(Y/E, (E \hookrightarrow K, K^\flat \simeq F)) \in \mathfrak{J}(X, E)_E$. These symmetries are the arithmetic analogs of the Virasoro action on certain extended moduli spaces of curves and Teichmüller spaces established in [Kontsevich, 1987], [Beilinson and Schechtman, 1988] (for an accessible account see [Frenkel and Ben-Zvi, 2001, Virasoro Uniformization Theorem]). The existence of this property in the present context reinforces my point that one is dealing with a Teichmüller Theory of the arithmetic sort in which arithmetic holomorphic structures change under certain Virasoro action type symmetries.

The key point in the demonstration is that an E -untilt $(E \hookrightarrow K, K^\flat \simeq F)$ of F can be viewed as a closed classical point on the Fargues-Fontaine curve $\mathcal{Y}_{F,E}$ constructed [Fargues and Fontaine, 2018]. A less precise version of this symmetry occurs, in the form of “Indeterminacy Ind2” in [Mochizuki, 2021a,b,c,d], where it plays a central role in [Mochizuki, 2021c, Theorem 3.11 and Corollary 3.12].

§ 5.13 Let E be a p -adic field and let π be a uniformizer for its ring of integers, let q be the cardinality of the residue field $\mathcal{O}_E/\pi$ of E . By a *Lubin-Tate formal group* $\mathcal{G}$ over $\mathcal{O}_E$, I mean a formal group constructed by [Lubin and Tate, 1965], using some polynomial $Q(T) \in \mathcal{O}_E[T]$ satisfying the following two hypothesis of [Lubin and Tate, 1965]:

- (1) $Q(T) = \pi T + O(T^2)$, and
- (2) $Q(T) = T^q \bmod \pi$.

By [Lubin and Tate, 1965] the formal groups determined by two such polynomials is naturally isomorphic. By the results of [Fargues and Fontaine, 2018, Chapter 1, 2] especially [Fargues and Fontaine, 2018, Proposition 2.1.7], the $\mathcal{O}_E$ -algebra $W_{\mathcal{O}_E}(\mathcal{O}_F)$ required in the construction of the Fargues-Fontaine curves is independent of the choice of the Lubin-Tate polynomial $Q(T)$ used to define $\mathcal{G}$ and in particular these constructions are independent of the choice of the Lubin-Tate group $\mathcal{G}$.

§ 5.14 Associated to a Lubin-Tate $\mathcal{O}_E$ -formal group $\mathcal{G}$ over $\mathcal{O}_E$, is a π -divisible group over $\mathcal{O}_E$, and its special fibre (over the residue field of $\mathcal{O}_E$) and also a π -divisible formal $\mathcal{O}_E$ -module. I will pass between these objects whenever needed (to invoke results of [Fargues and Fontaine, 2018]), but *beware that I will notationally conflate all of these objects as $\mathcal{G}$* . Hopefully there will be no confusion.

§ 5.15 I will also use $\mathcal{G}$ for the special fiber of $\mathcal{G}/\mathcal{O}_E$, I hope that readers will be able to unravel the usage from the context (in [Fargues and Fontaine, 2018], the special fiber is denoted by $\mathcal{G}_k$ where k is the residue field of F). This means for example where Fargues and Fontaine write $\mathcal{G}_k(\mathcal{O}_F)$, I will simply write $\mathcal{G}(\mathcal{O}_F)$. By [Fargues and Fontaine, 2018, Proposition 4.4.1] $\mathcal{G}(\mathcal{O}_F)$ is naturally a Banach space over E .

§ 5.16 Suppose E is a p -adic field and $\mathcal{O}_E$ its ring of p -adic integers, let K be an algebraically closed perfectoid field and let $\mathcal{G}$ be an Lubin-Tate group or a $\mathcal{O}_E$ -formal group equipped with an homomorphism $\mathcal{O}_E \rightarrow \text{End}_{\mathcal{O}_K}(\mathcal{G})$. Let $\pi \in \mathcal{O}_E$ be a uniformizer. Let

$$\widetilde{\mathcal{G}(\mathcal{O}_K)} = \varprojlim_{\text{mult. by } \pi} \mathcal{G}(\mathcal{O}_K),$$

this is naturally an E -vector space (and hence also an $\mathcal{O}_E$ -module). Note that $\widetilde{\mathcal{G}(\mathcal{O}_K)}$ is denoted as $X(\mathcal{G})(\mathcal{O}_K)$ [Fargues and Fontaine, 2018, Chap IV].

§ 5.17 Let me explicate this for $E = \mathbb{Q}_p$, $F = \mathbb{C}_p^\flat$ and $K = \mathbb{C}_p$, here $\mathcal{G}(\mathbb{C}_p^\flat) \simeq \widetilde{\mathcal{G}(\mathcal{O}_{\mathbb{C}_p})}$. In this case $\mathcal{G}(\mathcal{O}_{\mathbb{C}_p}) = \mathfrak{m}_{\mathcal{O}_{\mathbb{C}_p}}$ is considered as a $\mathbb{Z}_p$ -module via the Lubin-Tate action in which the endomorphism p acts on $\mathcal{G}(\mathcal{O}_{\mathbb{C}_p})$ by the endomorphism $z \mapsto z^p + pz$ of $\mathcal{G}(\mathcal{O}_{\mathbb{C}_p})$ and $\widetilde{\mathcal{G}(\mathcal{O}_{\mathbb{C}_p})}$ is the $\mathbb{Q}_p$ -vector space obtained by formally inverting the Lubin-Tate action of p on the group $\mathcal{G}(\mathcal{O}_{\mathbb{C}_p}) \simeq \mathfrak{m}_{\mathcal{O}_{\mathbb{C}_p}}$.

§ 5.18 Let E be a p -adic field and let F be an algebraically closed perfectoid field of characteristic $p > 0$. Recall that in [Fargues and Fontaine, 2018, Chapter 2] the construction of the Fargues-Fontaine curve proceeds via the construction of an auxiliary curve, denoted in loc. cit by $Y_{F,E}$ (resp. $X_{F,E}$) (or simply by Y if the choice of F, E is unambiguous) and denoted here by $\mathcal{Y}_{F,E}$ (resp. $\mathcal{X}_{F,E}$). More precisely $\mathcal{Y}_{F,E}$ is constructed as an adic space (but I will not use this fact here).

§ 5.19 Of particular interest to us are the sets of closed points of $|\mathcal{Y}_{F,E}|$ (resp. $|\mathcal{X}_{F,E}|$) of closed (classical points) of $\mathcal{Y}_{F,E}$ (resp. closed points of $\mathcal{X}_{F,E}$). By [Fargues and Fontaine, 2018, Corollaire 2.2.2 and Définition 2.3.1], the set of closed classical points of $\mathcal{Y}_{F,E}$ is in bijection $|\mathcal{Y}_{F,E}| \ni y \mapsto (E \hookrightarrow K_y, K_y^\flat \simeq F)$ with the set of untilts of F (up to isomorphisms of untilts) and where K_y is naturally identified with the residue field of $y \in \mathcal{Y}_{F,E}$. This will be used in what follows. The curve $\mathcal{Y}_{F,E}$ is equipped with a natural Frobenius morphism $\varphi : \mathcal{Y}_{F,E} \rightarrow \mathcal{Y}_{F,E}$ and $\mathcal{X}_{F,E} = \mathcal{Y}_{F,E}/\varphi^{\mathbb{Z}}$ is the (adic space) quotient by the infinite cyclic group generated by the Frobenius morphism φ . In the discussion which follows, I will habitually conflate $\mathcal{X}_{F,E}$ and $\mathcal{Y}_{F,E}$ with $|\mathcal{X}_{F,E}|$ (resp. $|\mathcal{Y}_{F,E}|$).

§ 5.20 By [Fargues and Fontaine, 2018, Théorème 6.5.2(4)], one has a canonical identification

$$|\mathcal{Y}_{F,E}|/\varphi^{\mathbb{Z}} \simeq |\mathcal{X}_{F,E}|,$$

given by $y \mapsto \{\varphi^n(y) : n \in \mathbb{Z}\}$. It is standard that $\mathcal{Y}_{F,E}/\varphi^{\mathbb{Z}} \rightarrow \mathcal{X}_{F,E}$ is in fact a morphism of adic spaces which provides the above identification on points.

The following theorem demonstrates how anabelomorphy of p -adic fields enters the theory of Fargues-Fontaine curves $\mathcal{Y}_{F,E}$ and $\mathcal{X}_{F,E}$.

Theorem 5.20.1. *Let E, E' be anabelomorphic p -adic fields and let F be algebraically closed perfectoid field of characteristic $p > 0$. Then any anabelomorphism $\alpha : G_E \rightarrow G_{E'}$ induces a homeomorphism of topological spaces compatible with the respective topological group actions*

$$G_E \curvearrowright |\mathcal{Y}_{F,E}| \simeq |\mathcal{Y}_{F,E'}| \curvearrowright G_{E'}$$

and

$$G_E \curvearrowright |\mathcal{X}_{F,E}| \simeq |\mathcal{X}_{F,E'}| \curvearrowright G_{E'}.$$

Proof. One has the natural identifications ([Fargues and Fontaine, 2018, Proposition 2.1.10])

$$(5.20.2) \quad |\mathcal{Y}_{F,E}| = (\mathcal{G}(\mathcal{O}_F) - \{0\})/\mathcal{O}_E^*,$$

and ([Fargues and Fontaine, 2018, Theorem 6.5.2])

$$(5.20.3) \quad |\mathcal{X}_{F,E}| = (\mathcal{G}(\mathcal{O}_F) - \{0\}) / E^*,$$

and similarly for E' . The topological space $(\mathcal{G}(\mathcal{O}_F) - \{0\})$ is independent of E, E' and the anabelomorphism $\alpha : G_E \rightarrow G_{E'}$ induces isomorphisms of the topological groups

$$\mathcal{O}_E^* \simeq \mathcal{O}_{E'}^*$$

and

$$E^* \simeq E'^*.$$

Hence all the assertions are immediate. $\square$

Remark 5.20.4. On the other hand, I have shown in [Joshi, 2020b] that, if E, E' are strictly anabelomorphic p -adic fields, then schemes $\mathcal{X}_{F,E}$ and $\mathcal{X}_{F,E'}$ are anabelomorphic but not isomorphic as schemes.

§ 5.21 With this preparation let me prove the following:

Theorem 5.21.1. *Let the notation and assumptions be as in the previous paragraph. Assume K is an algebraically closed perfectoid field with $K^\flat = F$.*

- (1) *The isomorphism class of the topological $\mathcal{O}_E$ -module $\widetilde{\mathcal{G}(\mathcal{O}_K)}$ is independent of K more precisely, there is a natural homeomorphism of $\mathcal{O}_E$ -modules*

$$\widetilde{\mathcal{G}(\mathcal{O}_K)} \simeq \mathcal{G}(\mathcal{O}_F),$$

(in fact this is an isomorphism of Banach E -vector spaces).

- (2) *The isomorphism class of the topological $\mathcal{O}_E$ -module $\mathcal{G}(\mathcal{O}_F)$ is independent of the choice of the Lubin-Tate $\mathcal{O}_E$ -formal group $\mathcal{G}$.*
- (3) *There is a natural action of the group $\text{Aut}_{\mathcal{O}_E}(\mathcal{G}(\mathcal{O}_F))$, of topological automorphisms of the $\mathcal{O}_E$ -module $\mathcal{G}(\mathcal{O}_F) \simeq \widetilde{\mathcal{G}(\mathcal{O}_K)}$, on the set of closed points of degree one of the Fargues-Fontaine curve $\mathcal{Y}_{F,E}$, arising from the natural identification*

$$|\mathcal{Y}_{F,E}| = (\mathcal{G}(\mathcal{O}_F) - \{0\}) / \mathcal{O}_E^*.$$

- (4) *Let $y \in \mathcal{Y}_{F,E}$ be a closed point of degree one. Then one has an action of $\text{Aut}_{\mathcal{O}_E}(\mathcal{G}(\mathcal{O}_F))$ on closed points of degree one of $\mathcal{X}_{F,E}$ via mapping*

$$\{\varphi^n(y) : n \in \mathbb{Z}\} \longmapsto \{\varphi^n(\sigma(y)) : n \in \mathbb{Z}\}.$$

- (5) *Thus given any topological $\mathcal{O}_E$ -linear automorphism*

$$\sigma : \mathcal{G}(\mathcal{O}_F) \xrightarrow{\simeq} \mathcal{G}(\mathcal{O}_F),$$

and a closed point $y \in \mathcal{Y}_{F,E}$ of degree one, giving an untilt $y = (E \hookrightarrow K_y, K_y^\flat \simeq F)$ of F with residue field K_y , there is an untilt $\sigma(y) = (E \hookrightarrow K_{\sigma(y)}, K_{\sigma(y)}^\flat \simeq F)$ given via (5.20.2) with residue (also perfectoid, algebraically closed field) $K_{\sigma(y)}$ and with isometries $K_y^\flat \simeq F \simeq K_{\sigma(y)}^\flat$ and an embedding $E \hookrightarrow K_{\sigma(y)}$.

(6) In particular $\text{Aut}_{\mathcal{O}_E}(\mathcal{G}(\mathcal{O}_F))$ acts naturally on $\mathfrak{J}(X, E)_F$ via

$$(Y, y = (E \hookrightarrow K, K^\flat \simeq F)) \longmapsto (Y, \sigma(y) = (E \hookrightarrow K_{\sigma(y)}, K_{\sigma(y)}^\flat \simeq F))$$

for all $\sigma \in \text{Aut}_{\mathcal{O}_E}(\mathcal{G}(\mathcal{O}_F))$.

Proof. Before proceeding to the proofs let me remark that items (1) and (2) are due to [Fargues and Fontaine, 2018] and I include them here for completeness. The assertion (1) is [Fargues and Fontaine, 2018, Proposition 4.5.11] (what I have denoted as $\widetilde{\mathcal{G}(\mathcal{O}_K)}$ is denoted by $X(\mathcal{G})(\mathcal{O}_K)$ in loc. cit.). The independence from the choice of the $\mathcal{O}_E$ -formal Lubin-Tate group $\mathcal{G}$ is clear from [Lubin and Tate, 1965] as the Lubin-Tate $\mathcal{O}_E$ -formal group over $\mathcal{O}_E$ is unique up to isomorphism by [Lubin and Tate, 1965].

The identification of $|\mathcal{Y}_{F,E}|$ with the $(\mathcal{G}(\mathcal{O}_F) - \{0\}) / \mathcal{O}_E^*$ is [Fargues and Fontaine, 2018, Proposition 2.1.10] and hence for any $\sigma \in \text{Aut}_{\mathcal{O}_E}(\mathcal{G}(\mathcal{O}_F))$, σ is evidently a bijection on $(\mathcal{G}(\mathcal{O}_F) - \{0\}) / \mathcal{O}_E^*$. Thus the claim (3) is immediate.

The proof of (4) is now clear now that (3) has been established.

The assertion (5) is self-evident and it implies (6). This completes the proof. $\square$

§ 5.22 For readers familiar with the Geometric Langlands Program [Beilinson and Drinfel'd, 2000] over $\mathbb{C}$, let me remark that the action of $\text{Aut}_{\mathbb{Z}_p}(\mathcal{G}(\mathcal{O}_{\mathbb{C}_p}))$ considered in Theorem 5.21.1 is the p -adic analog of the action of the Virasoro Algebra on moduli spaces of marked Riemann surfaces described in the Virasoro uniformization Theorem [Beilinson and Schechtman, 1988, Section 4], [Beilinson and Drinfel'd, 2000], [Frenkel and Ben-Zvi, 2001]. In the Geometric Langlands setting of [Beilinson and Drinfel'd, 2000], the Virasoro algebra plays a fundamental role and manifests itself via the action of the group scheme $\mathbb{C} \subset R \longmapsto \text{Aut}_{\text{cont}}(R((T)))$. As is described in [Beilinson and Schechtman, 1988, Section 4] or [Frenkel and Ben-Zvi, 2001, Theorem 17.3.2], this action also changes complex structures of marked Riemann surfaces (in general). As has been noted above, $\text{Aut}_{\mathbb{Z}_p}(\mathcal{G}(\mathcal{O}_{\mathbb{C}_p}))$ acts by changing the analytic structure of $(X \times_E K)^{\text{an}}$ and hence must be considered as the p -adic analog of the Virasoro action in the complex setting. As was remarked in [Beilinson and Schechtman, 1988], the Virasoro uniformization Theorem complements the Teichmüller Uniformization.

§ 5.23 An important consequence of this is that topological $\mathcal{O}_E$ -linear automorphisms of $\mathcal{G}(\mathcal{O}_F)$ can be used to change arithmetic holomorphic structures in the sense of Theorem 3.16.1. In [Mochizuki, 2021a,b,c,d], Mochizuki claims this assertion in the guise of “Indeterminacy Ind2” [Mochizuki, 2021c, Theorem 3.11].

§ 5.24 An important consequence of Theorem 5.21.1 and 10.5.1 is the following:

Theorem 5.24.1. *Let F be an algebraically closed, perfectoid field of characteristic $p > 0$. There is a natural action of $\text{Aut}_{\mathbb{Z}_p}(\mathcal{G}(\mathcal{O}_F)) \simeq \text{Aut}_{\mathbb{Z}_p}(\widehat{\mathbb{G}}_m(\mathcal{O}_F))$ on the closed points $|\mathcal{Y}_{F, \mathbb{Q}_p}|$*

of $\mathcal{Y}_{F, \mathbb{Q}_p}$, which provides an action of $\text{Aut}_{\mathbb{Z}_p}(\widehat{\mathbb{G}}_m(\mathcal{O}_F))$ by correspondences on $\mathfrak{J}(X, E)_F$. Explicitly this is given as follows.

(1) Let

$$\mathcal{G}(\mathcal{O}_F) \xrightarrow{\sigma} \mathcal{G}(\mathcal{O}_F)$$

equivalently, let

$$\widehat{\mathbb{G}}_m(\mathcal{O}_F) \xrightarrow{\sigma} \widehat{\mathbb{G}}_m(\mathcal{O}_F)$$

be a $\mathbb{Z}_p$ -linear isomorphism of topological groups.

(2) Let $\tau : |\mathcal{Y}_{F, E'}| \rightarrow |\mathcal{Y}_{F, \mathbb{Q}_p}|$ be the finite morphism given by mapping an E' -untilt $y \in |\mathcal{Y}_{F, E'}|$ of F to the underlying $\mathbb{Q}_p$ -untilt $\tau(y) = x \in |\mathcal{Y}_{F, \mathbb{Q}_p}|$ of F [Fargues and Fontaine, 2018, Proposition 2.3.20].

(3) Let $y_1, \dots, y_m \in |\mathcal{Y}_{F, E'}|$ be all the points lying over $\sigma(x) \in \mathcal{Y}_{F, \mathbb{Q}_p}$ i.e. $\tau^{-1}(\sigma(x)) = \{y_1, \dots, y_m\} \subset |\mathcal{Y}_{F, E'}|$.

Then the divisorial correspondence (i.e. one-to-many mapping between object class)

$$(5.24.2) \quad \underline{\mathfrak{J}}(X, E)_F \dashrightarrow \underline{\mathfrak{J}}(X, E)_F$$

is given by the rule

$$(Y/E', y = (E' \hookrightarrow K_y, K_y^\flat \simeq F)) \longmapsto ((Y/E', y_j = (E' \hookrightarrow K_{y_j}, K_{y_j}^\flat \simeq F)))_{j=1, \dots, m}.$$

Moreover, the number $m = [E' : \mathbb{Q}_p] = [E : \mathbb{Q}_p]$ depends only on E and not on the chosen object of $\underline{\mathfrak{J}}(X, E)$.

Proof. Let assertion $m = [E' : \mathbb{Q}_p]$ is [Fargues and Fontaine, 2018, Proposition 2.3.20]. The assertion $[E' : \mathbb{Q}_p] = [E : \mathbb{Q}_p]$ is immediate from the fact E' and E are anabelomorphic p -adic fields Definition 5.1.1(4), Proposition 5.8.1. The rest of the proof is clear from the method of proof of Theorem 5.21.1. $\square$

§ 5.25 The results of the preceding paragraphs can be assembled into the following theorem:

Theorem 5.25.1. *Let E be a p -adic field, let X/E be a geometrically connected, smooth, quasi-projective variety over E . Then there exists a category $\mathfrak{J}(X, E)$, called the Arithmetic Teichmüller Space associated to X/E , with the following properties:*

- (1) *objects of $\mathfrak{J}(X, E)$ are $(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{an})$ (Definition 5.1.1). Morphisms between these objects are defined in the obvious way.*
- (2) *The category $\mathfrak{J}(X, E)$ is an anabelian variation providing $\Pi = \Pi_{X/E}^{temp}$ (see § 13.8, Theorem 13.13.1) i.e. for any $(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{an}) \in \mathfrak{J}(X, E)$, one has an isomorphism of topological groups*

$$\Pi_{Y/E'}^{temp} \simeq \Pi_{X/E}^{temp},$$

inducing an isomorphism of their geometric tempered fundamental subgroups

$$\Pi_{Y/K}^{\text{temp}} \simeq \Pi_{X/K}^{\text{temp}},$$

(3) There are forgetful functors (see Proposition 4.1.10):

- (i) $(Y/E', (E' \hookrightarrow K, K^{\flat} \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{\text{an}}) \mapsto Y/\mathbb{Z}$ (i.e. to Schemes/ $\mathbb{Z}$).
- (ii) $(Y/E', (E' \hookrightarrow K, K^{\flat} \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{\text{an}}) \mapsto E'$ (i.e. to p -adic fields).
- (iii) $(Y/E', (E' \hookrightarrow K, K^{\flat} \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{\text{an}}) \mapsto K$ (i.e. to algebraically closed perfectoid fields of characteristic zero).
- (iv) $(Y/E', (E' \hookrightarrow K, K^{\flat} \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{\text{an}}) \mapsto K^{\flat}$ (i.e. to algebraically closed perfectoid fields of characteristic $p > 0$).

(4) There are functors to analytic spaces (see Proposition 4.1.10)

$$(Y/E', (E' \hookrightarrow K, K^{\flat} \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{\text{an}}) \mapsto (Y_{E'}^{\text{an}}),$$

and

$$(Y/E', (E' \hookrightarrow K, K^{\flat} \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{\text{an}}) \mapsto (Y_K^{\text{an}}),$$

and

$$(Y/E', (E' \hookrightarrow K, K^{\flat} \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{\text{an}}) \mapsto (Y_K^{\text{an}} \rightarrow Y_{E'}^{\text{an}}).$$

(5) There are functors to Mochizuki's anabelian landscape (see § 10.13): (one uses the given perfectoid field to compute algebraic closures)

$$(Y/E', (E' \hookrightarrow K, K^{\flat} \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{\text{an}}) \mapsto \Pi_{Y/E'}^{\text{temp}} \curvearrowright \mathcal{O}_{\overline{E}}^* \subset \mathcal{O}_K,$$

and also

$$(Y/E', (E' \hookrightarrow K, K^{\flat} \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{\text{an}}) \mapsto \Pi_{Y/E'}^{\text{temp}} \curvearrowright \mathcal{O}_{\overline{E}}^{\times \mu} \subset \mathcal{O}_K^*/\mu(K),$$

and similarly

$$(Y/E', (E' \hookrightarrow K, K^{\flat} \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{\text{an}}) \mapsto \Pi_{Y/E'}^{\text{temp}} \curvearrowright \mathcal{O}_{\overline{E}}^{\triangleright} \subset \mathcal{O}_K^{\triangleright}.$$

- (6) If $\dim(X) = 1$ and X is of Strict Belyi Type (this condition is defined in [Mochizuki, 2013, Definition 3.5]) then one has an action of $\text{Aut}(\Pi)$ on $\mathfrak{J}(X, E)$ (Proposition 5.9.1).
- (7) For a fixed algebraically closed, perfectoid field F of characteristic $p > 0$, there are categories $\mathfrak{J}(X, E)_F$ consisting of $(Y/E', (E' \hookrightarrow K, K^{\flat} \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{\text{an}})$ such that $K^{\flat} = F$.
- (8) Now fix an algebraically closed perfectoid field F of characteristic $p > 0$, a uniformizer π for E and let $\mathcal{G}/\mathcal{O}_E$ be the Lubin-Tate formal group. Then there is a natural action of

$\text{Aut}_{\mathcal{O}_E}(\mathcal{G}(\mathcal{O}_F))$ on $\mathfrak{J}(X, E)_F$ (Theorem 5.21.1). Notably for $F = \mathbb{C}_p^\flat$ one has a natural action (Corollary 5.24.1) of

$$\text{Aut}_{\mathbb{Z}_p}(\mathcal{G}(\mathcal{O}_{\mathbb{C}_p^\flat})) \simeq \text{Aut}_{\mathbb{Z}_p}(\widetilde{\mathcal{G}(\mathcal{O}_{\mathbb{C}_p^\flat})})$$

on $\mathfrak{J}(X, E)_{\mathbb{C}_p^\flat}$.

(9) The category $\mathfrak{J}(X, E)_{\mathbb{C}_p^\flat}$ is self-similar (Theorem 14.7.1).

Proof. The only assertion which remains to be proved is the last claim that $\mathfrak{J}(X, E)_{\mathbb{C}_p^\flat}$ is self-similar and this is Theorem 14.7.1 and will be proved in the next section. $\square$

6 Connectedness of Arithmetic Teichmüller Spaces and the abs. Grothendieck Conjecture

In this section I want to demonstrate that there is a natural notion of connectedness for $\mathfrak{J}(X, E)$ which is in fact related to the validity of the Absolute Grothendieck Conjecture for hyperbolic curves over p -adic fields. In particular, as Mochizuki has established this validity for curves of Strict Belyi Type, one obtains connectedness of $\mathfrak{J}^{hyp}(X, E)$ for X/E of strict Belyi Type.

§ 6.1 The Absolute Grothendieck Conjecture My formulation of the Absolute Grothendieck Conjecture over p -adic fields is based on [Mochizuki, 2007b, Corollary 2.3]. This conjecture asserts the following. Assume that E is a p -adic field, let X/E , (resp. Y/E') be a geometrically connected, smooth quasi-projective variety over a p -adic field E (resp. E'). Then any isomorphism of topological groups

$$\Pi_{X/E}^{temp} \xrightarrow{\simeq} \Pi_{Y/E'}^{temp}$$

arises from an isomorphism of $\mathbb{Z}$ -schemes

$$X \xrightarrow{\simeq} Y.$$

As stated this is, of course, *a very optimistic formulation*, and one expects that some additional hypothesis are required for the validity of this assertion. The only case in which this conjecture is presently known is the case of geometrically connected, smooth, hyperbolic curves of Strict Belyi Type over p -adic fields. This case was established by Mochizuki in [Mochizuki, 2007b, Corollary 2.12]. In [Joshi, 2020b], I have shown that this conjecture is false for Fargues-Fontaine curves (these are not of finite type over p -adic fields).

§ 6.2 Connectedness of $\mathfrak{J}(X, E)$ and the validity of the Absolute Grothendieck Conjecture There is a natural notion of connectedness in the theory of Arithmetic Teichmüller Spaces:

Definition 6.2.1. I will say that $\mathfrak{J}(X, E)$ (resp. any full subcategory $\mathfrak{J}$ of $\mathfrak{J}(X, E)$)—for example, if $\dim(X) = 1$, then one can take $\mathfrak{J} = \mathfrak{J}^{hyp}(X, E)$) is a *connected Arithmetic Teichmüller Space* (resp. is connected) if for every object $(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{an}) \in \mathfrak{J}(X, E)$ (resp. $(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{an}) \in \mathfrak{J}$), one has an isomorphism $X \simeq Y$ of schemes over $\mathbb{Z}$.

Remark 6.2.2. Since isomorphism of $\mathbb{Z}$ -schemes is an equivalence relation on schemes, one sees that the class of objects of $\mathfrak{J}(X, E)$ (resp. $\mathfrak{J}^{hyp}(X, E)$) can be partitioned, by this equivalence relation, into a disjoint union of connected components.

The relationship between connectedness of Arithmetic Teichmüller Spaces and the Absolute Grothendieck Conjecture is summarized in the following:

Theorem 6.2.3. *Let E be a p -adic field and let X/E be a geometrically connected, smooth, quasi-projective variety over E . Then*

- (1) *If the Absolute Grothendieck Conjecture (see section 6.1) holds true for any pair of objects of $\mathfrak{J}(X, E)$, then $\mathfrak{J}(X, E)$ is a connected Arithmetic Teichmüller Space.*
- (2) *In particular if X/E is additionally an hyperbolic curve of strict Belyi Type [Mochizuki, 2007a] then the Arithmetic Teichmüller Space $\mathfrak{J}^{SB}(X, E) = \mathfrak{J}^{hyp}(X, E)$ is connected.*

Proof. The first assertion is immediate from the formulation of the Absolute Grothendieck Conjecture in section 6.1. ones therefore sees that $X \simeq Y$ as $\mathbb{Z}$ -schemes and hence $\mathfrak{J}(X, E)$ is connected. This proves the first assertion.

To prove the second assertion, note that one has $\mathfrak{J}^{SB}(X, E) = \mathfrak{J}^{hyp}(X, E)$ by Proposition 5.8.2. By [Mochizuki, 2007b, Corollary 2.12], it is shown that the Absolute Grothendieck Conjecture holds for any pair of smooth, hyperbolic, anabelomorphic curves of Strict Belyi Type over any pair of p -adic fields. Hence $\mathfrak{J}^{hyp}(X, E)$ is connected and the second assertion follows. $\square$

§ 6.3 Existence of a field of definition of $\mathfrak{J}^{hyp}(X, E)$ for X/E of Strict Belyi Type Assume for this section that X/E is a geometrically connected, smooth hyperbolic curve of Strict Belyi Type. Then by Theorem 6.2.3, the Arithmetic Teichmüller Space $\mathfrak{J}^{hyp}(X, E)$ is connected. In this situation, I observe below that every object of $\mathfrak{J}(X, E)$ has a common field of definition.

In [Mochizuki, 2015, Theorem 1.9] Mochizuki provides a anabelian reconstruction theoretic proof of the existence of this common field over which every object of $\mathfrak{J}^{hyp}(X, E)$ is defined. The existence of this common field of definition is analogous to role played by the reflex field in the context of Shimura varieties.

Theorem 6.3.1. *Let X/E be a geometrically connected, smooth hyperbolic curve over a p -adic field E . Assume that X is of Strict Belyi Type (this implies that X is definable over some number field). Let L_{mod} be the field of moduli of X/E . Then for every $(Y/E', Y/K) \in \mathfrak{J}^{hyp}(X, E)$,*

- (1) *Y/E' is of strict Belyi Type,*
- (2) *and Y/E' is definable over the number field L_{mod} .*

In particular, there is a common number field L_{mod} which is determined by every object of $\mathfrak{J}^{hyp}(X, E)$.

Remark 6.3.2. Note that in the situation of Theorem 6.3.1, by [Joshi, 2022, Theorem 5.4(2)], one sees that $\mathfrak{J}(X, E)$ is a connected Arithmetic Teichmüller Space and hence this common number field L_{mod} should be considered as the field of definition of the Arithmetic Teichmüller Space $\mathfrak{J}(X, E)$ in a manner entirely analogous to the notion of the reflex field of a Shimura variety datum (see [Deligne, 1979]).

Proof. Let $(Y/E', E' \hookrightarrow K) \in \mathfrak{J}^{hyp}(X, E)$. This condition means that one has an anabelomorphism

$$(6.3.3) \quad \Pi_{Y/E'; K}^{temp} \simeq \Pi_{X/E, \mathbb{C}_p}^{temp}.$$

The first assertion is proved as follows. The isomorphism of tempered fundamental groups also forces an isomorphism of the corresponding étale fundamental groups (see [André, 2003b, Proposition 4.4.1]). One knows from the proof of [Mochizuki, 2007b, Proposition 2.4(iv)] that the property of being of strict Belyi Type is an amphoric property and hence Y/E' is also of strict Belyi Type. In particular Y/E' is also definable over a number field.

Now to deduce second assertion, one uses Mochizuki's proof of the validity of the Absolute Grothendieck Conjecture for curves of strict Belyi Type (by [Mochizuki, 2007b, Corollary 2.12] which was referred above in section 6.1). This furnishes us with an isomorphism of absolute schemes $Y \simeq X$ (over $\mathbb{Z}$). One also sees from Proposition 5.8.1 that X/E and Y/E' have the same genus and the same number of punctures (with respect to their respective, smooth proper models).

Since X/E and Y/E' are both of strict Belyi type, and hence both are definable over number fields, say L and L' respectively. The isomorphism $Y \simeq X$ as $\mathbb{Z}$ -schemes provides the same moduli point corresponding to the common isomorphism class of these curves in the moduli. Let L_{mod} be the field of moduli of X/E , then $L \supseteq L_{\text{mod}}$ and one has that Y/E' also has the field of moduli L_{mod} (so $L' \supseteq L_{\text{mod}}$). So the second assertion is proved. $\square$

7 Untilts of absolute Galois Groups and Fundamental Groups

§ 7.1 Untilts of fundamental groups of Riemann surfaces It will be useful to begin with a discussion of the classical case of Riemann surfaces. Let Σ be a connected Riemann surface, which one assumes to be hyperbolic to avoid trivialities. $\Pi = \pi_1^{top}(\Sigma)$ be the topological fundamental group of a connected Riemann surface Σ . Let $\mathcal{T}_\Sigma$ be the Teichmüller space of Σ .

§ 7.2 The assignment

$$\mathcal{T}_\Sigma \ni (\Sigma', f) \longmapsto \pi_1^{top}(\Sigma') \xrightarrow{\simeq} \Pi = \pi_1^{top}(\Sigma)$$

provides a function from the Teichmüller space $\mathcal{T}_\Sigma$ to the isomorphism class of the group Π . Hence $\pi_1^{top}(\Sigma')$ is an isomorph of Π with $(\Sigma', f) \in \mathcal{T}_\Sigma$ serving as a geometrically distinguishable feature of this copy of Π .

I will write $\Pi_{(\Sigma', f)}$ for the isomorph of $\Pi = \Pi_\Sigma$ provided by the Teichmüller datum (Σ', f) and I will say $\Pi_{(\Sigma', f)}$ is an *untilt of the fundamental group* $\Pi = \Pi_\Sigma$ of the Riemann surface Σ .

For the anabelian contexts of [Mochizuki, 2021a,b,c,d], the construction of the Arithmetic Teichmüller Space $\mathfrak{J}(X, E)$ provides similar, intrinsic arithmetic-geometric labels for isomorphs of the tempered fundamental group.

§ 7.3 Untilts of absolute Galois groups of p -adic fields Let E be a p -adic field. Recall that to define the absolute Galois group G_E of E , one requires an algebraic closure of E .

Definition 7.3.1. Let F be an algebraically closed, perfectoid field of characteristic $p > 0$. Let $(K \supset E, K^\flat \simeq F)$ be a characteristic zero E -untilt of F . Let $\overline{E} \subset K$ be the algebraic closure of E contained in K (note that K is algebraically closed, perfectoid of characteristic zero, and $\overline{E}$ is equipped with the valuation induced from K). I will refer to $\overline{E}$ as *the algebraic closure of E provided by the given E -untilt $(K \supset E, K^\flat \simeq F)$ of F* . Let

$$G_{E,(K \supset E, K^\flat \simeq F)} = \text{Gal}(\overline{E}/E) \simeq G_E$$

be the absolute Galois group of E computed using the algebraic closure provided by the untilt E -untilt. I will refer to $G_{E,(K \supset E, K^\flat \simeq F)} = \text{Gal}(\overline{E}/E)$ as the *untilt of the absolute Galois group G_E of E provided by the E -untilt $(K \supset E, K^\flat \simeq F)$ of F* .

Remark 7.3.2. For brevity, I will habitually shorten the notation

$$G_{E,(K \supset E, K^\flat \simeq F)}$$

to

$$G_{E;K}.$$

The following is immediate from Definition 7.3.1.

Proposition 7.3.3. *Let E be a p -adic field. Then*

- (1) *Any pair of untilts of G_E are isomorphic as topological groups, but there is no canonical isomorphism between them.*
- (2) *There is a category, denoted $\mathcal{G}\mathcal{A}\mathcal{L}_E$, of all untilts of G_E which has for its objects the collection of untilts of G_E given by*

$$\{G_{E,(K \supset E, K^\flat \simeq F)} : F \text{ is some algebraically closed perfectoid field of } \text{char}(F) = p > 0.\}$$

and whose morphisms are isomorphisms of topological groups.

- (3) *The category $\mathcal{G}\mathcal{A}\mathcal{L}_E$ comes equipped with functors*

$$\begin{array}{ccccc} & G_{E,(K \supset E, K^\flat \simeq F)} & & & \\ & \swarrow \quad \downarrow \quad \searrow & & & \\ F & & K & & (K \supset E, K^\flat \simeq F) \end{array}$$

to the category of perfectoid fields of characteristic $p > 0$, the category of perfectoid fields of characteristic zero, the category of E -untilts of some perfectoid field of characteristic $p > 0$ respectively.

Proof. The proof is clear from the construction of untilts of G_E . □

§ 7.4 Untilts of tempered fundamental groups Now I am ready to define untilts of tempered fundamental groups.

Definition 7.4.1. Let X/E be a geometrically connected, smooth, quasi-projective variety over a p -adic field E . Let $(X/E; (K \supset E, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an})$ be an arithmetic holomorphic structure on X/E . An *untilt of the tempered fundamental group* $\Pi = \Pi_{X/E}^{temp}$ is the isomorph

$$(7.4.2) \quad \Pi_{(X/E; (K \supset E, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an})}^{temp} = \pi_1^{temp}(X_E^{an}, *_K : \mathcal{M}(K) \rightarrow X_E^{an})$$

of the tempered fundamental group $\Pi_{X/E}^{temp}$ computed using the geometric base-point provided by the given arithmetic holomorphic structure on X/E .

To understand the Anabelian consequences of the theory of Arithmetic Holomorphic Structures, let me begin with the following assertion.

Theorem 7.4.3. *Let X/E be a geometrically connected, smooth, quasi-projective variety over a p -adic field E . Let $\mathbb{C}_p$ be the completion of a fixed algebraic closure of $\mathbb{Q}_p$. Let $\Pi_{X/E}^{temp}$ be the tempered fundamental group of X/E computed using any geometric base-point $\mathcal{M}(\mathbb{C}_p) \rightarrow X_E^{an}$. Then*

- (1) *Non-isomorphic arithmetic holomorphic structures on X/E (Definition 4.1.1) exist (by Lemma 4.1.7).*
- (2) *For any algebraically closed perfectoid field F of characteristic $p > 0$, the topological isomorphism class of the pro-discrete group*

$$\Pi_{(X/E; (K \supset E, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an})}^{temp}$$

is independent of the choice of an arithmetic holomorphic structure

$$(X/E; (K \supset E, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an})$$

on X/E .

- (3) *Hence each arithmetic holomorphic structure provides an isomorph (7.4.2) of the topological groups $\Pi_{X/E}^{temp} \supset \Pi_{X/\mathbb{C}_p}^{temp}$ and an exact sequence (4.1.5).*
- (4) *Any choice of an arithmetic holomorphic structure (in the sense of Definition 4.1.1) provides an arithmetic holomorphic structure in the sense of Mochizuki [Mochizuki, 2021b, Example 1.8]:*

$$\Pi_{(X/E; (K \supset E, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow X_E^{an})}^{temp} \twoheadrightarrow G_{E;K}$$

of i.e. an isomorph of the surjection

$$\Pi_{X/E}^{temp} \twoheadrightarrow G_E.$$

- (5) *If F is an algebraically closed, perfectoid field of characteristic $p > 0$, then one has continuous families of isomorphs of $\Pi_{X/E}^{temp}$ parameterized by Fargues-Fontaine curves $\mathcal{Y}_{F,E}$ and $\mathcal{X}_{F,E}$.*

(6) In particular one has continuous families of arithmetic holomorphic structures (in the sense of Definition 4.1.1) and also in the sense of [Mochizuki, 2021a,b,c,d] which are parameterized by Fargues-Fontaine curves $\mathcal{Y}_{F,E}$ and $\mathcal{X}_{F,E}$ for any arbitrary algebraically closed perfectoid field F of characteristic $p > 0$.

Proof. The assertion (1) follows from Proposition 4.1.7. The assertions (2,3) is immediate from the independence of the tempered fundamental groups on geometric base-points [André, 2003b]. The assertion (4) is immediate from the exact sequence (4.1.5) and [Mochizuki, 2021b, Example 1.8]. Now (5) is immediate from Definition 4.1.1 and the constructions of $\mathcal{Y}_{F,E}$ [Fargues and Fontaine, 2018]. The last assertion (6) is immediate from (4). $\square$

Remark 7.4.4. Theorem 7.4.3, especially Theorem 7.4.3(6), represents a substantial strengthening of the anabelian philosophy espoused by Mochizuki in [Mochizuki, 2021a,b,c,d], as the existence of continuous parameter spaces for isomorphisms of tempered fundamental groups is not suggested by [Mochizuki, 2021a,b,c,d].

Remark 7.4.5. One may think of an untilt of $\Pi_{X/E}^{temp}$ as an isomorphism equipped an arithmetic-geometric label (namely the datum of the arithmetic holomorphic structure giving birth to the said isomorphism). This is similar to labeling procedure discussed in the classical case Section 7.1.

Corollary 7.4.6. *Let X/E be a geometrically connected, smooth, quasi-projective variety over a p -adic field E . Then the class of untilds of the tempered fundamental group of X/E is the class of (isomorphic) topological groups*

$$(7.4.7) \quad \left\{ \Pi_{(X/E; (K \supset E, K^b \simeq F), *_{K}: \mathcal{M}(K) \rightarrow X_E^{an})}^{temp} \right\}$$

where F is some algebraically closed, perfectoid field of characteristic $p > 0$, and where $(X/E; (K \supset E, K^b \simeq F), *_{K}: \mathcal{M}(K) \rightarrow X_E^{an})$ is some arithmetic holomorphic structure on X/E .

Remark 7.4.8. The above corollary is the precise version of Mochizuki's Key Principle of Inter-Universality [Mochizuki, 2021a, § I3, Pages 25–26] which according to Mochizuki lies the foundation of the anabelian theory described in [Mochizuki, 2021a,b,c,d].

Remark 7.4.9. In this paper, for simplicity of notation, I will habitually contract the notation

$$\Pi_{(X/E; (K \supset E, K^b \simeq F), *_{K}: \mathcal{M}(K) \rightarrow X_E^{an})}^{temp}$$

to

$$\Pi_{X/E; K}^{temp}$$

8 Galois cohomology and Arithmetic Teichmüller Spaces

In this section I want to outline some consequences, for Galois cohomology, of the existence of Arithmetic Teichmüller Space $\mathfrak{T}(X, E)$ constructed in § 5. As an application one obtains a robust version of Mochizuki's idea, in [Mochizuki, 2021c, Section 3], of collating Galois cohomology classes arising from distinct arithmetic holomorphic structures. This collation

process is central for the formulation of [Mochizuki, 2021c, Corollary 3.12] and for the volume computations of [Mochizuki, 2021d], [Joshi, 2024b]. However, the present section is devoted to developing the local formalism which facilitates this collation of Galois cohomology classes, while the adelic case (which is needed for arithmetic applications) is detailed in [Joshi, 2023a], [Joshi, 2024c]. This allows me to independently and explicitly demonstrate many assertions of [Mochizuki, 2021a,b,c,d] dealing with this collation process—notably [Mochizuki, 2021c, Corollary 3.12].

§ 8.1 Let X/E be a geometrically connected, smooth, quasi-projective variety over a p -adic field. Then by Definition 5.1.1 one has the arithmetic Teichmüller space $\mathfrak{J}(X, E)$ associated with X/E . For applications to [Mochizuki, 2021c, Corollary 3.12], one assumes additionally that X/E is a hyperbolic curve. Choose an arithmetic holomorphic structure $(X/E, (\mathbb{C}_p \supset E, \mathbb{C}_p^b \xrightarrow{id} \mathbb{C}_p^b), *_{\mathbb{C}_p} : \mathcal{M}(\mathbb{C}_p) \rightarrow X_E^{an})$ where $\mathbb{C}_p^b \xrightarrow{id} \mathbb{C}_p^b$ is the identity isometry of $\mathbb{C}_p^b$. The E -untilt $(\mathbb{C}_p \supset E, \mathbb{C}_p^b \xrightarrow{id} \mathbb{C}_p^b)$ is a closed classical point of $\mathcal{Y}_{\mathbb{C}_p^b, E}$ lying over the canonical point of $\mathcal{X}_{\mathbb{C}_p^b, \mathbb{Q}_p}$ for the composite morphism $\mathcal{Y}_{\mathbb{C}_p^b, E} \rightarrow \mathcal{X}_{\mathbb{C}_p^b, E} \rightarrow \mathcal{X}_{\mathbb{C}_p^b, \mathbb{Q}_p}$. I will refer to this as the *standard arithmetic holomorphic structure on X/E* .

§ 8.2 Let $(Y/E', (E' \hookrightarrow K, K^b \simeq F), *_{\mathcal{K}} : \mathcal{M}(K) \rightarrow Y_{E'}^{an}) \in \mathfrak{J}(X, E)$. This comes equipped with a preferred algebraic closure $\overline{E}'$ of E' namely the algebraic closure $\overline{E}' \subset K$ contained in the given algebraically closed perfectoid field K . Hence one also has $G_{E'; K} = \text{Gal}(\overline{E}'/E')$, of course (as abstract topological groups one has $G_{E'; K} \simeq G_{E'}$ but I am emphasizing the choice of the preferred algebraic closure $\overline{E}'$). By Proposition 5.5.1 one has an isomorphism of tempered fundamental groups $\Pi_{Y/E'; K}^{temp} \simeq \Pi_{X/E; \mathbb{C}_p}^{temp}$ and also of geometric tempered fundamental subgroups $\Pi_{Y/K}^{temp} \simeq \Pi_{X/\mathbb{C}_p}^{temp}$ and an isomorphism $G_{E'} \simeq G_E$ (i.e. E' and E are anabelomorphic p -adic fields).

§ 8.3 Let $\mu(\overline{E}') \subset \overline{E}' \subset K$ the (multiplicative) group of all the roots of unity contained in $\overline{E}'$ and let $\mu_p(\overline{E}') \subset \overline{E}' \subset K$ the (multiplicative) group of all the p -power roots of unity contained in $\overline{E}'$. Write $\widehat{\mathbb{Z}}(1)_K \subset K$ (resp. $\mathbb{Z}_p(1)_K$) for the $G_{E'; K}$ -module of roots of unity (resp. the $G_{E'; K}$ -module of p -power roots of unity) contained in $\overline{E}'_K \subset K$. The notation $\widehat{\mathbb{Z}}(1)_K \subset K$ is to emphasize that this $G_{E'; K}$ -module is contained in the $G_{E'; K}$ -module $\overline{E}'_K$. Similarly one has the $G_{E'; K}$ -modules $\mathbb{Q}(1)_K$ and $\mathbb{Q}_p(1)_K$. One may consider these $G_{E'; K}$ -modules as $\Pi_{Y/E'; K}^{temp}$ -modules through the quotient homomorphism $\Pi_{Y/E'; K}^{temp} \twoheadrightarrow G_{E'; K}$. Write

$$G_{E'; K} \curvearrowright \mathbb{Z}(1)_K, G_{E'; K} \curvearrowright \mathbb{Z}_p(1)_K, \Pi_{X/E'; K}^{temp} \curvearrowright \mathbb{Z}(1)_K, \Pi_{X/E'; K}^{temp} \curvearrowright \mathbb{Z}_p(1)_K$$

etc. for these modules. These modules in turn provide continuous cohomology groups

$$H^i(G_{E'; K}, \mathbb{Z}(1)_K), H^i(G_{E'; K}, \mathbb{Z}_p(1)_K), H^i(\Pi_{X/E'; K}^{temp}, \mathbb{Z}(1)_K), \text{etc.}$$

I will refer to these modules as the *Kummer Theory and Galois cohomology for the p -adic field E' provided by the arithmetic holomorphic structure $(Y/E', (E' \hookrightarrow K, K^b \simeq F), *_{\mathcal{K}} : \mathcal{M}(K) \rightarrow Y_{E'}^{an}) \in \mathfrak{J}(X, E)$* . These considerations may also be applied to various subspaces of these cohomology groups $H_e^1(G_{E'}, -)$, $H_f^1(G_{E'}, -)$, $H_g^1(G_{E'}, -)$ considered in [Bloch and Kato, 1990].

Now let $(X/E, (\mathbb{C}_p \supset E, \mathbb{C}_p^\flat \xrightarrow{id} \mathbb{C}_p^\flat), *_{\mathbb{C}_p} : \mathcal{M}(\mathbb{C}_p) \rightarrow X_E^{an})$ be the standard arithmetic holomorphic structure chosen above. Then I will omit the reference to the algebraically closed, perfectoid field from the above notations and simply write $G_E = \text{Gal}(\overline{E}/E)$, and $\widehat{\mathbb{Z}}(1) \subset \mathbb{C}_p$ (resp. $\mathbb{Z}_p(1) \subset \mathbb{C}_p$) and $H^i(G_E, \mathbb{Z}(1))$, $H^i(G_E, \mathbb{Z}_p(1))$ etc. and refer to this as the *standard Kummer Theory and Galois cohomology* etc. Thus each arithmetic holomorphic structure in $\mathfrak{J}(X, E)$ provides us with a distinguished Kummer Theory. To summarize:

Proposition 8.3.1. *Given any arithmetic holomorphic structure $(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_{\mathbb{C}_p} : \mathcal{M}(K) \rightarrow Y_{E'}^{an}) \in \mathfrak{J}(X, E)$ one has, for each integer $i \geq 0$, a Kummer Theory and continuous Galois cohomology given by the assignment:*

$$(8.3.2) \quad G_{E';K} \curvearrowright \mathbb{Z}(1)_K, G_{E';K} \curvearrowright \mathbb{Z}_p(1)_K, \Pi_{X/E';K}^{temp} \curvearrowright \mathbb{Z}(1)_K, \Pi_{X/E';K}^{temp} \curvearrowright \mathbb{Z}_p(1)_K$$

$$(8.3.3) \quad (Y/E', Y^{an}/K) \longmapsto H^i(\Pi_{Y/E';K}^{temp}, \widehat{\mathbb{Z}}(1)_K)$$

and also

$$(8.3.4) \quad (Y/E', Y^{an}/K) \longmapsto H^i(G_{E';K}, \widehat{\mathbb{Z}}(1)_K),$$

and a similar construction for the subspaces $H_e^1(G_{E'}, -)$, $H_f^1(G_{E'}, -)$, $H_g^1(G_{E'}, -)$.

§ 8.4 One has the following:

Theorem 8.4.1. *Let X/E be a geometrically connected, smooth, quasi-projective variety over a p -adic field E . Let $(Y/E', (E' \hookrightarrow K, K^\flat \simeq F), *_{\mathbb{C}_p} : \mathcal{M}(K) \rightarrow Y_{E'}^{an}) \in \mathfrak{J}(X, E)$ and let $(X/E, (\mathbb{C}_p \supset E, \mathbb{C}_p^\flat \xrightarrow{id} \mathbb{C}_p^\flat), *_{\mathbb{C}_p} : \mathcal{M}(\mathbb{C}_p) \rightarrow X_E^{an})$ be the standard arithmetic holomorphic structure on X/E . Then*

- (1) Any anabelomorphism $\Pi_{X/E;K}^{temp} \simeq \Pi_{X/E;\mathbb{C}_p}^{temp}$ provides, for all integers $i \geq 0$, an isomorphism of continuous cohomology groups

$$H^i(\Pi_{Y/E';K}^{temp}, \widehat{\mathbb{Z}}(1)_K) \simeq H^i(\Pi_{X/E;\mathbb{C}_p}^{temp}, \widehat{\mathbb{Z}}(1))$$

and also an isomorphism

$$H^i(G_{E';K}, \widehat{\mathbb{Z}}(1)_K) \simeq H^i(G_E, \widehat{\mathbb{Z}}(1))$$

and also an isomorphism (of $\mathbb{Q}_p$ -vector spaces)

$$H^i(G_{E';K}, \mathbb{Q}_p(1)_K) \simeq H^i(G_E, \mathbb{Q}_p(1)),$$

- (2) and a similar isomorphism holds for the subgroups (resp. subspaces) H_e^1, H_f^1, H_g^1 of these groups (resp. subspaces), for example:

$$H_f^i(G_{E';K}, \mathbb{Q}_p(1)_K) \simeq H_f^i(G_E, \mathbb{Q}_p(1)).$$

Proof. It is enough to remark that by [Hoshi, 2021, Proposition 4.2(iv)] (or [Mochizuki, 2004, Proposition 1.2.1]) the anabelomorphism $G_{E';K} \simeq G_E$ induces an isomorphism of $\widehat{\mathbb{Z}}(1)_K \simeq \widehat{\mathbb{Z}}(1)$ which is compatible with $G_{E';K}$ (resp. G_E -action) on either side. This proves the first assertion. The second assertion is proved similarly noting that these subgroups (resp. subspaces) are amphoric. $\square$

§ 8.5 Collation of cohomology classes The following corollary of Theorem 8.4.1 will play a crucial role. It asserts that Galois cohomology classes provided by distinct arithmetic holomorphic structure may be collated in the standard Galois cohomology:

Corollary 8.5.1. *Let $i \geq 0$ be an integer. Let $\mathcal{S}$ be a collection of objects of $\mathfrak{J}(X, E)$. For each $s \in \mathcal{S}$, let*

$$\Psi_s^i \subset H^i(G_{E,s}, \widehat{\mathbb{Z}}(1)_s) \quad (\Psi_s^i \subset H_f^i(G_{E,s}, \widehat{\mathbb{Z}}(1)_s) \text{ resp.})$$

be a collection of Galois cohomology classes provided by each object $s \in \mathcal{S}$. Then these classes may be assembled into a subset

$$\Psi \subset H^i(G_E, \widehat{\mathbb{Z}}(1)) \quad (\Psi_f^i \subset H_f^i(G_E, \widehat{\mathbb{Z}}(1)) \text{ resp.})$$

of the relevant standard Galois cohomology.

Proof. Let $\Psi \subset H^i(G_E, \widehat{\mathbb{Z}}(1))$ ($\Psi_s^i \subset H_f^i(G_{E,s}, \widehat{\mathbb{Z}}(1)_s)$ resp.) be the union of the images of each element of Ψ_s (for each $s \in \mathcal{S}$) under all the isomorphisms provided by Theorem 8.4.1(2). $\square$

Remark 8.5.2. The adelic version of this corollary (established in [Joshi, 2023a], [Joshi, 2024c]) is the key principle which underlies Mochizuki's construction of his Θ -values set in [Mochizuki, 2021c]. For [Mochizuki, 2021c, Corollary 3.12], one uses a specific choice of $\mathcal{S}$ (for each prime of a fixed number field) whose construction requires the notions of Θ_{gau} -links and log-links, both of which I have established in [Joshi, 2023b].

9 Adelic Arithmetic Teichmüller Space

In this section, I want to indicate how the theory of arithmetic Teichmüller Spaces extends to an adelic theory in the presence of a number field. The constructions presented here are complemented and strengthened by the fact that a fixed number field itself has an Arithmetic Teichmüller Space. The adelic theory of a number field is glued together by means of the product formula and this formula comes to play here as well. The construction of an arithmetic Teichmüller Theory of a fixed number field is detailed in [Joshi, 2023a] and it clearly demonstrates the role the product formula of [Artin and Whaples, 1945] plays in the theory. This aspect also comes into play in [Joshi, 2024c] which applies the said global+adelic theory to demonstrate the assertions of [Mochizuki, 2021c,d] using the theory of Arithmetic Teichmüller Spaces developed here and [Joshi, 2023a].

§ 9.1 Let X/L be a geometrically connected, hyperbolic, smooth and quasi-projective variety over a number field L and let $\mathfrak{p}$ denote a non-archimedean prime of L and let $\infty_1, \dots, \infty_n$ be all the archimedean primes of L . Let $L_{\mathfrak{p}}$ (resp. L_{∞_i}) be the completions of L at $\mathfrak{p}$ (resp. ∞_i).

By an *local arithmetic-geometric anabelian variation of fundamental group of X/E at $\mathfrak{p}$* is the following collection:

(1) If $\mathfrak{p}$ is a non-archimedean place of L , then let

$$\mathfrak{J}(X, L, \mathfrak{p}) = \mathfrak{J}(X, L_{\mathfrak{p}}), \text{ and,} \\ \mathfrak{J}(X, L, \mathfrak{p})_F = \mathfrak{J}(X, L_{\mathfrak{p}})_F.$$

(2) If $\mathfrak{p} = \infty_i$ for some i (i.e. $\mathfrak{p} = \infty_i$ is an archimedean place of L) then let

$$\mathfrak{J}(X, L, \infty_i) = \mathfrak{J}(X, L_{\infty_i}).$$

Similar definition can be made of $\mathfrak{J}(X, L, \mathfrak{p})_F$. For $\mathfrak{p} = \infty_i$ one takes $\mathfrak{J}(X, L, \mathfrak{p})_F = \mathfrak{J}(X, L, \infty_i)$ purely for notational symmetry.

§ 9.2 Properties of $\mathfrak{J}(X, L, \mathfrak{p}) = \mathfrak{J}(X, L_{\mathfrak{p}})$ (resp. $\mathfrak{J}(X, L, \mathfrak{p})_F = \mathfrak{J}(X, L_{\mathfrak{p}})_F$) can be gleaned from the results established in Section 5, Section 6.

Definition 9.2.1. Let X/L be a geometrically connected, smooth, quasi-projective variety over a number field L with no real embeddings (I will make this restriction to avoid notational complexity). Then the *adelic tempered fundamental group of X/L* is the group (equipped with product topology):

$$\widetilde{\Pi}_{X/L}^{temp} = \prod_{0 \neq \mathfrak{p} \in \text{Spec}(\mathcal{O}_L)} \Pi_{X/L_{\mathfrak{p}}}^{temp} \times \prod_{i=1}^n \pi_1^{top}(X/L_{\infty_i}).$$

The adelic Arithmetic Teichmüller Space will be an anabelian variation providing the adelic tempered fundamental group $\widetilde{\Pi}_{X/L}^{temp}$.

§ 9.3 In the global i.e. number situation, the local p -adic Teichmüller Spaces can be assembled into a global category.

Theorem 9.3.1. Let X/L be a geometrically connected, smooth, quasi-projective, hyperbolic variety over a number field L . Assume L has no real embeddings. Then there exists a category $\widetilde{\mathfrak{J}(X/L)}$, called the Arithmetic Teichmüller Space associated to X/L which has the following properties:

(1) $\widetilde{\mathfrak{J}(X/L)}$ is given as a product category:

$$\widetilde{\mathfrak{J}(X/L)} = \prod_{\mathfrak{p}} \mathfrak{J}(X, L_{\mathfrak{p}})$$

where $\mathfrak{p}$ runs over all the inequivalent, non-trivial valuations of L and where $\mathfrak{J}(X, L_{\mathfrak{p}})$ is the p -adic Teichmüller Space associated to $X/L_{\mathfrak{p}}$ constructed in Theorem 5.25.1.

(2) $\widetilde{\mathfrak{J}(X/L)}$ is an anabelian variation providing the adelic tempered fundamental group

$$\widetilde{\Pi_{X/L}^{temp}} = \prod_{0 \neq \mathfrak{p} \in \text{Spec}(\mathcal{O}_L)} \Pi_{X/L_{\mathfrak{p}}}^{temp} \times \prod_{i=1}^n \pi_1^{top}(X/L_{\infty_i})$$

i.e. each object of $\widetilde{\mathfrak{J}(X/L)}$ provides an isomorph of $\widetilde{\Pi_{X/L}^{temp}}$ and two such isomorphs may be distinguished from each other by means of the arithmetic-geometric labels provided by the objects of $\widetilde{\mathfrak{J}(X/L)}$ which give rise to the isomorphs.

§ 9.4 Adelic Theory and Diamonds This is a continuation of Section 4.3 and should be read along with it. Let X/L be a geom. connected quasi-projective variety over a number field L and for a prime $v \in \mathbb{V}_L$, let L_v be the v -adic completion of L and let $X_{L_v}^{an}$ be the analytic space associated to X/L_v . Then the Arithmetic Teichmüller Space (Definition 5.1.1) contains sets of the form

$$\left\{ (X_{L_v}^{an}, (L_v \hookrightarrow K_v, \iota_v : K_v^b \simeq F_v), *_{(K_v, K_v^b \simeq F_v^b)} : \mathcal{M}(K_v) \rightarrow X_{L_v}^{an}) \right\} / \text{suitable equivalence},$$

where F_v is an algebraically closed, perfectoid field of characteristic $p_v > 0$ if v is non-archimedian otherwise $F_v = \mathbb{C}$. The precise relationship between this and $X_{L_v}^{\diamond}$ is given by Proposition 4.3.1. Thus working with the adelic Arithmetic Teichmüller Space $\mathfrak{J}(X, L)$ is approximately akin to considering

$$\prod_v X_{L_v}^{\diamond}.$$

But beware that Definition 5.1.1 is more general than $X_{L_v}^{\diamond}$.

10 Relationship to Mochizuki's Inter-Universal Teichmüller Theory

§ 10.1 A more detailed treatment of the relationship between the theory of the present paper and that of [Mochizuki, 2021a,b,c,d] is found in the 'Rosetta Stone' provided in [Joshi, 2024c]. In this brief section, I show that the Arithmetic Teichmüller Theory of the present series of papers ([Joshi, 2020a, 2021a, 2022, 2023b,a]) is fully compatible with [Mochizuki, 2021a,b,c,d]. Notably, the central deficiency [Mochizuki, 2021a,b,c] is that it provides no way of distinguishing between two distinct arithmetic holomorphic structures. As one knows from Theorem 7.4.3 this is not an issue in the present theory.

§ 10.2 Mochizuki's Key Principle of Inter-Universality Mochizuki's [Mochizuki, 2021a,b,c,d] asserts the existence of a Teichmüller Theory in the arithmetic context. The central principle which is key to [Mochizuki, 2021a,b,c,d] is Mochizuki's Key Principle of Inter-Universality [Mochizuki, 2021a, § I3, Pages 25–26]. This principle asserts that the theory of [Mochizuki, 2021a,b,c,d] is a theory which works with the tempered fundamental groupoid (rather than the tempered fundamental group).

More precisely, Mochizuki's Key Principle of Inter-Universality [Mochizuki, 2021a, § I3, Pages 25–26] requires one to work with tempered fundamental groups arising from arbitrary

geometric base-points and also asserts that the domains and the codomains of the key operations of [Mochizuki, 2021a,b,c,d] are required to have distinct geometric base-points (in general).

In the notation of this paper, Mochizuki's Key Principle of Inter-Universality requires one to work with a category whose objects are

$$(\Pi_{X/E;K}^{temp}, K, \text{"Frobenius Picture"}),$$

where the algebraically closed perfectoid field is provided by the K -geometric base-point used to calculate $\Pi_{X/E;K}^{temp}$. The 'Frobenius Picture' aspect is detailed in the 'Rosetta Stone' [Joshi, 2024c, § 8.3] and will not be recalled here (at this juncture it will suffice to say that this aspect is naturally incorporated in Definition 4.1.1). On the other hand, the perfectoid fields K are not amenable to the anabelian reconstruction methods of [Mochizuki, 2012, 2013, 2015], [Mochizuki, 2021a,b,c], and are completely ignored (while requiring them). This is the second important deficiency of [Mochizuki, 2021a,b,c]—namely [Mochizuki, 2021a,b,c] provides no clear way of disambiguating between distinct arithmetic holomorphic structures (while [Mochizuki, 2021c, Theorem 3.11, Corollary 3.12] clearly rest upon disambiguating such structures). [This central difficulty of [Mochizuki, 2021a,b,c] disappears with a cleaner formulation of this notion provided by Definition 4.1.1.] It should be clear to readers familiar with the theory of algebraically closed perfectoid fields that dealing with the missing information *necessarily* requires one to take the path considered in the present series of papers ([Joshi, 2020a, 2021a, 2022, 2023b,a]).

In the present paper, I work with the category $\mathfrak{J}(X, E)$ and more precise objects $(Y/E', (E' \hookrightarrow K, K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{an}) \in \mathfrak{J}(X, E)$ provided by Definition 5.1.1. By Corollary 7.4.6 one obtains the natural class of topological groups

$$\Pi_{(Y/E'; (K \supset E', K^b \simeq F), *_K : \mathcal{M}(K) \rightarrow Y_{E'}^{an})}^{temp},$$

all of which are isomorphic to $\Pi_{X/E}^{temp}$, and its relationship to Mochizuki's Theory and his Key Principle of Inter-Universality is established by Theorem 7.4.3.

The present paper works with more precise geometric object of $\mathfrak{J}(X, E)$ (Definition 5.1.1) and untills of fundamental groups provided by Definition 7.4.1 from which the functor to Mochizuki's category is obvious.

These results should make applications of the theory of the the present series of papers ([Joshi, 2020a, 2021a, 2022, 2023b,a]) to [Mochizuki, 2021a,b,c,d] unsurprising. *However there are some important differences between the two approaches—these are discussed in § 10.9.*

§ 10.3 Let me remark that in [Mochizuki, 2021a,b,c,d] Mochizuki works with multiplicative groups as the anabelian approach considered in loc. cit. is inherently multiplicative. On the other hand, the approach taken in Theorem 5.21.1 via [Fargues and Fontaine, 2018] is necessarily additive and not multiplicative. In the next few paragraphs I provide a translation between the two. This allows one to construct functors from $\mathfrak{J}(X, E)$ to various categories used in [Mochizuki, 2021a,b,c,d].

§ 10.4 For the multiplicative description let me fix some notations. Let $E = \mathbb{Q}_p$, let $\mathcal{G}/\mathbb{Z}_p$ be the Lubin-Tate formal group with formal logarithm given by $\sum_{n=0}^{\infty} \frac{T^n}{p^n}$, $F = \mathbb{C}_p^b$. Let $\overline{E} = \overline{\mathbb{Q}}_p$

be the algebraic closure of $E = \mathbb{Q}_p$ in $\mathbb{C}_p$.

§ 10.5 The Artin-Hasse Exponential provides the following:

Lemma 10.5.1. *Let $\mathcal{G}$ be the Lubin-Tate formal group over $\mathbb{Z}_p$ with logarithm $\sum_{n=0}^{\infty} \frac{T^{p^n}}{p^n}$. Let $\text{Exp}_{AH}(T)$ be the Artin-Hasse exponential function. Then the homomorphism $a \mapsto \text{Exp}_{AH}(a)$ provides a natural isomorphism of topological $\mathbb{Z}_p$ -modules*

$$\text{Exp}_{AH} : \mathcal{G}(\mathcal{O}_{\mathbb{C}_p^\flat}) \simeq \widehat{\mathbb{G}}_m(\mathcal{O}_{\mathbb{C}_p^\flat})$$

and hence also of

$$\widetilde{\mathcal{G}(\mathcal{O}_{\mathbb{C}_p})} \simeq \mathcal{G}(\mathcal{O}_{\mathbb{C}_p^\flat}) \simeq \widehat{\mathbb{G}}_m(\mathcal{O}_{\mathbb{C}_p^\flat}) \simeq \widetilde{\widehat{\mathbb{G}}_m(\mathcal{O}_{\mathbb{C}_p})}.$$

Proof. See [Fargues and Fontaine, 2018, Example 4.4.7]. $\square$

§ 10.6 Let $\widehat{\mathbb{G}}_m/\mathbb{Z}_p$ be the multiplicative formal group. Then one has for the multiplicative formal $\widehat{\mathbb{G}}_m$ one has

$$(10.6.1) \quad \widehat{\mathbb{G}}_m(\mathcal{O}_{\mathbb{C}_p}) = 1 + \mathfrak{m}_{\mathcal{O}_{\mathbb{C}_p}}$$

where $1 + \mathfrak{m}_{\mathcal{O}_{\mathbb{C}_p}} \subset \mathcal{O}_{\mathbb{C}_p}^*$ is the subgroup of units congruent to 1 modulo the maximal ideal $\mathfrak{m}_{\mathcal{O}_{\mathbb{C}_p}} \subset \mathcal{O}_{\mathbb{C}_p}$, and one also has from this that

$$(10.6.2) \quad \widetilde{\widehat{\mathbb{G}}_m(\mathcal{O}_{\mathbb{C}_p})} = \left\{ (x_n)_{n \in \mathbb{Z}} : x_n \in \widehat{\mathbb{G}}_m(\mathcal{O}_{\mathbb{C}_p}) = 1 + \mathfrak{m}_{\mathbb{C}_p}, x_{n+1}^p = x_n \forall n \in \mathbb{Z} \right\}$$

This fits into an exact sequence of $\mathbb{Q}_p$ -Banach spaces ([Fargues and Fontaine, 2018, Proposition 4.5.14])

$$0 \rightarrow T_p(\widehat{\mathbb{G}}_m) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \rightarrow \widetilde{\widehat{\mathbb{G}}_m(\mathcal{O}_{\mathbb{C}_p})} \xrightarrow{\log_{\widehat{\mathbb{G}}_m}} \mathbb{C}_p \rightarrow 0,$$

where $T_p(\widehat{\mathbb{G}}_m)$ is the p -adic Tate-module of $\widehat{\mathbb{G}}_m$ (note that $T_p(\widehat{\mathbb{G}}_m)$ is a rank one free $\mathbb{Z}_p$ -module computed using the p -power roots of unity contained in $\mathbb{C}_p$), and where $\log_{\widehat{\mathbb{G}}_m}$ is the logarithm of the formal group $\widehat{\mathbb{G}}_m$. Explicitly $\log_{\widehat{\mathbb{G}}_m}$ is given in terms of the p -adic logarithm as follows ([Fargues and Fontaine, 2018, Proof of Proposition 4.5.9]). Let $x = (x_n)_{n \in \mathbb{Z}} \in \widetilde{\widehat{\mathbb{G}}_m(\mathcal{O}_{\mathbb{C}_p})}$ then

$$\log_{\widehat{\mathbb{G}}_m}(x) = \log(x_0),$$

where

$$\log : \mathcal{O}_{\mathbb{C}_p}^* \rightarrow \mathbb{C}_p$$

is the p -adic logarithm.

§ 10.7 For a valued field $K \supset \mathbb{Q}_p$ let where

$$\mu(K) \subset \mathcal{O}_K^*$$

be the (topological) subgroup of roots of unity in K and write

$$(10.7.1) \quad \mathcal{O}_K^{\times \mu} = \mathcal{O}_K^* / \mu(K)$$

for the quotient of $\mathcal{O}_K^*$ by $\mu(K)$. Let

$$(10.7.2) \quad \mathcal{O}_K^\times = \mathcal{O}_K - \{0\}$$

be the multiplicative monoid of non-zero elements of $\mathcal{O}_K$. Both these notations are introduced and used extensively in [Mochizuki, 2021a,b,c,d].

§ 10.8 Let $\mu(\mathbb{C}_p) \subset \mathbb{C}_p^*$ be the subgroup of roots of unity contained in $\mathbb{C}_p$. Then one has the exact sequence of topological groups

$$0 \rightarrow \mu(\mathbb{C}_p) \rightarrow \mathcal{O}_{\mathbb{C}_p}^* \xrightarrow{\log} \mathbb{C}_p \rightarrow 0.$$

Remark 10.8.1. In [Mochizuki, 2021a,b,c,d] especially [Mochizuki, 2021c], Mochizuki works with

$$\mathcal{O}_{\mathbb{Q}_p}^{\times\mu} = \mathcal{O}_{\mathbb{Q}_p}^* / \mu(\bar{\mathbb{Q}}_p)$$

and the exact sequence

$$1 \rightarrow \mu(\bar{\mathbb{Q}}_p) \rightarrow \mathcal{O}_{\bar{\mathbb{Q}}_p}^* \xrightarrow{\log} \bar{\mathbb{Q}}_p \rightarrow 0.$$

Since one is dealing with p -adic logarithms, I prefer to work with the corresponding sequence obtained for the complete field $\mathbb{C}_p$ as opposed to $\bar{\mathbb{Q}}_p$. From the anabelian reconstruction point of view taken in [Mochizuki, 2021a,b,c,d], complete, algebraically closed valued fields are not determinable from the absolute Galois group. This is one key difference between the theory of the present paper and Mochizuki's work.

Lemma 10.8.2. *The inclusions $\mathcal{O}_{\mathbb{Q}_p}^* \subset \mathcal{O}_{\mathbb{C}_p}^*$ and $\mathcal{O}_{\mathbb{Q}_p}^{\times\mu} \subset \mathcal{O}_{\mathbb{C}_p}^{\times\mu}$ are dense inclusions.*

Proof. The first assertion is standard and it implies the second assertion as $\mu(\bar{\mathbb{Q}}_p) = \mu(\mathcal{O}_{\mathbb{C}_p})$. $\square$

Lemma 10.8.3. *One has an exact sequence of topological $G_{\mathbb{Q}_p}$ -modules*

$$1 \rightarrow \mu_p(\bar{\mathbb{Q}}_p) \rightarrow (1 + \mathfrak{m}_{\bar{\mathbb{Q}}_p}) \rightarrow \mathcal{O}_{\bar{\mathbb{Q}}_p}^{\times\mu} \rightarrow 1.$$

Proof. Let $\mu'(\bar{\mathbb{Q}}_p) \subset \mu(\bar{\mathbb{Q}}_p)$ (resp. $\mu_p(\bar{\mathbb{Q}}_p) \subset \mu(\bar{\mathbb{Q}}_p)$) be the subgroup of roots of unity with orders coprime to p (resp. the subgroup of roots of unity of order a power of p). Then one has

$$\mu'(\bar{\mathbb{Q}}_p) \times \mu_p(\bar{\mathbb{Q}}_p) \simeq \mu(\bar{\mathbb{Q}}_p).$$

Note that for every $n \geq 1$,

$$1 - \zeta_{p^n} \equiv 0 \pmod{\mathfrak{m}_{\mathbb{C}_p}}.$$

So any p -power root of unity is contained in the group of 1-units $1 + \mathfrak{m}_{\bar{\mathbb{Q}}_p}$ and hence

$$\mu_p(\bar{\mathbb{Q}}_p) \subset 1 + \mathfrak{m}_{\mathbb{C}_p}.$$

Hence

$$\mathcal{O}_{\bar{\mathbb{Q}}_p}^* = \mu'(\bar{\mathbb{Q}}_p) \times (1 + \mathfrak{m}_{\bar{\mathbb{Q}}_p}),$$

and hence by definition

$$\mathcal{O}_{\mathbb{Q}_p}^*/\mu(\bar{\mathbb{Q}}_p) \simeq (1 + \mathfrak{m}_{\bar{\mathbb{Q}}_p})/\mu_p(\bar{\mathbb{Q}}_p) \simeq \mathcal{O}_{\mathbb{Q}_p}^{\times\mu},$$

and this provides the asserted exact sequence. $\square$

§ 10.9 Let me explain the key difference between the theory described here and that of [Mochizuki, 2021a,b,c,d]. In [Mochizuki, 2021a,b,c,d] Mochizuki works with the pair G_E and its action on $\mathcal{O}_{\mathbb{Q}_p}^{\times\mu}$, that is with

$$G_E \curvearrowright \mathcal{O}_{\mathbb{Q}_p}^{\times\mu}.$$

In [Mochizuki, 2021a,b,c,d], roughly speaking, algorithms of Anabelian Reconstruction Theory, automorphisms of $G_E \curvearrowright \mathcal{O}_{\mathbb{Q}_p}^{\times\mu}$, the theory of log-link and theta-link are used to produced variation in the data of arithmetic line bundles.

The present paper can also be read in the multiplicative context using the isomorphism (§ 10.5)

$$G_E \curvearrowright \widetilde{\widehat{\mathbb{G}}_m(\mathcal{O}_{\mathbb{C}_p})} \simeq G_E \curvearrowright \widehat{\mathbb{G}}_m(\mathcal{O}_{\mathbb{C}_p^\flat}),$$

and variation of the data of arithmetic line bundles arises from existence Arithmetic Teichmüller Spaces (Theorem 5.25.1) which arise from existence of deformations of analytic structure of $X_{\mathbb{C}_p}^{an}$ via deformations of $\mathbb{C}_p$.

§ 10.10 The Virasoro type symmetry action is detailed in Theorem 5.21.1 and Theorem 5.24.1; its relationship to Mochizuki's Indeterminacy 'Ind2' is discussed in the 'Rosetta Stone' [Joshi, 2024c].

§ 10.11 Let me now show that Theorem 3.10.1, Theorem 3.16.1, Theorem 5.21.1 provide functors to various anabelian categories considered in [Mochizuki, 2021a,b,c,d]. Notably these theorems show that there are geometrically distinguishable isomorphisms of the tempered fundamental groups. The Arithmetic Teichmüller Space $\mathfrak{J}(X, E)$ constructed in Theorem 5.25.1 includes all hyperbolic curves with tempered fundamental group topologically isomorphic to that of a given hyperbolic curve X/E .

§ 10.12 Let Y/E' be a smooth, quasi-projective variety over a p -adic field E' and let $\overline{E'}$ be an algebraic closure of E' .

§ 10.13 This section is taken from 'Rosetta Stone' [Joshi, 2024c] where readers will find a detailed discussion of [Mochizuki, 2021a,b,c,d] and its relationship to the the present series of papers ([Joshi, 2020a, 2021a, 2022, 2023b,a]). Consider an arbitrary triple $(Y/E', E' \hookrightarrow K) \in \mathfrak{J}(X, E)$. Then as K is an algebraically closed, perfectoid field of characteristic zero, consider the algebraic closure $\overline{E'} \subset K$ (as a valued fields) of $E' \hookrightarrow K$. Let $G_{E';K}$ be the absolute Galois group of $\overline{E'}/E$. The one has the tautological action of $G_{E';K}$ on the various multiplicative submonoids of $\overline{E'}$. One has the following functors from $\mathfrak{J}(X, E)$ to various categories (called *multiradial environments*) given by [Mochizuki, 2021b, Example 1.8, Page 247] and used throughout [Mochizuki, 2021a,b,c,d]:

(1)

$$(Y/E', E' \hookrightarrow K) \longmapsto \Pi_{X/E;K}^{temp} \twoheadrightarrow G_{E';K}$$

(2) and

$$(Y/E', E' \hookrightarrow K) \longmapsto \Pi_{Y/E'; K}^{temp} \overset{K}{\curvearrowright} \mathcal{O}_{\overline{E}}^{\times \mu},$$

(3) and

$$(Y/E', E' \hookrightarrow K) \longmapsto \Pi_{Y/E'; K}^{temp} \overset{K}{\curvearrowright} \mathcal{O}_{\overline{E}}^{\triangleright},$$

(4) and

$$(Y/E', E' \hookrightarrow K) \longmapsto \Pi_{Y/E'; K}^{temp} \overset{K}{\curvearrowright} \overline{E}'^*.$$

[Here $\Pi_{Y/E'; K}^{temp} \overset{K}{\curvearrowright} \mathcal{O}_{\overline{E}}^*$ means that the field K is used to compute the algebraic closure of E' , and the action of $\Pi_{Y/E'; K}^{temp}$ on $\mathcal{O}_{\overline{E}}^*$ through the quotient $\Pi_{Y/E'}^{temp} \rightarrow G_{E'}$ is computed using the algebraic closure $\overline{E}' \subset K$.]

Note that one has an isomorphism of topological groups $\Pi_{Y/E'; K}^{temp} \simeq \Pi_{X/E}^{temp}$.

Remark 10.13.1. More details can be found in [Joshi, 2023b], [Joshi, 2024c].

11 Appendix: Existence of algebraically closed perfectoid fields

In the early days of my investigations of arithmetic Teichmüller Theory ([Joshi, October 2020], [Joshi, 2021a]), my ideas were shaped by my reading of [Schmidt, 1933] and [Kaplansky, 1942, Theorem 8]. Eventually I replaced [Schmidt, 1933] by [Kedlaya and Temkin, 2018] which is rather fundamental to my theory as it asserts the existence of topologically non-isomorphic perfectoid fields with tilts isometric with $\mathbb{C}_p^b$. Nevertheless, an understanding of [Schmidt, 1933] will still be useful to many readers of my theory. A more natural way of thinking about the theorem below is to think in terms of the absolute Riemann-Zariski Space $RZ(K, \mathbb{Z})$ of an algebraically closed, complete valued field K , but I will take the more prosaic approach of [Schmidt, 1933].

§ 11.1 The following is an essentially refined (but less general) version of [Schmidt, 1933, Satz 2]. The proof given here is similar to the one given there:

Theorem 11.1.1. *Fix a prime number p . Let $(K, |-\|_K)$ be an algebraically closed, complete, rank-one valued field equipped and suppose that $(\mathbb{Q}, |-\|_K)$ is equivalent to the standard p -adic valuation on $\mathbb{Q}$ (for example $K = \mathbb{C}_p$). Then there exist uncountably many rank-one valuations, $(K, |-\|_v)$ on K , with respect to each of which K is complete, algebraically closed but each of these valuations $|-\|_v$ is not equivalent to the given valuation $(K, |-\|_K)$, while its restriction $(\mathbb{Q}, |-\|_v)$ is equivalent to the standard p -adic valuation on $\mathbb{Q}$. [In particular each $(K, |-\|_v)$ is an algebraically closed, perfectoid field with an isometric embedding of $\mathbb{Q}_p$.]*

Proof. Since K is complete with respect to a non-trivial valuation, so K is an uncountable field. The hypothesis implies that $K \supset \mathbb{Q}$. Choose a transcendence basis, $T = \{t_\alpha\}_\alpha$, for the extension $K/\mathbb{Q}$. By replacing each $t_\alpha \in T$, by a multiple of the form $p^{c_\alpha} \cdot t_\alpha$ for a suitable $c_\alpha \in \mathbb{Z}$, one can assume additionally that $|t_\alpha|_K \neq 1$ (such a scaling evidently still provides a

transcendence basis). Then $K \supset \mathbb{Q}(T) \supset \mathbb{Q}$ with $K \supset \mathbb{Q}(T)$ an algebraic extension while $\mathbb{Q}(T) \supset \mathbb{Q}$ is a purely transcendental extension.

Since any valuation on a field extends to any of its algebraic extensions, it suffices to construct valuation $|\cdot|_v$ of $\mathbb{Q}(T) \supset \mathbb{Q}$ and extend it to a valuation $|\cdot|_v$ of K .

Let $S \subset T$ be any non-empty subset. Clearly, for any choice of $\emptyset \neq S \subset T$, there exist a choice of integers $\{n_s \in \mathbb{Z} : s \in S\}$ such that for at least one $s \in S$, one has $n_s < 0$.

Define an automorphism $\sigma : \mathbb{Q}(T) \rightarrow \mathbb{Q}(T)$ by

$$\sigma(s) = s^{n_s} \quad \forall s \in S,$$

and

$$\sigma(t) = t \quad \text{whenever } t \in T - S.$$

Let $|\cdot|_v = |\cdot|_{v_{\sigma, S}} : \mathbb{Q}(T) \rightarrow \mathbb{R}$ be the pull-back, via σ , of the valuation $|\cdot|_K$ of $\mathbb{Q}(T)$. By construction, for any $x \in \mathbb{Q}(T)$:

$$|x|_v = |\sigma(x)|_K.$$

From this one has

$$|x \cdot y|_v = |\sigma(x \cdot y)|_K = |\sigma(x)|_K \cdot |\sigma(y)|_K = |x|_v \cdot |y|_v,$$

and

$$|x + y|_v = |\sigma(x + y)|_K = |\sigma(x) + \sigma(y)|_K \leq \max(|\sigma(x)|_K, |\sigma(y)|_K) = \max(|x|_v, |y|_v).$$

So $|\cdot|_v$ is indeed a valuation on K . Moreover, since σ is a $\mathbb{Q}$ -automorphism of $\mathbb{Q}(T)$, σ is identity on $\mathbb{Q}$, so for every $x \in \mathbb{Q}$, one has

$$|x|_v = |\sigma(x)|_K = |x|_K.$$

So the restriction of $|\cdot|_v$ to $\mathbb{Q}$ is equivalent to the standard p -adic valuation. Also one has for each $s \in S$:

$$|s|_v = |\sigma(s^{n_s})|_{\mathbb{Q}(T)} = |s|_{\mathbb{Q}(T)}^{n_s}.$$

Now $|s|_K \neq 1$ for all $s \in S$ and $n_s < 0$ for at least one $s \in S$. Thus, for some $s \in S$, one has either $|s|_v < 1$ and $|s|_K > 1$ or vice versa. Hence by [Lang, 2002, Chap XII, Proposition 1.1], the two valuations $|\cdot|_v$ and $|\cdot|_K$ are not equivalent on $\mathbb{Q}(T)$ and hence on K . The completion of $(K, |\cdot|_v)$ has the same cardinality and the same characteristic as K and both are algebraically closed, and hence the completion can be identified with the field K and with this identification K is algebraically closed, complete with respect to both $|\cdot|_v$ and $|\cdot|_K$ and these two valuations on K are inequivalent. At any rate, each v provides a complete, algebraically closed field equipped with an isometric embedding of $\mathbb{Q}_p$. That such algebraically closed, fields are perfectoid is immediate from Lemma 3.2.1. This proves the assertion. $\square$

12 Appendix: Anabelian Hyperbolic Varieties

This section may be skipped on the initial reading. Reader may simply work with hyperbolic curves instead. The discussion of hyperbolic varieties of dimension bigger than one is included only to illustrate my point that the construction provided in this paper works in higher dimensions.

§ 12.1 Let $E \supset \mathbb{Q}$ be any complete valued field, which is either an archimedean or a non-archimedean with a rank one valuation inducing a p -adic valuation on $\mathbb{Q}$, and $\widehat{E}$ be its completed algebraic closure. Let X/E be a geometrically connected, smooth projective variety over E . I will say that X/E is a *hyperbolic variety* if the analytic space $X^{an}/\widehat{E}$, is a (Brody) hyperbolic variety (see [Lang, 1986] for the archimedean case, [Javanpeykar and Vezzani, 2018] for the non-archimedean case).

§ 12.2 If $\dim(X) = 1$ then X/E is hyperbolic (in the above sense) if and only if $X \times_E \mathbb{C}$ is a hyperbolic Riemann surface.

§ 12.3 In [Lang, 1986], Serge Lang has conjectured that any hyperbolic variety X defined over a number field E has a finite number of E -rational points.

Remark 12.3.1.

- (1) For concrete applications, beyond $\dim(X) = 1$, one may also need to assume that X/E is a $K(\pi, 1)$ -space in some suitable sense.
- (2) As an aside let me say that I have used the case of *hyperbolic varieties* here because of their relevance in Diophantine Geometry via Lang's Conjecture [Lang, 1986] and [Faltings, 1991]. One can also use other related hypothesis as a substitute for the hyperbolic hypothesis, for example, one can work with *groupless varieties* instead of hyperbolic varieties (see [Javanpeykar, 2020] for other related hypothesis which may be used here instead).

§ 12.4 An important property in many anabelian considerations is the following: a profinite group Π is said to be a *slim profinite group* (or simply Π is *slim*) if every open subgroup of Π has trivial center. By [Mochizuki, 2004, Def. 0.1 and Remark 0.1.3], Π is slim if and only if the centralizer of any open subgroup of Π is trivial.

§ 12.5 A hyperbolic variety over a finitely generated field has the following anabelian property.

Proposition 12.5.1. *Let X/E be a geometrically connected, smooth, hyperbolic variety over a finitely generated field E of characteristic zero or a p -adic field. Then there exists a basis of Zariski open subsets $\{U\}$ of X such that*

- (1) every $U \neq \emptyset$ in this basis is hyperbolic, and
- (2) for every pair U, V nonempty opens in this basis one has

$$\mathrm{Isom}_E(U, V) \simeq \mathrm{Isom}_{G_E}^{\mathrm{out}}(\pi_1(U/E), \pi_1(V/E)),$$

(3) and for every $U \neq \emptyset$, $\pi_1(U/E)$ is slim.

Proof. From [Schmidt and Stix, 2016, Corollary 1.7], if E is finitely generated field and by [Hoshi, 2014, Theorem C], if E is a p -adic field, it follows that a basis of Zariski open sets satisfying property (2) exist. From [Lang, 1986] it is immediate that every non-empty open subvariety of X is hyperbolic. The last assertion is proved by induction: for $\dim(U) = 1$, U is a hyperbolic curve and so the slim-ness property is immediate from the aforementioned results. For $\dim(U) > 1$ one uses induction on dimension and the proofs of [Schmidt and Stix, 2016] or [Hoshi, 2014] from the fibration structure U is equipped with by the construction of this basis. $\square$

Definition 12.5.2. Let X/E be an hyperbolic variety over a p -adic field or a finitely generated field E . Let $\overline{E}$ be an algebraic closure of E . Then X is said to be an *anabelian hyperbolic variety* if the étale fundamental group $\pi_1(X/E)$ and the geometric tempered étale fundamental group $\pi_1(X/\overline{E})$ are both slim.

Remark 12.5.3. *There exist hyperbolic varieties which are not anabelian hyperbolic non-slim (see [Ihara and Nakamura, 1997]).* As an addendum to [Ihara and Nakamura, 1997], which will not be used in the rest of this paper, let me remark that if A/E is a simple abelian variety over a number field E , and suppose $D \subset A$ over E is an irreducible, smooth, ample divisor, then by [Debarre, 1995, Theorem 4.1], one has an isomorphism $\pi_1(D \times \overline{E}) \simeq \pi_1(X \times \overline{E})$ of the geometric fundamental groups. In particular $\pi_1(D \times \overline{E})$ is not slim. But as A is simple, D is not a translate of an a sub-abelian variety of A , and hence by [Lang, 1986], D is hyperbolic. *Hence D is not an anabelian hyperbolic variety.* So this provides another class of examples of non-slim hyperbolic varieties complementing the non-slim examples of [Ihara and Nakamura, 1997]. Note that by [Faltings, 1991, Theorem 1], D has finitely many rational points.

Proposition 12.5.4. *If X/E is a hyperbolic curve over a p -adic field or a finitely generated field E then X/E is an anabelian hyperbolic variety.*

Proof. This is immediate from Definition 12.5.2 and [Mochizuki, 2004, Corollary 1.3.3 and Lemma 1.3.1]. $\square$

§ 12.6 The absolute Grothendieck Conjecture for anabelian hyperbolic varieties is the following:

Conjecture 12.6.1. Let X/L be any smooth, quasi-projective and anabelian hyperbolic variety over a finitely generated field or a p -adic field L . If Y/L' is a smooth, quasi-projective anabelian hyperbolic variety which is anabelomorphic to X/L the one has an isomorphism $Y \simeq X$ of $\mathbb{Z}$ -schemes.

Remark 12.6.2.

- (1) This is a formulation, for anabelian hyperbolic varieties, of the celebrated anabelian conjecture of Grothendieck.
- (2) For $\dim(X) = 1$, Conjecture 12.6.1 is a well-known theorem of [Mochizuki, 1996], [Tamagawa, 1997].

13 Appendix: Anabelian variations providing Π

§ 13.1 I want to present some elementary considerations which will prove useful in understanding the problem of constructing Arithmetic Teichmüller Spaces using the tempered fundamental group.

§ 13.2 Fix a pro-discrete group Π (for example $\Pi = \Pi_{X/E}^{temp}$). Let $\mathcal{P}ro\mathcal{D}_\Pi$ be the isomorphism class of the pro-discrete group Π , and for each pro-discrete group $H \in \mathcal{P}ro\mathcal{D}_\Pi$, let $\hat{H}$ be its profinite completion. One can make $\mathcal{P}ro\mathcal{D}_\Pi$ into a category with the class $\mathcal{P}ro\mathcal{D}_\Pi$ as objects and morphisms between any pair of objects are isomorphisms. This makes the category $\mathcal{P}ro\mathcal{D}_\Pi$ into a groupoid whose every object is isomorphic to Π . Let $\mathcal{Sch}_\mathbb{Z}$ be the category of schemes. Let $\mathcal{An}_{\mathbb{Q}_p}$ (resp. $\mathcal{An}_\mathbb{C}$) be the category of $\mathbb{Q}_p$ -analytic spaces in the sense of Theorem 3.10.1 (resp. $\mathbb{C}$ -analytic spaces).

§ 13.3 Let S be one of categories $\mathcal{Sch}_\mathbb{Z}, \mathcal{An}_\mathbb{C}, \mathcal{An}_{\mathbb{Q}_p}$. Suppose $\mathcal{C}$ is a category and Π is a fixed pro-discrete group. I will say that $\mathcal{C}$ is an *anabelian variation providing Π with base S* if the following conditions are satisfied:

(1) For every V in $\mathcal{C}$ there exists an isomorphism of pro-discrete groups $\alpha_V : \Pi_V \xrightarrow{\simeq} \Pi$ i.e. one is given a functor

$$\text{ob}(\mathcal{C}) \rightarrow \mathcal{P}ro\mathcal{D}_\Pi$$

written $V \mapsto \Pi_V$ from the class of objects of $\mathcal{C}$ to the category $\mathcal{P}ro\mathcal{D}_\Pi$ of Π .

(2) There is a functor $\mathcal{C} \rightarrow S$ denoted $V \mapsto [V] \in S$. In this case $[V]$ called the scheme (resp. $\mathbb{C}$ -analytic space, $\mathbb{Q}_p$ -analytic space) underlying $V \in \mathcal{C}$.

§ 13.4 I will often simply say “anabelian variation providing Π ” instead of “anabelian variation providing Π with base S .” Hopefully there will be no confusion. The categories constructed here will come with functors to all the three values of S .

§ 13.5 I will say that an anabelian variation providing Π is a *trivial anabelian variation providing Π* if any pair of objects $V, V' \in \mathcal{C}$ are isomorphic. Obviously one is interested in constructing non-trivial anabelian variations providing Π .

§ 13.6 Suppose $\mathcal{C}$ is a non-trivial anabelian variation providing Π . Then the function $V \mapsto \Pi_V$ can be thought of as providing labeled isomorphisms of Π and the function $\text{ob}(\mathcal{C}) \rightarrow \mathcal{P}ro\mathcal{D}_\Pi$ ($V \mapsto \Pi_V$) will be called the *labeling function of $\mathcal{C}$* .

§ 13.7 The classical Teichmüller Space of any connected, hyperbolic Riemann surface Σ is a non-trivial anabelian variation providing Π , the topological fundamental group of X . Indeed suppose Σ is a connected, hyperbolic Riemann surface with $\Pi = \pi_1(\Sigma)$ be its étale fundamental group and consider the Teichmüller space T_Σ of Σ [Lehto, 1987, Chapter V]. Indeed one may think of T_Σ as a category consisting of pairs $(\Sigma, f : \Sigma \rightarrow \Sigma')$ where f is a quasi-conformal mapping of Σ onto a Riemann surface Σ' . Then one has $(\Sigma, f) \mapsto \pi_1(\Sigma') \simeq \pi_1(\Sigma) = \Pi$; and as any connected hyperbolic Riemann surface is obviously a $\mathbb{C}$ -analytic space so one has an obvious functor to $T_\Sigma \rightarrow \mathcal{An}_\mathbb{C}$ given by $(\Sigma, f : \Sigma \rightarrow \Sigma') \mapsto \Sigma'$. Hence T_Σ is an anabelian variation providing Π with base $\mathcal{An}_\mathbb{C}$.

§ 13.8 I will say that an anabelian variation providing Π with base S is a *geometric anabelian variation providing Π with base S* if there exists $V, V' \in \mathcal{C}$ such that $[V] \neq [V']$ (in S).

§ 13.9 Obviously any geometric anabelian variation providing Π is non-trivial. Hence if one constructs geometric anabelian variations providing Π then one automatically gets a non-trivial anabelian variation providing Π .

§ 13.10 The moduli $\mathcal{M}_g/\mathbb{C}$ stack of smooth, proper curves of genus $g \geq 2$ over $\mathbb{C}$ is an example of a geometric anabelian variation providing Π . Indeed it is clear that this is an anabelian variation providing $\Pi = \pi_1^{\text{top}}(X(\mathbb{C}))$ is the topological fundamental group of the Riemann surface $X(\mathbb{C})$ where $X/\mathbb{C}$ is any smooth curve, proper curve of genus g . This is also a geometric anabelian variation providing Π because one can obviously find two non-isomorphic smooth, proper curves of genus g over $\mathbb{C}$. Such a pair of curves cannot be isomorphic as $\mathbb{Z}$ -schemes as well. In particular $\mathcal{M}_g$ is a non-trivial anabelian variation providing Π .

§ 13.11 These examples should convince the reader that, when a non-trivial (or even a geometric) anabelian variation providing Π exists, then it can serve as an *anabelian stand-in for a variation of (mixed) Hodge structures*.

§ 13.12 In some sense the presence of this sort of a structure (i.e. an anabelian variation providing Π with base S) should be understood as a manifestation of Kodaira-Spencer classes! Constructing similar structures in the p -adic setting leads to an *p -adic Teichmüller Landscape* or an *p -adic Teichmüller Theory* presented here. Assembling such data for each valuation of a number field, leads to a global *Arithmetic Teichmüller Landscape* or an *Arithmetic Teichmüller Theory* (also presented here) in which one can hope to contemplate applications to global Diophantine problems as is done in [Mochizuki, 2021a,b,c,d].

§ 13.13 Now the results of Section 5 and the preceding lead to the following:

Theorem 13.13.1. *Let X/E be a geometrically connected, smooth, quasi-projective variety over a p -adic field E or let $E = \mathbb{C}$. If E is a p -adic field, let $S = \underline{\mathfrak{J}}(X, E)$ and if $E = \mathbb{C}$, let S be the category of complex Banach spaces with isometries as morphisms. If E is a p -adic field, let $\Pi = \Pi_{X/E}^{\text{temp}}$ be the tempered fundamental group and if $E = \mathbb{C}$ let Π be the topological fundamental group.*

(1) *If E is a p -adic field, then one has a natural functor (given by forgetting the geometric base-point)*

$$\mathfrak{J}(X, E) \rightarrow S = \underline{\mathfrak{J}}(X, E),$$

and the tempered fundamental group functor:

$$\mathfrak{J}(X, E) \rightarrow \mathcal{P}r\mathcal{O}\mathcal{D}_{\Pi}.$$

(2) *If $E = \mathbb{C}$ one has the Royden algebra functor $\mathfrak{J}(X, E) \rightarrow S$ and the topological fundamental group functor*

$$\mathfrak{J}(X, E) \rightarrow \mathcal{P}r\mathcal{O}\mathcal{D}_{\Pi}.$$

In both the cases, these two functors make $\mathfrak{J}(X, E)$ into a non-trivial anabelian variation providing Π .

Proof. The proof is clear from the results of Section 5. $\square$

14 Appendix: Self similarity of $\mathbb{C}_p^b$ and its consequences

§ 14.1 Let me begin with the following reformulation of an important result of [Matignon and Reversat, 1984, Théorème 2 and §3 Remarque 2] (and also [Kedlaya and Temkin, 2018]).

Theorem 14.1.1. *Let p be any prime number, $\mathbb{C}_p = \widehat{\overline{\mathbb{Q}_p}}$ be the completion of an algebraic closure of $\mathbb{Q}_p$. Let $\mathbb{C}_p^b$ be the tilt of $\mathbb{C}_p$. Then*

- (1) *There exists an isomorphism $\mathbb{C}_p^b \simeq \widehat{\overline{\mathbb{F}_p((x))}}$.*
- (2) *There exists $y \in \widehat{\overline{\mathbb{F}_p((x))}}$ such that $\widehat{\overline{\mathbb{F}_p((y))}} \subsetneq \widehat{\overline{\mathbb{F}_p((x))}}$ (more precisely $x \notin \widehat{\overline{\mathbb{F}_p((y))}}$).*
- (3) *Hence $\widehat{\overline{\mathbb{F}_p((x))}}$ is a self-similar valued field.*
- (4) *Hence $\mathbb{C}_p^b$ is a self-similar valued field i.e. it contains infinitely many proper subfields which are topologically isomorphic to $\widehat{\overline{\mathbb{F}_p((t))}} \simeq \mathbb{C}_p^b$ for some variable t .*

Proof. The first assertion is standard (see [Scholze, 2012]). The second assertion is a consequence of the main theorem of [Matignon and Reversat, 1984, Théorème 2 and §3 Remarque 2] or the proof of [Kedlaya and Temkin, 2018, Theorem 1.2]. The remaining assertions are immediate from the first two. $\square$

§ 14.2 By a labeled copy of $\mathbb{C}_p^b$ will mean an identification $\mathbb{C}_p^b \simeq \widehat{\overline{\mathbb{F}_p((t))}}$ for some variable t . I will write $\mathbb{C}_{p,t}^b$ for a copy of $\mathbb{C}_p^b$ labeled by the variable t .

§ 14.3 A fundamental consequence of Theorem 14.1.1 is the following:

Theorem 14.3.1. *The Fargues-Fontaine curve $\mathcal{Y}_{\mathbb{C}_{p,y}^b, \mathbb{Q}_p}$ is a self-similar curve. More precisely, for every pair of elements $x, y \in \mathbb{C}_p^b$ as in Theorem 14.1.1, there exists infinitely many strict inclusions*

$$\left| \mathcal{Y}_{\mathbb{C}_{p,y}^b, \mathbb{Q}_p} \right| \hookrightarrow \left| \mathcal{Y}_{\mathbb{C}_{p,x}^b, \mathbb{Q}_p} \right|$$

arising from the strict isometric inclusions $\mathbb{C}_{p,y}^b \hookrightarrow \mathbb{C}_{p,x}^b$.

Proof. Since $\left| \mathcal{Y}_{\mathbb{C}_{p,y}^b, \mathbb{Q}_p} \right|$ is identified by [Fargues and Fontaine, 2018, Théorème 2.4.1 and Corollaire 2.4.2] with the set of primitive degree one elements of $W(\mathcal{O}_{\mathbb{C}_p^b})$. By [Fargues and Fontaine, 2018, Corollaire 2.2.9] any primitive element of degree one in $W(\mathcal{O}_{\mathbb{C}_{p,y}^b})$ can be written, up to multiplication by a unit in $W(\mathcal{O}_{\mathbb{C}_{p,y}^b})$, as

$$[\alpha] - p,$$

for some element $\alpha \in \mathcal{O}_{\mathbb{C}_{p,y}^b}$ with $v(\alpha) > 0$ and $\mathcal{Y}_{\mathbb{C}_{p,y}^b, \mathbb{Q}_p} \hookrightarrow \mathcal{Y}_{\mathbb{C}_{p,x}^b, \mathbb{Q}_p}$ is given by sending the primitive element $[\alpha] - p \in W(\mathcal{O}_{\mathbb{C}_{p,y}^b})$ to the primitive degree one element $[\alpha] - p \in W(\mathcal{O}_{\mathbb{C}_{p,x}^b})$ and at the level of ideals

$$([\alpha] - p)W(\mathcal{O}_{\mathbb{C}_{p,y}^b}) \mapsto ([\alpha] - p)W(\mathcal{O}_{\mathbb{C}_{p,x}^b}).$$

One has

$$W(\mathcal{O}_{\mathbb{C}_{p,y}^b}) \subsetneq W(\mathcal{O}_{\mathbb{C}_{p,x}^b}),$$

as $x \notin \mathbb{C}_{p,y}^b$ so $[x] - p \notin W(\mathcal{O}_{\mathbb{C}_{p,y}^b})$ and hence there is a primitive element of degree one of $W(\mathcal{O}_{\mathbb{C}_{p,x}^b})$ which is not contained in the set of primitive elements of degree one of $W(\mathcal{O}_{\mathbb{C}_{p,y}^b})$.

So the inclusion of $\left| \mathcal{Y}_{\mathbb{C}_{p,y}^b, \mathbb{Q}_p} \right| \hookrightarrow \left| \mathcal{Y}_{\mathbb{C}_{p,x}^b, \mathbb{Q}_p} \right|$ is strict. This proves the assertion. $\square$

§ 14.4 Before proceeding it will be useful to understand this self-similarity in terms of Classical Teichmüller Theory. In the classical Teichmüller Theory (i.e Teichmüller Theory for Riemann Surfaces), the Teichmüller space is tiled by isomorphs of a fundamental domain for the mapping class group or modular group actions. To put it differently the Teichmüller space is equipped with a self-similar tiling (not unique in general).

§ 14.5 So the question arises if there is a theory of fundamental domains in the Arithmetic Teichmüller Theory constructed here. The answer to this question is yes and is given by Theorem 14.7.1. The notion of fundamental domains in Arithmetic Teichmüller Theory and this arises precisely from the fact that $\mathbb{C}_p^b$ is a self-similar valued field i.e. $\mathbb{C}_p^b$ (see Theorem 14.1.1 and Theorem 14.3.1).

§ 14.6 Let E be a p -adic field and let X/E be a smooth, geometrically connected, smooth, quasi-projective variety. Let $\mathfrak{J}^{\text{arith}}(X, E)$ be the category of $\mathfrak{J}(X, E)$ defined by Definition 5.6.2 for $F = \mathbb{C}_p^b$ i.e.

$$(14.6.1) \quad \mathfrak{J}^{\text{arith}}(X, E) = \mathfrak{J}(X, E)_{\mathbb{C}_p^b}$$

§ 14.7 A fundamental domain for $\mathfrak{J}^{\text{arith}}(X, E)$ is the full subcategory $\mathfrak{J}^{\text{arith}}(X, E)_{\mathbb{C}_{p,x}^b}$ for some variable x . The following is now a tautology:

Theorem 14.7.1. *Let X be a geometrically connected, smooth, quasi-projective variety over a p -adic field E .*

- (1) *Any object of $\mathfrak{J}^{\text{arith}}(X/E)$ belongs to some fundamental domain $\mathfrak{J}(X, E)_{\mathbb{C}_{p,t}^b}$.*
- (2) *The identification $y = x$ (of variables) provides a tautological equivalence of categories*

$$\mathfrak{J}^{\text{arith}}(X, E)_{\mathbb{C}_{p,y}^b} \simeq \mathfrak{J}^{\text{arith}}(X, E)_{\mathbb{C}_{p,x}^b}.$$

- (3) *In particular $\mathfrak{J}^{\text{arith}}(X, E)$ is tiled with copies of the fundamental domains $\mathfrak{J}(X, E)_{\mathbb{C}_{p,x}^b}$.*

§ 14.8 The self-similarity of $\mathbb{C}_p^b$ discussed in Section 14 at once implies that the group $\mathcal{G}(\mathcal{O}_{\mathbb{C}_p^b})$ is self-similar and this in turn has the following important consequence whose proof is clear from Theorem 14.1.1 and the preceding definitions and discussion:

Theorem 14.8.1. *Assume $F = \mathbb{C}_p^b$. Then*

- (1) $\tilde{\Theta}_{X,\ell} \subset B^{\varphi=p} \simeq \mathcal{G}(\mathcal{O}_F)$.

(2) The set $\tilde{\Theta}_{X,\ell}$ is a self-similar subset of $B^{\varphi=p}$: more precisely let

$$\tilde{\Theta}_{X,\ell,t} = \{z \in \mathcal{G}(\mathcal{O}_F) : \eta_K(z) \in f_\theta(C[\ell] - \{O\}) \subset K \text{ for some perfectoid field with } K^\flat = \mathbb{C}_{p,t}^\flat\}$$

Then

$$\tilde{\Theta}_{X,\ell} = \bigcup_{\mathbb{C}_{p,t}^\flat} \tilde{\Theta}_{X,\ell,t}$$

where the union runs over all the (isometric) identifications $\mathbb{C}_p^\flat = \mathbb{C}_{p,t}^\flat$.

(3) To put it colloquially, the θ -torsion value sets $\tilde{\Theta}_{X,\ell}$ form a fractal in $B^{\varphi=p} \simeq \mathcal{G}(\mathcal{O}_F)$.

Proof. After Theorem 14.1.1 it is enough to note that any $z \in \tilde{\Theta}_{X,\ell}$ lives in some $\tilde{\Theta}_{X,\ell,t}$ and so the assertion is immediate. $\square$

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